The Sharp Logarithmic Cost of Khatri–Rao Subspace Embeddings

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Abstract

For fixed tensor order $d \geq 2$, accuracy, confidence, and directional sub-Gaussian parameters, we establish the worst-case Khatri–Rao subspace embedding dimension $\Theta(r \log^{d-1}(er))$. A general upper bound and matching Gaussian, Rademacher, and spherical lower bounds determine the sharp rank logarithm. In contrast, bounded directional fourth moments suffice for $(1+\varepsilon)$-relative-error low-rank approximation and regression with $O(r)$ samples at fixed parameters. This distinction holds on the same input: approximation can succeed with high probability while the embedding condition fails. On specified Gaussian support classes, range-based outputs have exactly the dense-Gaussian distribution. Our analysis also yields explicit finite-sample moment certificates, polylogarithmic confidence bounds for raw regression under sub-Gaussian factors, and angular-moment bounds for expected full-range error.

Keywords: Khatri–Rao sketches; subspace embeddings; relative-error approximation; matrix moments; tensor geometry.

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1 Introduction

Khatri–Rao sketches represent each randomized measurement probe as a rank-one tensor $z = \bigotimes_{j=1}^{d} g^{(j)}$. Storing the small factor vectors instead of an ambient random matrix reduces storage, while Kronecker, canonical polyadic (CP), or tensor-train structure can make the sketch efficient to apply. These advantages motivate applications to tensor compression, polynomial kernels, regression, and eigensolvers [1, 31, 12, 7, 33].

The computational advantages raise two sample-complexity questions: how many probes are needed to preserve every vector in a fixed r-dimensional subspace, and must downstream approximation tasks use the same number? Beretta–Musco [5, Theorem 1.3] establish the nearly linear embedding bound $O(r \log^{d+1}(er))$ for sub-Gaussian factors at fixed parameters. We determine the sharp worst-case order $\Theta(r \log^{d-1}(er))$, saving two rank logarithms, and prove that the remaining logarithmic factor is necessary for Gaussian, Rademacher, and spherical factors.

Relative-error approximation has a different sample requirement. Embedding demands two-sided spectral control, including an upper bound that can fail because of a single large projected probe. The approximation guarantees instead follow from lower injection and control of the average interaction with the residual. Bounded directional fourth moments suffice for these two ingredients, yielding $(1+\varepsilon)$-relative-error low-rank approximation and regression with $O(r)$ samples at fixed parameters. We prove that approximation can succeed while embedding fails on the same input, and identify Gaussian supports on which range-based outputs have exactly the dense-Gaussian law.

1.1 Setting and guarantees

Let $N = \prod_{j=1}^{d} n_j$ and let the tensor order satisfy $d \geq 2$. We draw iid probes with independent isotropic factors:

$$z_i = \bigotimes_{j=1}^{d} g_i^{(j)} \in \mathbb{R}^N, \qquad \mathbb{E} g_i^{(j)} g_i^{(j)\mathsf{T}} = I_{n_j}, \qquad \Omega = m^{-1/2}[z_1 \ \cdots \ z_m], \quad \Pi = \Omega^\top. \tag{1.1}$$

The sketch size m counts probe vectors, equivalently columns of Ω or rows of Π. Coordinates within a factor may be dependent unless specified otherwise. Every input and subspace is deterministic before sampling. For two-sided embedding, we assume the directional sub-Gaussian condition

$$\mathbb{E} \exp\left(\frac{\langle u, g^{(j)} \rangle^2}{{K_{\mathrm{sg}}}^2} \right) \leq 2 \qquad (\|u\| = 1), \tag{1.2}$$

in addition to the isotropy and independence in (1.1). This fixes our directional sub-Gaussian convention. The approximation theorem needs only the weaker ambient condition

$$\mathbb{E} z z^\top = I_N, \qquad \mathbb{E} |\langle a, z \rangle|^4 \leq C_4 \|a\|^4 \quad (a \in \mathbb{R}^N). \tag{1.3}$$

Definition 1.1 (Embedding, injection, and relative approximation)**.** A sketch distribution is an *oblivious subspace embedding* (OSE) with squared-norm distortion ε and failure probability δ if, for every fixed isometry $U \in \mathbb{R}^{N \times r}$,

$$\mathbb{P}\{ \|U^\top \Omega \Omega^\top U - I_r\| \leq \varepsilon \} \geq 1 - \delta. \tag{1.4}$$

A *lower injection* with parameter $\alpha > 0$ requires only $U^\top \Omega \Omega^\top U \succeq \alpha I_r$ with that probability. For a fixed matrix A, a *relative approximation* satisfies $\|A - \widehat{A}\|_{\mathrm{F}}^2 \leq (1+\varepsilon)\|A - A_r\|_{\mathrm{F}}^2$, where A_r is a best rank-at-most-r approximation. For regression, relative error refers to the original least-squares residual, as stated in Theorem 4.5.

Table 1 separates the embedding and approximation guarantees. All rank-dependence comparisons fix tensor order, factor-law parameters, accuracy, and confidence.

Result	Factor law	Sketch size / guarantee
Two-sided embedding		
Bujanović et al. [7]	Gaussian, $d = 2$	$O(r^{3/2})$
Beretta–Musco [5]	Sub-Gaussian	$O(r \log^{d+1}(er))$
This work: matching bounds	Sub-Gaussian upper; Gaussian, Rademacher, spherical lower	$\Theta(r \log^{d-1}(er))$
Approximation		
Camaño et al. [10]	Fourth-moment product laws	$O(r)$, constant factor
Saibaba et al. [33]	Sub-Gaussian entries	$O(r^d)$, constant factor
Theorem 4.5	Bounded fourth moments	$O(r)$, $1+\varepsilon$ error
Proposition 4.12	Gaussian, specified supports	Exact Gaussian law despite failure on the leading subspace

Table 1: Rank dependence at fixed parameters. The Θ statement is worst-case over fixed subspaces and admissible mode dimensions; it does not assert a lower bound for every factor law. For $r \geq 2$, $\log(er)$ and $\log r$ give the same asymptotic order. The last row is a distributional comparison on the specified support class.

1.2 Why the logarithm appears

Write $y_i = U^\top z_i$ and $M = U^\top \Omega\Omega^\top U = m^{-1}\sum_i y_i y_i^\top$ for the Gram matrix restricted to the subspace. Since every summand is positive semidefinite,

$$\lambda_{\max}(M) \geq \frac{\max_i \|y_i\|^2}{m}.$$

Thus a single projected probe with $\|y_i\|^2 > (1+\varepsilon)m$ is enough to violate the embedding condition. The common-factor Gaussian witness makes the source of such a large probe explicit. For $Ue_a = e_a \otimes e_1 \otimes \cdots \otimes e_1$, the projected vector is

$$y_i = \left(\prod_{j=2}^{d} \eta_{i,j}\right) \xi_i, \qquad \xi_i \sim N(0, I_r), \quad \eta_{i,j} \sim N(0,1),$$

with all the displayed Gaussian variables independent. The scalar weight $w_i = \prod_{j=2}^{d} \eta_{i,j}^2$ multiplies the squared norm of the entire vector: $\|y_i\|^2 = w_i \|\xi_i\|^2$. A large value of w_i therefore amplifies all r coordinates simultaneously. At fixed d, the maximum of these weights has order $(\log m)^{d-1}$, while $\|\xi_i\|^2$ concentrates around r. This produces the necessary term $r \log^{d-1} r$ in the witness constructions. Independent Rademacher coordinates reproduce the same mechanism through rare alignment with a shared direction in modes of dimension proportional to $\log r$ (Proposition 2.4).

The upper bound must control these extremes uniformly over the target subspace, as well as the collective covariance fluctuations. Under the present moment assumptions, the covariance route gives, at its stated confidence,

$$\|M - I_r\| \lesssim_{d,K_{\mathrm{sg}}} \sqrt{r/m} + \frac{\max_i \|y_i\|^2}{m}.$$

Gaussian averaging bounds the largest squared projected row norm by $O_{d,K_{\mathrm{sg}}}(rs^{d-1} + s^d)$ with $s = \log(m/\beta)$ and failure probability β. The resulting term $(rt^{d-1} + t^d)/\varepsilon$ in (1.5) reflects the

large-row scale. Its term rt/ε^2 includes a logarithm from the trace-moment proof, rather than an unavoidable dense-Gaussian fluctuation cost: the covariance route reduces it to r/ε^2 when $\delta \geq 2/r$. These estimates are bounds, not an additive formula for the top eigenvalue; matching extremes are asserted only for the constructed witnesses.

Subspace geometry can reduce the shared fluctuation. In the order-two Gaussian common-Schmidt model, the weight becomes $w = \sum_b \lambda_b h_b^2$, and $L = \max_b \lambda_b$ gives the scale $1 + L \log m$ for its maximum, up to constants at fixed confidence. Spreading the weights therefore explains the transition $r[1 + L \log(er)]$.

Why approximation needs less. Relative-error approximation is controlled by two quantities: the inverse $\|M^{-1}\|$ on the leading subspace and a mean-square interaction between that subspace and the residual. Lower injection bounds the inverse, while fourth moments bound the interaction in Frobenius norm without estimating a sample maximum. Neither step requires an upper bound on $\lambda_{\max}(M)$, which allows $O(r)$ samples at fixed accuracy and confidence. This is a statement about the requirements of the error bound: a large leading eigenvalue can still accompany a large residual interaction, and the realized approximation error need not be monotone in the sample weights.

1.3 Main results

1. **Sharp embedding dimension.** With $t = \log(4r/\delta)$, Theorem 2.1 gives

$$m \geq C_{d,K_{\mathrm{sg}}} \left[\frac{rt}{\varepsilon^2} + \frac{rt^{d-1} + t^d}{\varepsilon} \right]. \tag{1.5}$$

   The covariance proof replaces rt/ε^2 by r/ε^2 when $\delta \geq 2/r$. Gaussian, Rademacher, and spherical constructions match the fixed-parameter rank order; the latter two use common-mode dimensions growing as $\log r$. Compared with Beretta–Musco [5, Theorem 1.3], the bound saves two rank logarithms and gives the large-row term accuracy cost ε^{-1}.

2. **Relative approximation with linear sketch size.** Theorem 4.5 gives rangefinding, fixed-rank output, and regression guarantees under bounded directional fourth moments, with

$$m = O\left(C_4 r + C_4 \log(2/\delta) + \frac{C_4 r}{\varepsilon\delta} \right). \tag{1.6}$$

   For range-based objectives, concatenating independent sketches and using range containment improves the confidence dependence to $O(C_4 r \varepsilon^{-1}[1 + \log(1/\delta)])$. For raw regression, the inverse-confidence cost is necessary over general fourth-moment laws, while sub-Gaussian product laws admit polylogarithmic confidence cost (Theorem 4.8).

3. **A separation on the same input.** At $m = O(r)$, Gaussian and Rademacher sketches achieve relative error on fixed inputs whose leading-space Gram matrices have a bounded lower edge and an excessive upper edge (Corollary 4.10). On Gaussian common-factor supports, Proposition 4.12 strengthens this to an exact dense-Gaussian law for range-based outputs, including the classical expected-error bound.

4. **Subspace geometry.** Theorem 3.2 identifies the order-two common-Schmidt transition $r[1 + L \log(er)]$, where L is the largest Schmidt weight. Input-dependent approximation bounds retain the leading-space fourth moment and its interaction with the residual. Spherical normalization improves moment constants while retaining the worst-case logarithm.

Section 2 proves the sharp embedding order, giving the covariance proof and stating a separate explicit moment certificate; the detailed rank-one argument is in Appendix A. Section 3 explains the effect of subspace geometry and spherical normalization. Section 4 develops approximation, the same-input separation, and exact range laws. Numerics and discussion follow in Sections 5–6.

1.4 Related work

Tensor sketches. Independent sign-product sketches are analyzed by Meister–Sarlós–Woodruff [26]; direct product and recursive polynomial-kernel sketches are analyzed by Ahle et al. [1]; Rakhshan–Rabusseau [31] study tensorized random projections. Fast Kronecker transforms and Kronecker sign constructions appear in [22, 3], with sampling laws different from the independent product probes here. The Gaussian OSE analysis of Bujanović et al. [7], the rangefinding and tensor algorithms of Saibaba et al. [33], and Beretta–Musco's general OSE theorem [5] give the direct comparisons in Table 1. The present upper and lower bounds determine the fixed-parameter worst-case rank dependence. TTStack interpolates between Khatri–Rao and Gaussian tensor-train sketches and gives OSE and injection budgets linear in tensor order through a richer sketch family [11]. These guarantees do not remove the obstruction for the independent rank-one product probes studied here.

Covariance and moments. Rank-one covariance approximation is developed in [32, 29]; scalar moment inequalities for sums are studied in [24]. Gaussian pairings and trace inequalities are classical [20, 21, 37]. Our matrix argument retains the rank-one direction, preserves centering in the first two insertion orders, and uses only a logarithmic terminal moment order. The covariance proof applies Tikhomirov's covariance theorem [35]; Bousquet's empirical-process inequality [6] supplies arbitrary confidence. Gaussian smoothing and Catoni's influence function give the optimized lower-covariance profile [30, 42]; its rate and method are classical. Matrix-chaos inequalities also include Khatri–Rao applications [4]. Sparse-chaos estimates of Dai–He–Wang–Zhu [15] account for sparsity and input geometry in general; their scope differs from the dense worst-case rank dependence considered here.

Approximation. Lower covariance and small-ball arguments appear in [30, 23]; Camaño et al. [10, Theorem 5.1] give fourth-moment Khatri–Rao injection. The range, inverse-moment, and regression reductions come from [34, 13, 18, 17, 14, 38]; see also tensor regression and rank-one estimation in [12, 8]. Range invariance for Gaussian and spherical factors is noted in [10, Section 5.4]. We use these ingredients to obtain explicit fourth-moment relative guarantees and a same-input embedding separation, and identify support classes on which the full range output has the dense-Gaussian law.

The August 2026 update to the Simons workshop problems [2, Problems 5.1–5.2] records the counterexamples of Townsend–Wang [36] and asks for convenient additional assumptions. Our fourth-moment theorem supplies one sufficient condition for product probes; it does not identify the weakest possible condition.

1.5 Notation and scope

Vectors use the Euclidean norm and matrices the spectral norm; $\|\cdot\|_{\mathrm{F}}$ is the Frobenius norm. Traces are unnormalized, $\preceq$ is the Loewner order, and the dagger denotes the Moore–Penrose inverse. $P_{\mathcal{L}}$ is the orthogonal projector onto $\mathcal{L}$.

Symbol	Meaning
d, n_j, N	Tensor order, mode dimensions, and ambient dimension.
r, m	Target rank and sketch size.
Ω, Π	Normalized probe matrix and sketching map $\Pi = \Omega^\top$.
K_{sg}, C_4	Directional sub-Gaussian and fourth-moment parameters.
$q, \mathfrak{g}_q$	Half of the terminal trace-moment order and the scalar Gaussian moment $(2q-1)!!$.
Z, Δ_Ω	Restricted covariance error and ambient covariance error.
H_i, X_i	Projected covariance sample and its centered version.
$C_Q, \kappa(Q, W)$	Leading-space fourth moment and residual interaction.
$\mathfrak{m}_{\mathrm{inj}}$	Sample budget for lower injection.
$\tau_r^{\mathrm{F}}(A), \tau_r^{\mathrm{tr}}(K)$	Squared Frobenius tail $\|A - A_r\|_{\mathrm{F}}^2$ and PSD trace tail $\sum_{j>r} \lambda_j(K)$.
$\nu, \mathfrak{r}$	Variance bound and higher-moment radius in the appendix comparison.
λ_b, L, ρ	Schmidt weights, their maximum, and their purity $\sum_b \lambda_b^2$.

Constants depend only on their displayed subscripts. Inputs are fixed before sampling, and the guarantees concern exact arithmetic.

2 Sharp embedding dimension

Two effects govern the embedding dimension: accumulated covariance fluctuations and unusually large projected rows. The covariance route controls both and proves the sharp fixed-parameter order. A separate rank-one moment certificate supplies explicit constants and a complementary high-confidence bound; its proof is deferred to Appendix A. Gaussian and Rademacher witnesses show that a single large row can force the logarithmic overhead; spherical factors are treated in Section 3.

Work in the tensor model of Section 1.1, under the sub-Gaussian condition (1.2). Fix an isometry $U : \mathbb{R}^r \to \mathcal{H} = \bigotimes_{j=1}^d \mathbb{R}^{n_j}$, and set

$$y_i = U^\mathsf{T} \bigotimes_{j=1}^{d} g_i^{(j)}, \qquad H_i = y_i y_i^\mathsf{T}, \qquad X_i = H_i - I_r, \qquad Z = \frac{1}{m}\sum_{i=1}^{m} X_i. \tag{2.1}$$

Independence across the factors gives $\mathbb{E}H_i = I_r$. The sketch with rows $m^{-1/2}(\bigotimes_j g_i^{(j)})^\mathsf{T}$ embeds $\mathrm{range}(U)$ with squared-norm distortion at most ε precisely when $\|Z\| \le \varepsilon$.

Theorem 2.1 (Sharp dimension order)**.** *For every fixed $d \ge 2$ and K_{sg}, there is a constant $C_{d,K_{\mathrm{sg}}}$ such that, for $0 < \varepsilon, \delta < 1$ and $t = \log(4r/\delta)$,*

$$m \ge C_{d,K_{\mathrm{sg}}} \left[\frac{rt}{\varepsilon^2} + \frac{rt^{d-1} + t^d}{\varepsilon}\right] \quad \Longrightarrow \quad \mathbb{P}\{\|Z\| > \varepsilon\} \le \delta. \tag{2.2}$$

For each of the standard Gaussian and independent-coordinate Rademacher factor laws, there are deterministic r-dimensional subspaces and $c_d > 0$ for which

$$\inf_{1 \le m \le c_d r(\log r)^{d-1}} \mathbb{P}\{\|Z\| > 1/2\} \longrightarrow 1 \qquad (r \longrightarrow \infty). \tag{2.3}$$

For either law, and for any sub-Gaussian class containing one of them, the worst-case embedding dimension over admissible mode dimensions and deterministic r-dimensional subspaces is $\Theta_{d,K_{\mathrm{sg}},\varepsilon,\delta}(r\log^{d-1}(er))$ as $r \to \infty$, with $d, K_{\mathrm{sg}}, \varepsilon, \delta$ fixed.

Upper bound and conclusion of Theorem 2.1. For the upper bound, take $q = \lceil t \rceil$ in Proposition 2.6. Its parameters satisfy $h_q \le \sqrt{2q}$, $\eta_q \le \sqrt{e}$, and $c_q \le \sqrt{q}$ (the last inequality is AM–GM on $1, 3, \ldots, 2q-1$), with $q \le 2t$. Thus, outside an event of probability δ,

$$\|Z\| \le \sqrt{e}\sqrt{vq/m} + \sqrt{2e}\, q\mathfrak{r}_q/m.$$

Here $v = O_{d,K_{\mathrm{sg}}}(r)$ and $q\mathfrak{r}_q = O_{d,K_{\mathrm{sg}}}(rq^{d-1} + q^d)$. The square-root term controls covariance fluctuations, while the term proportional to m^{-1} accounts for large projected rows. Choosing the constant in (2.2) so that each term is at most $\varepsilon/2$ proves the claimed upper bound. For $d = 2$, the term rt/ε is absorbed by rt/ε^2. At fixed accuracy and confidence, this gives $O_{d,K_{\mathrm{sg}},\varepsilon,\delta}(r \log^{d-1}(er))$.

The Gaussian construction in Section 2.2.1 and the Rademacher construction in Proposition 2.4 prove the lower bound. In fact, their covariance error is at least one with probability tending to one uniformly over the indicated widths. They therefore rule out every fixed $\varepsilon < 1$ and $\delta < 1$, which proves the matching worst-case order. □

Section 2.1 gives an independent covariance proof of the fixed-parameter upper order, followed by the matching lower bounds in Section 2.2. Section 2.3 closes the section with the explicit moment certificate used above; its rank-one comparison argument is developed in Appendix A. The moment route improves the ε^{-1} term of the general-confidence covariance bound when $\delta < 2/r$.

2.1 Directional moments and covariance bounds

Isotropy fixes the quadratic moment of every unit projection. Subtracting this known contribution, together with the constant term, from the exponential moment leaves a smaller bound for the higher moments. Set

$$\theta = 1 - {K_{\mathrm{sg}}}^{-2}, \qquad \beta_p = \sup_{x>0} \frac{x^p}{e^x - 1 - x} \quad (p \geq 2). \tag{2.4}$$

Here $\beta_2 = 2$ and $\beta_p \leq p!$. Retaining the full exponential remainder, instead of a single Taylor coefficient, improves every higher scalar budget. Jensen's inequality gives ${K_{\mathrm{sg}}}^2 \geq 1/\log 2$, so θ is positive. The proof then applies the same scalar estimate successively to the independent tensor factors.

Lemma 2.2 (Isotropy-anchored moments)**.** *For an isotropic factor g satisfying (1.2), every $H \succeq 0$ and integer $p \geq 2$ satisfy*

$$\mathbb{E}(g^{\mathsf{T}} H g)^p \leq \theta \beta_p {K_{\mathrm{sg}}}^{2p} (\operatorname{tr} H)^p. \tag{2.5}$$

Consequently, for every deterministic tensor $a \in \mathcal{H}$,

$$\mathbb{E} \left| \langle a, \bigotimes_{j=1}^{d} g^{(j)} \rangle \right|^{2p} \leq [\theta \beta_p {K_{\mathrm{sg}}}^{2p}]^d \|a\|^{2p}. \tag{2.6}$$

Proof. If $H \neq 0$, its normalized eigenvalues are convex weights. Convexity of the exponential and (1.2) give $\mathbb{E}e^Q \leq 2$, where $Q = g^{\mathsf{T}} H g / ({K_{\mathrm{sg}}}^2 \operatorname{tr} H)$. Isotropy gives $\mathbb{E}Q = {K_{\mathrm{sg}}}^{-2}$. Thus $\mathbb{E}(e^Q - 1 - Q) \leq \theta$, and $Q^p \leq \beta_p(e^Q - 1 - Q)$ proves (2.5). The case $H = 0$ is immediate.

More generally, for a linear map T from $\mathcal{H}$ to a Euclidean space, induction on the number of factors proves

$$\mathbb{E}\|T(\bigotimes_{j} g^{(j)})\|^{2p} \leq [\theta \beta_p {K_{\mathrm{sg}}}^{2p}]^d \|T\|_{\mathrm{F}}^{2p}.$$

Indeed, after conditioning on the last factor, apply the induction hypothesis to the partially contracted map T_g. Its squared Frobenius norm is $g^{\mathsf{T}} H g$ for a positive semidefinite H with $\operatorname{tr} H = \|T\|_{\mathrm{F}}^2$. Apply (2.5) to finish the induction. A scalar target gives (2.6). □

For Gaussian averaging, write the scalar and radial moments as follows.

$$\begin{aligned} \mathfrak{g}_p &= \mathbb{E}G^{2p} = (2p-1)!!, & G &\sim N(0,1), \quad \mathfrak{g}_0 = 1, \\ D_{r,p} &= \mathbb{E}\|G_r\|^{2p} = \prod_{h=0}^{p-1} (r + 2h), & G_r &\sim N(0, I_r), \quad D_{r,0} = 1. \end{aligned} \tag{2.7}$$

The covariance argument. A classical covariance estimate proves the sharp fixed-parameter upper order.

Theorem 2.3 (Sample covariance bounds)**.** *Under the assumptions of Theorem 2.1, put* $t = \log(4r/\delta)$ *and* $u = \log(8/\delta)$*. There is a constant* $C_{d,K_{\mathrm{sg}}}$ *such that, if* $\delta \geq 2/r$*, then*

$$m \geq C_{d,K_{\mathrm{sg}}} \left[\frac{r}{\varepsilon^2} + \frac{rt^{d-1} + t^d}{\varepsilon} \right] \quad \Longrightarrow \quad \mathbb{P}\{\|Z\| > \varepsilon\} \leq \delta. \tag{2.8}$$

For every $0 < \varepsilon, \delta < 1$*, one also has*

$$m \geq C_{d,K_{\mathrm{sg}}} \left[\frac{r+u}{\varepsilon^2} + \frac{u(rt^{d-1} + t^d)}{\varepsilon} \right] \quad \Longrightarrow \quad \mathbb{P}\{\|Z\| > \varepsilon\} \leq \delta. \tag{2.9}$$

The covariance input is Tikhomirov's theorem [35, Corollary 2]; Bousquet's inequality [6, Theorem 2.3] supplies the second confidence bound. The proof of both bounds is in Appendix A.1. At fixed confidence, (2.8) removes the logarithm from the variance term. At arbitrary confidence, (2.9) pays an additional confidence logarithm in the term proportional to ε^{-1}. The finite-moment certificate (2.17) provides explicit constants.

Covariance proof of the fixed-parameter upper bound. We prove the upper order in Theorem 2.1 directly. Let $G \sim N(0, I_r)$ be independent of y. Integrate $|\langle G, y\rangle|^{2q}$ in the two orders using Lemma 2.2. Since $\mathfrak{g}_q \geq q!$ and $D_{r,q} \leq (r+2q)^q$,

$$(\mathbb{E}\|y\|^{2q})^{1/q} \leq {K_{\mathrm{sg}}}^{2d} q^{d-1}(r+2q), \qquad q \geq 2.$$

Markov and a union bound with $q \asymp \log(m/\delta)$ control the maximum of the m squared row norms. Independent random signs center the rows without changing their Gram matrices. Their directional sixth moments are bounded, so Tikhomirov's Corollary 2, for $m \geq 2r$, gives

$$\|Z\| \leq C_{d,K_{\mathrm{sg}}} \left[\sqrt{r/m} + m^{-1} \max_i \|y_i\|^2 \right]$$

outside an event of probability $1/r$. At fixed ε, δ, the choice $m \asymp r \log^{d-1}(er)$ makes this at most ε with failure at most δ for all sufficiently large r. A larger fixed constant handles the finitely many smaller ranks. Appendix A.1 closes the sample-size logarithm uniformly in accuracy and gives both displayed confidence bounds, using Bousquet for the second one. □

Where the two logarithms are saved. Beretta–Musco's proof first bounds squared row norms by $O(r \log^d(m/\delta))$, then applies matrix Chernoff, paying another $\log(r/\delta)$ [5, Section 2]. Gaussian averaging above divides the directional moment by $\mathfrak{g}_q$, reducing the leading row scale to $rq^{d-1} + q^d$. The covariance estimate then uses the maximum row divided by m, without a further rank logarithm. Proposition 2.6 gives the complementary explicit moment certificate.

2.2 Why the logarithms are necessary

Both constructions separate two regimes. When $m < r$, rank deficiency forces covariance error at least one. When $m \geq r$, a single large sample suffices: positivity gives

$$\lambda_{\max}\left(m^{-1} \sum_i y_i y_i^\top \right) \geq m^{-1} \max_i \|y_i\|^2.$$

On a common-factor subspace, y_i is an isotropic r-vector multiplied by a product of scalar projections. Gaussian extremes amplify this common scalar weight. For signs, prescribing all h coordinates in one factor to be $+1$ has probability 2^{-h} and produces a squared projection of size h. Choosing h proportional to $\log r$ makes simultaneous alignment across the common factors likely among $m \geq r$ rows, at the same logarithmic scale as the Gaussian extreme.

2.2.1 Gaussian factors

Gaussian lower bound in Theorem 2.1. Choose $Ue_a = e_a \otimes e_1 \otimes \cdots \otimes e_1$. For standard Gaussian factors the projected row has the form

$$y_i = \left(\prod_{\ell=1}^{d-1} \eta_{i\ell} \right) \xi_i, \qquad \xi_i \sim N(0, I_r), \quad \eta_{i\ell} \sim N(0,1),$$

with all variables independent. Write $w_i = \prod_{\ell=1}^{d-1} \eta_{i\ell}^2$. If $m < r$, the sample covariance has a zero eigenvalue, so $\|Z\| \geq 1$.

For $m \geq r$ sufficiently large, put $a = \sqrt{\log m/(d-1)}$. Integration of the Gaussian density on $[a, a + 1/a]$ gives

$$\mathbb{P}\{\eta^2 \geq a^2\} \geq c_d m^{-1/[2(d-1)]} (\log m)^{-1/2}.$$

Indeed, the interval has length $1/a$, and its minimum density is $(2\pi)^{-1/2} \exp[-(a+1/a)^2/2]$; the remaining factor is bounded below for $a \geq 1$. Independence implies

$$\mathbb{P}\left\{ \max_i w_i < \left(\frac{\log m}{d-1} \right)^{d-1} \right\} \leq \exp\left[-c'_d \frac{\sqrt{m}}{(\log m)^{(d-1)/2}} \right].$$

Choose a maximizing index J using only the weights. Conditional on the weights, ξ_J is still a standard Gaussian vector. Moreover,

$$\mathbb{P}\{\|\xi_J\|^2 < r/2\} \leq e^{r/4} 2^{-r/2} \leq e^{-r/16},$$

by Markov applied to $e^{-\|\xi_J\|^2/2}$. Positivity of every summand consequently gives, outside these two failure events,

$$\lambda_{\max}\left(\frac{1}{m} \sum_i H_i \right) \geq \frac{r}{2m} \left(\frac{\log m}{d-1} \right)^{d-1}.$$

For $r \leq m \leq [4(d-1)^{d-1}]^{-1} r (\log r)^{d-1}$, this is at least 2, so $\|Z\| \geq 1$. Both exceptional probabilities tend to zero uniformly in this range. For example, throughout the range, $\sqrt{m} \geq \sqrt{r}$ and $\log m \leq 2 \log r$ for large r. This proves (2.3). Finally, $\mathbb{E}e^{G^2/{K_{\mathrm{sg}}}^2} = (1 - 2/{K_{\mathrm{sg}}}^2)^{-1/2} \leq 2$ precisely when ${K_{\mathrm{sg}}}^2 \geq 8/3$, so this witness belongs to every factor class with such a Gaussian-admitting parameter. □

2.2.2 Rademacher factors

Proposition 2.4 (A bounded-entry obstruction)**.** *Fix $d \geq 2$. For every sufficiently large integer r, there is a deterministic isometry $U_r : \mathbb{R}^r \to \bigotimes_{j=1}^d \mathbb{R}^{n_j}$, with $n_1 = r$ and $n_2 = \cdots = n_d = O_d(\log r)$, such that independent factors with independent Rademacher coordinates satisfy*

$$\inf_{1 \leq m \leq c_d r (\log r)^{d-1}} \mathbb{P}\left\{ \left\| \frac{1}{m} \sum_{i=1}^m y_i y_i^{\mathsf{T}} - I_r \right\| > \frac{1}{2} \right\} \geq 1 - \exp(-\sqrt{r}), \qquad y_i = U_r^{\mathsf{T}} \bigotimes_{j=1}^d g_i^{(j)}, \tag{2.10}$$

where the infimum is over integer m and one may take

$$c_d = \frac{1}{2[4(d-1)\log 2]^{d-1}}.$$

These factors obey (1.2) for every ${K_{\mathrm{sg}}}^2 \geq 8/3$. Consequently, the worst-case dimension order in Theorem 2.1 is also necessary when the factor law is restricted to independent Rademacher coordinates.

Proof. For $m < r$, rank deficiency makes the covariance deviation at least 1 deterministically. It remains to consider $m \geq r$. Take $r \geq 2^{4(d-1)}$, and put

$$h = \left\lfloor \frac{\log_2 r}{2(d-1)} \right\rfloor, \qquad b = h^{-1/2}(1, \dots, 1)^{\mathsf{T}} \in \mathbb{R}^h, \qquad U_r e_a = e_a \otimes b^{\otimes(d-1)} \quad (1 \leq a \leq r).$$

The images of the coordinate vectors are orthonormal. Set $n_1 = r$ and $n_j = h$ for $j \geq 2$. With $\xi_i = g_i^{(1)}$ and $s_{ij} = \langle b, g_i^{(j)} \rangle$, the projected samples are

$$y_i = \left(\prod_{j=2}^{d} s_{ij} \right) \xi_i, \qquad \|\xi_i\|^2 = r.$$

Let E_i be the event that all h coordinates of $g_i^{(j)}$ are $+1$ for every $2 \leq j \leq d$. The events E_i are independent, and

$$\mathbb{P}(E_i) = 2^{-h(d-1)} \geq r^{-1/2}.$$

On E_i one has $s_{ij} = \sqrt{h}$, hence $\|y_i\|^2 = rh^{d-1}$. Every sample covariance summand is positive semidefinite, so the occurrence of any E_i implies

$$\lambda_{\max} \left(\frac{1}{m} \sum_{i=1}^{m} y_i y_i^{\mathsf{T}} \right) \geq \frac{rh^{d-1}}{m}. \tag{2.11}$$

The assumption on r gives $h \geq \log r / [4(d-1)\log 2]$. Thus the right side of (2.11) is at least 2 throughout $m \leq c_d r (\log r)^{d-1}$. The probability that no E_i occurs is at most

$$(1 - r^{-1/2})^m \leq \exp(-m/\sqrt{r}) \leq \exp(-\sqrt{r}).$$

Thus the covariance deviation is at least 1 except on an event of probability at most $\exp(-\sqrt{r})$. Together with the rank-deficient regime, this proves the uniform bound.

Finally, Rademacher vectors are isotropic and satisfy $\mathbb{E}e^{t\langle g,u\rangle} \leq e^{t^2/2}$ for unit u by Hoeffding's bound [19]; Gaussian integration gives $\mathbb{E}e^{\langle g,u\rangle^2/{K_{\mathrm{sg}}}^2} \leq (1 - 2/{K_{\mathrm{sg}}}^2)^{-1/2} \leq 2$ when ${K_{\mathrm{sg}}}^2 \geq 8/3$, verifying (1.2). □

Remark 2.5. The witness uses only entries of absolute value one. Its logarithmic overhead comes from directional alignment across tensor factors: $\langle b, g \rangle$ can attain $\sqrt{h}$ when its h coordinates align. The prescribed simultaneous alignment is rare for an individual row, but occurs with probability tending to one among $m \geq r$ independent rows for the chosen h. The dimensions of the common factors grow with r; the proposition does not address a restriction that keeps those mode dimensions fixed. Coordinatewise boundedness therefore does not supply a bound $\|y_i\|^2 \leq C_d r$ uniform in r.

An accuracy-sensitive lower obstruction. The heavy-row argument can retain the unit background instead of discarding it. Proposition A.12 proves that, for Gaussian factors,

$$m = \Omega_d \left(\frac{r(\log r)^{d-1}}{\varepsilon_r} \right)$$

is necessary at fixed confidence whenever $\varepsilon_r r (\log r)^{d-1} \to \infty$. Its finite failure bound and proof are in Appendix A.6. This refinement does not determine the full joint dependence on confidence and accuracy.

2.3 An explicit moment certificate

We finish with numerical constants for the moment bound used in the upper-bound argument for Theorem 2.1. The covariance bound in Theorem 2.3 has a smaller variance term when $\delta \geq 2/r$; the moment argument avoids the additional confidence logarithm in the large-row term of (2.9).

Set

$$s_r = \frac{r + 2 + \sqrt{3r(r+2)}}{6}, \qquad v = [2\theta {K_{\mathrm{sg}}}^4]^d s_r. \tag{2.12}$$

A quadratic Gaussian averaging argument, proved in Appendix A.2, gives

$$\mathbb{E}H^2 \preceq vI_r, \qquad V := \mathbb{E}(H - I_r)^2 \preceq (v-1)I_r. \tag{2.13}$$

Here $s_1 = 1$ and $(1+\sqrt{3})r/6 \leq s_r \leq r$. The sharper fourth-moment constant in Lemma A.2 is used in the individual moment profiles and approximation bounds; the all-order factorial certificate retains the normalization v.

Proposition 2.6 (Explicit finite-moment certificate)**.** *Under the factor assumptions of Theorem 2.1, use v from (2.12). For an integer $q \geq 1$ and $0 < \delta < 1$, put*

$$\mathcal{R}_q = \tfrac{1}{2}{K_{\mathrm{sg}}}^{2d}(2q)^{d-2}(r+2q), \qquad \mathfrak{r}_q = \mathcal{R}_q + 1 + \frac{4}{3v}, \tag{2.14}$$

$$h_1 = 0, \qquad h_q = \frac{2(q-1)}{\sqrt{2q-3}} \quad (q \geq 2),$$

$$c_q = \mathfrak{g}_q^{1/(2q)}, \qquad \eta_q = (r/\delta)^{1/(2q)}. \tag{2.15}$$

Then

$$\mathbb{P}\left\{\|Z\| > \eta_q c_q \left[\sqrt{v/m} + h_q \mathfrak{r}_q/m\right]\right\} \leq \delta. \tag{2.16}$$

In particular, writing $\mathsf{a}_q = \eta_q c_q \sqrt{v}$ and $\mathsf{b}_q = \eta_q c_q h_q \mathfrak{r}_q$, squared-norm distortion ε is ensured with failure at most δ by

$$m \geq \left\lceil \left(\frac{\mathsf{a}_q + \sqrt{\mathsf{a}_q^2 + 4\varepsilon \mathsf{b}_q}}{2\varepsilon} \right)^2 \right\rceil. \tag{2.17}$$

The integer q may be selected to minimize this deterministic certificate.

The proof is in Appendix A.4, after the moment estimates in Appendices A.2–A.3. It retains the matrix direction when converting scalar to rank-one moments. Centering removes single insertions; two insertions retain the variance, and higher insertions are bounded by a positive moment envelope. Gaussian pairing coefficients solve the resulting scalar recurrence. Only directional moments through order $4q$ are needed. The upper-bound argument following Theorem 2.1 makes the choice $q = \lceil \log(4r/\delta) \rceil$ to obtain its all-confidence bound. The constants above belong to this moment certificate, rather than to the covariance theorem.

For Gaussian, independent-sign, or spherical factors, Proposition A.10 gives the alternative $v_{\mathrm{G}} = 3^d r$ and the smaller law-specific radius (A.37). Appendix A.5 also retains individual moment budgets, zero initial moments, and centered terminal insertions. Its discrete and integral certificates retain these finer inputs without new probabilistic assumptions.

3 Geometry beyond the worst case

The following supports isolate the scalar amplification behind the worst-case logarithm. On a common-factor support, alignment in the shared modes amplifies every projected coordinate at once. The common-Schmidt construction replaces this product weight by a weighted sum of squared contractions. For order-two Gaussian probes, the largest Schmidt weight determines how much of the logarithmic penalty remains.

3.1 Common-factor and common-Schmidt supports

Definition 3.1 (Common-factor and common-Schmidt supports)**.** Let $z = g \otimes h$, where $g \sim N(0, I_{n_1})$ is independent of $h = h_2 \otimes \cdots \otimes h_d$ with independent standard Gaussian factors. For the following supports the projected probe, conditional on h, is $N(0, w(h)I_R)$ with $w(h) > 0$. Its line is therefore uniform.

1. For orthonormal $a_1, \dots, a_R$ and fixed unit vectors $b_2, \dots, b_d$, the common-factor basis
$$v_a = a_a \otimes b_2 \otimes \cdots \otimes b_d$$
has $w(h) = \prod_{j=2}^{d} \langle b_j, h_j \rangle^2 > 0$ almost surely.
2. More generally, let $f_1, \dots, f_\ell$ be orthonormal vectors in the remaining tensor modes, let $\lambda_b > 0$ with $\sum_b \lambda_b = 1$, and let the $R\ell$ vectors $e_{a,b}$ be jointly orthonormal in the first mode. Then
$$v_a = \sum_{b=1}^{\ell} \sqrt{\lambda_b}\, e_{a,b} \otimes f_b, \qquad a = 1, \dots, R,$$
are orthonormal and have $w(h) = \sum_b \lambda_b \langle f_b, h \rangle^2 > 0$ almost surely.

Projected row law. In the first case $\langle v_a, z \rangle = \langle a_a, g \rangle \prod_{j=2}^{d} \langle b_j, h_j \rangle$. In the second, conditional covariance between coordinates a, a' is

$$\sum_{b,c} \sqrt{\lambda_b \lambda_c} \langle f_b, h \rangle \langle f_c, h \rangle \langle e_{a,b}, e_{a',c} \rangle = \mathbf{1}_{\{a=a'\}} \sum_b \lambda_b \langle f_b, h \rangle^2.$$

The coordinates are jointly Gaussian conditional on h. Positivity holds because the contraction with any nonzero f_b is a nonzero polynomial in Gaussian factors: product tensors span the ambient tensor space, and a nonzero polynomial has a Gaussian zero set of measure zero.

3.2 The common-Schmidt transition

The worst-case logarithm can be located more precisely on the order-two common-Schmidt supports of Definition 3.1. Write their Schmidt weights as $\lambda_b \geq 0$, $\sum_b \lambda_b = 1$, and put $L = \max_b \lambda_b$ and $\rho = \sum_b \lambda_b^2$. Their projected Gaussian row has the distribution

$$y = \sqrt{w}\, g, \qquad w = \sum_b \lambda_b h_b^2, \qquad g \sim N(0, I_r), \quad h_b \sim N(0, 1),$$

with g independent of the weight. Direct Gaussian moments give

$$\mathbb{E}(yy^\top - I_r)^2 = \big[r + 1 + 2(r+2)\rho\big] I_r. \tag{3.1}$$

The variance depends on the quadratic weight sum ρ, but the embedding width also depends on extreme rows. The largest Schmidt weight L controls the size of a rare row that can violate the upper spectral bound.

Theorem 3.2 (Common-Schmidt transition)**.** *For independent rows of the order-two common-Schmidt law, a universal constant C has the following property. For $0 < \varepsilon, \delta < 1$, write $\tau_\delta = r + \log(8/\delta)$. Then*

$$m \geq C \left[\frac{\tau_\delta}{\varepsilon^2} + \frac{L\tau_\delta}{\varepsilon} \log \frac{e\tau_\delta}{\varepsilon\delta} \right] \quad \Longrightarrow \quad \mathbb{P}\left\{ \left\| m^{-1} \sum_i y_i y_i^\top - I_r \right\| > \varepsilon \right\} \leq \delta. \tag{3.2}$$

In particular, for fixed accuracy and confidence, the sufficient width is $C_{\varepsilon,\delta} r[1 + L\log(er)]$. Conversely, for a universal $c > 0$, the probability of exceeding the same fixed distortion ε tends to one uniformly over the Schmidt weights and $1 \leq m \leq cr[1 + L\log(er)]$ as $r \to \infty$.

The proof in Appendix A.7 first derives a conditional weighted-Gaussian upper bound and then uses one extreme row for the matching lower bound. At $L = 1$, only one Schmidt weight is nonzero, so the support reduces to a common-factor support and the theorem recovers the sharp $\Theta_{\varepsilon,\delta}(r \log(er))$ width. For flat weights $\lambda_b = 1/s$, the scalar weight w is an average of s independent squared Gaussians, and the sharp width is $\Theta_{\varepsilon,\delta}(r[1+\log(er)/s])$. Thus spreading the Schmidt spectrum removes the rank logarithm once $s \gtrsim \log r$, at fixed accuracy and confidence.

3.3 Spherical normalization

Spherical factors remove Gaussian radial randomness. Averaging it out can only reduce convex trace moments. Alignment of directions remains, which is why the worst-case logarithm survives.

For independent uniform unit-sphere vectors $u^{(j)} \in \mathbb{R}^{n_j}$, use isotropic factors $x^{(j)} = \sqrt{n_j}\, u^{(j)}$. Gaussian polar decomposition and conditional Jensen transfer every centered even trace moment of their projected sample covariance from the corresponding Gaussian tensor model. Consequently the upper bound in Theorem 2.1 holds for these spherical factors, with the Gaussian constants. The directional fourth constant of one factor improves to $3n_j/(n_j + 2)$.

This finite-dimensional gain does not change the worst-case rank order: for fixed $d \geq 2$ there are common-factor supports with $n_1 = r$ and $n_2 = \cdots = n_d = O_d(\log r)$ for which spherical factors require $\Omega_d(r \log^{d-1} r)$ rows. Their first-mode row length is constant, but rare common coordinates still produce a large common weight. Appendix A.7 gives the moment transfer and the matching obstruction. Uniformity of the individual factors does not imply uniformity of the direction after projection onto an arbitrary tensor subspace.

4 Approximation without embedding

For each fixed input, bounded directional fourth moments suffice for relative-error rangefinding, low-rank approximation, and regression with sketch size linear in the target rank at fixed accuracy and confidence. Use iid isotropic probes with the normalization (1.1). The first results assume (1.3); Corollary 4.6 later replaces this global condition by moments on the leading space and its residual interaction. The mechanism is visible in SVD coordinates. If the leading block Ω_1 has full row rank, the rank-r competitor $B_* = (A\Omega)\Omega_1^{\dagger} V_1^{\top}$ satisfies

$$\|A - B_*\|_{\mathrm{F}}^2 = \|A - A_r\|_{\mathrm{F}}^2 + \|\Sigma_2 \Omega_2 \Omega_1^{\dagger}\|_{\mathrm{F}}^2.$$

Lower injection controls the inverse factor, while orthogonal fourth moments control the interaction between the leading and trailing singular directions. These estimates play separate roles: only $\|M^{-1}\|$ is used, with M the leading Gram matrix, so no upper spectral bound on M is needed. The residual interaction is controlled in mean square and may be dependent on M. The deterministic reduction and approximate-matrix-product step are classical [34, 13, 18, 14]. The contribution here is a single product fourth-moment hypothesis that supplies both lower injection and orthogonal residual control, with explicit constants.

Lemma 4.1 (Product fourth moments)**.** *If independent isotropic factors x_j obey directional fourth-moment bounds c_j, their product $z = x_1 \otimes \cdots \otimes x_d$ obeys (1.3) with $C_4 = \prod_j c_j$. Independent centered, unit-variance coordinates with fourth moment μ_4 have $c_j = \max\{3, \mu_4\}$, by expanding the fourth power. Thus Gaussian and sign factors admit $c_j = 3$, including four-wise independent sign coordinates, while standardized sparse signs of density ζ_j admit $c_j = \max\{3, \zeta_j^{-1}\}$. Under (1.2), one may instead use $c_j = c_{\mathrm{sg}}$ from Lemma A.2. For fully independent signs and normalized spherical factors, the respective exact constants are $3 - 2/n_j$ and $3n_j/(n_j + 2)$; their higher moments are given in Appendix A.5.3.*

Proof. Use the Hilbert-valued extension: for coefficient vectors a_b,

$$\mathbb{E}\left(\sum_b \langle a_b, x_j\rangle^2\right)^2 \leq c_j \left(\sum_b \|a_b\|^2\right)^2,$$

by Cauchy–Schwarz on each pair of components. Successive contractions of the coefficient tensor, conditioning on one independent factor at a time, give the claimed product constant; the same induction gives isotropy. For centered independent coordinates ξ_b with unit variance and $\mathbb{E}\xi_b^4 \le \mu_4$, expansion gives

$$\mathbb{E}\Big(\sum_b a_b\xi_b\Big)^4 = 3\Big(\sum_b a_b^2\Big)^2 + \sum_b(\mathbb{E}\xi_b^4 - 3)a_b^4 \le \max\{3,\mu_4\}\|a\|^4.$$

The same expansion uses only fourth-order joint moments, which also justifies the four-wise-independent sign example. □

For $s > 0$ and $0 < \alpha, \eta < 1$, define

$$A_s^{\mathrm{inj}}(\beta) = 1 + \frac{2(s+2)}{\beta} + \frac{4s(s+2)}{3\beta^2}, \qquad B_\eta^{\mathrm{inj}}(\beta) = \frac{\beta}{2} + 2\log\frac{2}{\eta}, \tag{4.1}$$

$$\mathfrak{m}_{\mathrm{inj}}(C, s; \alpha, \eta) = \frac{2C}{(1-\alpha)^2}\min_{\beta>0} A_s^{\mathrm{inj}}(\beta)B_\eta^{\mathrm{inj}}(\beta). \tag{4.2}$$

Set $\mathfrak{m}_{\mathrm{inj}}(C, 0; \alpha, \eta) = 0$. The minimum exists for $s > 0$; the choice $\beta = 2(s+2)$ gives

$$\mathfrak{m}_{\mathrm{inj}}(C, s; \alpha, \eta) \le \frac{14C}{3(1-\alpha)^2}\left(s + 2 + 2\log\frac{2}{\eta}\right). \tag{4.3}$$

Lemma 4.2 (Explicit lower injection)**.** *For a fixed isometry $Q \in \mathbb{R}^{N\times r}$ and $0 < \alpha, \eta < 1$, $m \ge \mathfrak{m}_{\mathrm{inj}}(C_4, r; \alpha, \eta)$ implies*

$$\mathbb{P}\{Q^\top\Omega\Omega^\top Q \succeq \alpha I_r\} \ge 1 - \eta.$$

In particular, half injection with failure at most $\delta/2$ holds if

$$m \ge \mathfrak{m}_{\mathrm{inj}}(C_4, r; 1/2, \delta/2). \tag{4.4}$$

A sufficient count requiring no minimization is $\frac{56}{3}C_4[r + 2 + 2\log(4/\delta)]$.

The proof in Appendix B.1 uses Gaussian smoothing and Catoni's logarithmic influence function [30, 42]. At fixed η,

$$\mathfrak{m}_{\mathrm{inj}}(C, r; 1/2, \eta) = \left[8\left(1 + \frac{2}{\sqrt{3}}\right) + o(1)\right]Cr \qquad (r \to \infty).$$

The lower-covariance method and rate are classical; the profile supplies finite constants for the raw iid sketch.

After lower injection, the remaining quantity is the interaction of the leading subspace with its orthogonal residual. The next bound extracts the required mixed moment from the directional fourth-moment assumption; Lemma 4.4 then controls the mean-square Frobenius interaction by averaging over the independent probes.

Lemma 4.3 (Orthogonal fourth moments)**.** *Let $y \in \mathbb{R}^s$ be isotropic and satisfy $\mathbb{E}\langle u, y\rangle^4 \le c\|u\|^4$ for all u, with $c \ge 1$. Define*

$$\gamma(c) = \min\{c/2, c - 1\}. \tag{4.5}$$

For orthogonal unit vectors u, v,

$$\mathbb{E}\langle u, y\rangle^2\langle v, y\rangle^2 \le \gamma(c). \tag{4.6}$$

Consequently,

$$\mathbb{E}(yy^\mathsf{T})^2 \preceq [c + (s-1)\gamma(c)]I_s. \tag{4.7}$$

Proof. Put $A = \langle u, y\rangle$, $B = \langle v, y\rangle$, and $h = \mathbb{E}A^2B^2$. The fourth-moment bounds in the directions $(u+v)/\sqrt{2}$ and $(u-v)/\sqrt{2}$ give

$$\mathbb{E}A^4 + \mathbb{E}B^4 + 6h \le 4c.$$

Cauchy–Schwarz gives $\mathbb{E}A^4 + \mathbb{E}B^4 \ge 2h$, hence $h \le c/2$. Isotropy and Jensen also give $\mathbb{E}A^4 + \mathbb{E}B^4 + 2h = \mathbb{E}(A^2 + B^2)^2 \ge 4$, hence $h \le c - 1$. To prove the matrix assertion, extend a unit test vector to an orthonormal basis. Its own fourth moment contributes at most c to $\mathbb{E}\|y\|^2\langle u, y\rangle^2$, and each of the other $s - 1$ coordinates contributes at most $\gamma(c)$. □

For the approximation theorems, let B_r bound the aggregate interaction with a unit residual direction:

$$\mathbb{E}\|Q^\top z\|^2\langle v, z\rangle^2 \le B_r \quad (Q^\top Q = I_r,\ \|v\| = 1,\ Q^\top v = 0). \tag{4.8}$$

The following choices follow from the bilinear and Wick calculations in Appendix B.2; take the minimum of applicable bounds.

Probe law	Aggregate bound B_r
Directional fourth moment C_4	$r\gamma(C_4)$
Gaussian or four-wise-independent sign factors	$r(3^d - 1)/2$
Independent factors with four-wise-independent coordinates	$r(C_{\text{coord}} - 1)/2$
Order two, coordinate fourth moments at most three	$\Xi_r = \min\{4r, 3r + 2\sqrt{r}\}$
Two normalized spherical factors	$\varsigma_2\Xi_r$

In the coordinate row, coordinates are centered with variance one and $C_{\text{coord}} = \prod_j \max\{3, \mu_{4,j}\}$, where $\mu_{4,j}$ bounds their fourth moments. Factors are mutually independent throughout. For spheres, $\varsigma_2 = \prod_{j=1}^2 n_j/(n_j + 2)$. The leading coefficient three in Ξ_r is sharp for Gaussian factors. These are aggregate bounds; dividing B_r by r need not give a valid bound for each individual pair of directions.

Lemma 4.4 (Orthogonal cross moment)**.** *Let $Q \in \mathbb{R}^{N\times r}$ be an isometry and let $Q^\top W = 0$. For $W \ne 0$ define*

$$\kappa(Q, W) = \frac{\mathbb{E}[\|Q^\top z\|^2\|W^\top z\|^2]}{\|W\|_{\mathrm{F}}^2}, \qquad \Omega = m^{-1/2}[z_1\ \cdots\ z_m]. \tag{4.9}$$

For iid isotropic z_i, whenever the displayed expectation is finite,

$$\mathbb{E}\|Q^\top(\Omega\Omega^\top - I_N)W\|_{\mathrm{F}}^2 = \frac{\kappa(Q, W)}{m}\|W\|_{\mathrm{F}}^2. \tag{4.10}$$

Under (4.8), $\kappa(Q, W) \le B_r$, so

$$\mathbb{E}\|Q^\top(\Omega\Omega^\top - I_N)W\|_{\mathrm{F}}^2 \le \frac{B_r}{m}\|W\|_{\mathrm{F}}^2. \tag{4.11}$$

Proof. The matrix $(Q^\top z)(z^\top W)$ has mean zero and squared Frobenius norm $\|Q^\top z\|^2\|W^\top z\|^2$. Independence across samples gives (4.10). Apply (4.8) to each nonzero residual column $w_b/\|w_b\|$ and sum to obtain the upper bound. The case $W = 0$ is immediate. □

The same calculation retains subspace geometry at second order. For two Gaussian factors, Lemma B.5 gives an exact formula using the partial traces and partial transpose of the head projector, and the uniform centered bound

$$\mathbb{E}(H - I_r)^2 \preceq \min\{4r + 4, 3r + 3 + 2\sqrt{r}\}I_r.$$

These smaller variance inputs enter the finite-profile recurrence without changing its higher-order budgets.

Theorem 4.5 (Relative error for rangefinding and regression)**.** *Assume* (1.3)*, fix* $r \geq 1$ *and* $\varepsilon, \delta \in (0,1)$*, and suppose*

$$m \geq \max\left\{ \mathfrak{m}_{\mathrm{inj}}(C_4, r; 1/2, \delta/2), \frac{8B_r}{\varepsilon\delta} \right\}. \tag{4.12}$$

For each of the following fixed inputs, the corresponding conclusion holds with probability at least $1-\delta$*.*

1. *For* $A \in \mathbb{R}^{n\times N}$*, put* $Y = A\Omega$ *and let* P_Y *be its orthogonal range projector. If* A_r *is a best rank-at-most-*r *approximation and* $\tau = \|A - A_r\|_{\mathrm{F}}^2$*, then*

$$\|A - P_Y A\|_{\mathrm{F}}^2 \leq (1+\varepsilon)\tau, \qquad \|A - \widehat{A}_r\|_{\mathrm{F}}^2 \leq (1+\varepsilon)\tau, \tag{4.13}$$

   where $\widehat{A}_r$ *minimizes the Frobenius error over matrices of rank at most* r *whose columns lie in* $\operatorname{range} Y$*. If* Q_Y *is an orthonormal basis of that range, one may compute* $\widehat{A}_r = Q_Y(Q_Y^\top A)_r$*.*

2. *For a full-column-rank design* $A \in \mathbb{R}^{N\times r}$ *and any fixed multiple-response matrix* $B \in \mathbb{R}^{N\times s}$*, let*

$$\widetilde{X} \in \arg\min_X \|\Pi(AX - B)\|_{\mathrm{F}}^2.$$

   Then the residual in the original space obeys

$$\|A\widetilde{X} - B\|_{\mathrm{F}}^2 \leq (1+\varepsilon) \min_X \|AX - B\|_{\mathrm{F}}^2. \tag{4.14}$$

   The failure bound is independent of the number of response columns.

Proof. For the first assertion, initially suppose $\operatorname{rank} A > r$ and write a reduced SVD $A = U_1\Sigma_1 V_1^\top + U_2\Sigma_2 V_2^\top$, with the first block of size r. Define $\Omega_j = V_j^\top \Omega$, $\Delta_\Omega = \Omega\Omega^\top - I_N$, $M = \Omega_1\Omega_1^\top$, and

$$\mathcal{E} = \|\Sigma_2 V_2^\top \Delta_\Omega V_1\|_{\mathrm{F}}^2.$$

Lemma 4.4, with $W = V_2\Sigma_2$, gives $\mathbb{E}\mathcal{E} \leq B_r\tau/m$. On $M \succeq I_r/2$, the matrix Ω_1 has full row rank and

$$\Omega_2\Omega_1^\dagger = V_2^\top \Delta_\Omega V_1 M^{-1}.$$

The matrix $B_* = (A\Omega)\Omega_1^\dagger V_1^\top$ has rank at most r and columns in $\operatorname{range} Y$. Its two error blocks have orthogonal right singular spaces, so

$$\|A - B_*\|_{\mathrm{F}}^2 = \tau + \|\Sigma_2\Omega_2\Omega_1^\dagger\|_{\mathrm{F}}^2 \leq \tau + 4\mathcal{E}.$$

Both errors in (4.13) are at most this competitor's error. Lemma 4.2 bounds failure of the lower event by $\delta/2$, while Markov's inequality gives

$$\mathbb{P}\{4\mathcal{E} > \varepsilon\tau\} \leq \frac{4B_r}{m\varepsilon} \leq \delta/2.$$

A union bound proves the claim, without independence of these two events. If $\operatorname{rank} A \leq r$, apply the injection lemma to its actual row space: the range is captured exactly on that event and both errors vanish.

For regression, write $A = QR$ with Q orthonormal and R invertible, and put $E = (I_N - QQ^\top)B$. The optimum original residual is $\|E\|_{\mathrm{F}}^2$. On $M = Q^\top\Omega\Omega^\top Q \succeq I_r/2$, the sketched problem has a unique solution and its normal equations give

$$R\widetilde{X} = Q^\top B + M^{-1}G, \qquad G = Q^\top \Delta_\Omega E.$$

Orthogonality yields

$$\|A\widetilde{X} - B\|_{\mathrm{F}}^2 = \|E\|_{\mathrm{F}}^2 + \|M^{-1}G\|_{\mathrm{F}}^2 \leq \|E\|_{\mathrm{F}}^2 + 4\|G\|_{\mathrm{F}}^2.$$

Lemma 4.4 and Markov give failure at most $\delta/2$ for the second term exceeding $\varepsilon\|E\|_{\mathrm{F}}^2$; the lower event contributes another $\delta/2$. When $E = 0$, the normal equations already give exact original residual zero and no division by $\|E\|_{\mathrm{F}}^2$ is needed. □

Corollary 4.6 (Input-dependent relative error)**.** *In the rangefinding assertion of Theorem 4.5, suppose* $\operatorname{rank} A > r$ *and take* $Q = V_1$, $W = V_2\Sigma_2$ *from its SVD. In the regression assertion take* Q *to span* $\operatorname{range} A$ *and* $W = (I - QQ^\top)B$*. For the relevant fixed pair define*

$$C_Q = \sup_{\|u\|=1} \mathbb{E}\langle Qu, z\rangle^4.$$

For iid isotropic probes, the same relative-error conclusion holds if $C_Q < \infty$, $\kappa(Q, W) < \infty$, *and*

$$m \geq \max\left\{\mathfrak{m}_{\mathrm{inj}}(C_Q, r; 1/2, \delta/2), \frac{8\kappa(Q,W)}{\varepsilon\delta}\right\}. \tag{4.15}$$

When $W = 0$*, only the injection condition on the relevant row or design space is needed, and the residual is exactly zero.*

Proof. Apply the injection lemma to $Q^\top z$ and replace the cross estimate in the preceding proof by (4.10). No fourth-moment bound outside the leading space and its residual interaction is used. □

If the common-factor or common-Schmidt support contains both spaces, the projected Gaussian probe is $\sqrt{w}\, g$, with g independent of w. Orthogonal Gaussian coordinates give

$$C_Q = 3\mathbb{E}w^2, \qquad \kappa(Q, W) = r\mathbb{E}w^2.$$

For the order-two common-Schmidt law, the interaction is therefore $r(1 + 2\rho)$. The universal fourth-moment bound would give the larger allowance $3r(1 + 2\rho)/2$.

Optimizing the injection level. For a deterministic failure allocation $0 < \eta < \delta$, set

$$h_\eta = \tfrac{1}{4}\mathfrak{m}_{\mathrm{inj}}(C_4, r; 1/2, \eta), \qquad k_\eta = \frac{B_r}{\varepsilon(\delta - \eta)}.$$

The proof of Theorem 4.5, with lower level α, requires $m \geq \max\{h_\eta/(1-\alpha)^2, k_\eta/\alpha^2\}$. For $B_r > 0$, the minimizing level $\alpha = \sqrt{k_\eta}/(\sqrt{h_\eta} + \sqrt{k_\eta})$ gives

$$m \geq (\sqrt{h_\eta} + \sqrt{k_\eta})^2. \tag{4.16}$$

The allocation η can also be optimized before sampling. The same formula applies to Corollary 4.6 with (C_4, B_r) replaced by $(C_Q, \kappa(Q, W))$; zero interaction requires only injection.

For a positive semidefinite matrix K, write

$$\mathcal{N}_K(\Omega) = K\Omega(\Omega^\top K\Omega)^\dagger\Omega^\top K.$$

Applying the range theorem to $K^{1/2}$ gives the Nyström trace guarantee, including fixed-rank truncation (Corollary B.7). Range containment also improves confidence.

Corollary 4.7 (Confidence amplification by range containment)**.** *Let*

$$m_0 = \left\lceil \max\left\{ \mathfrak{m}_{\mathrm{inj}}(C_4, r; 1/2, 1/8),\ \frac{32B_r}{\varepsilon} \right\} \right\rceil, \qquad b = \left\lceil \frac{\log(1/\delta)}{\log 4} \right\rceil.$$

A single raw iid sketch with $m = bm_0$ columns satisfies (4.13) and (B.11) with probability at least $1-\delta$, for their respective fixed inputs. In particular, the sufficient count is

$$O\left(\frac{C_4 r}{\varepsilon}[1 + \log(1/\delta)] \right).$$

Its proof in Appendix B.3 partitions one iid sketch into independent groups. The full range contains every group range, so all groups must fail before the combined approximation fails. This argument does not apply to the minimizer of one stacked regression objective. Under only a fourth-moment assumption, this distinction is necessary: Proposition B.10 proves a $1/(\varepsilon\delta)$ lower bound for the raw solve, already at rank one. It does not assert this lower bound for a fixed Gaussian tensor law.

4.1 High confidence under sub-Gaussian factors

Fourth moments alone give the inverse-confidence cost in Theorem 4.5. Higher directional moments give a polylogarithmic confidence certificate for the same raw regression solve. The underlying Hilbert-space moment principle is classical [47, 24]; the following formulation keeps the tensor moment budgets and the orthogonal interaction explicit.

Fix an integer $q \geq 1$. Let A_j be any valid directional moment bound $\mathbb{E}|\langle u, z\rangle|^{2j} \leq A_j\|u\|^{2j}$, for $2 \leq j \leq 2q$. One may use Lemma 2.2 or the law-specific profiles in Appendix A.5.3. Let $\Gamma_\perp > 0$ bound the mixed fourth moment of any two orthogonal unit directions. Choose an aggregate budget B_r as in (4.8), taking its minimum with $r\Gamma_\perp$ if necessary. One may take $\Gamma_\perp = \gamma(C_4)$ whenever that number is positive, or use the coordinate or Gaussian constants above. If $B_r = 0$, every orthogonal interaction vanishes almost surely and only the lower-injection condition below is needed; omit the formulas dividing by B_r. Otherwise set

$$\nu_H = B_r, \qquad b_j^H = r^{j/2}A_j\ (3 \leq j \leq 2q), \qquad \mathfrak{r}_q^H = \max_{3\leq j\leq 2q}\left(\frac{2b_j^H}{j!\nu_H}\right)^{1/(j-2)}. \tag{4.17}$$

For $q = 1$ the empty maximum is zero. Write $c_q = \mathfrak{g}_q^{1/(2q)}$ and h_q as in Theorem A.5.

Theorem 4.8 (A confidence certificate for the raw solve)**.** *Assume (1.3), fix $r \geq 1$ and $\varepsilon, \delta \in (0,1)$, and suppose the directional budgets above hold. The rangefinding, rank-r, and regression conclusions of Theorem 4.5 hold with probability at least $1-\delta$ if*

$$m \geq \mathfrak{m}_{\mathrm{inj}}(C_4, r; 1/2, \delta/2),$$
$$(2/\delta)^{1/(2q)}c_q\left(\sqrt{\nu_H/m} + h_q\mathfrak{r}_q^H/m\right) \leq \sqrt{\varepsilon}/2. \tag{4.18}$$

In particular, in the sub-Gaussian tensor model, with $t = \log(2/\delta)$, a sufficient sample count is

$$m \geq C_{d,K_{\mathrm{sg}}}\left[\frac{rt}{\varepsilon} + \sqrt{\frac{r}{\varepsilon}}\,t^d\right]. \tag{4.19}$$

This statement concerns each fixed input and has no dependence on the number of response columns.

Appendix B.4 proves the certificate and gives a second comparison that keeps the weak Hilbert variance separately. Its leading term is at most $\sqrt{(B_r + \Gamma_\perp(q-1))/m}$ before the confidence multiplier, compared with $\mathfrak{g}_q^{1/(2q)}\sqrt{B_r/m}$ above. The two higher-order radii need not be ordered, so the smaller valid allowance may be used.

4.2 An angular-moment bound for the full sampled range

For Gaussian or spherical product columns, a complementary expected-error bound uses moments near order two rather than directional fourth moments. The use of entropy-scale moments for product probes is motivated by Holden's product-probe note [45]; the scalar residual recursion follows the RPCholesky analysis of Chen–Epperly–Tropp–Webber [46]. We give the fractional-moment conversion explicitly. The normalization step is the Gaussian/spherical range invariance recorded in [10, Section 5.4].

For positive integer mode dimensions $n_1, \dots, n_d$, define

$$\begin{aligned} c_n^{\mathrm{sph}}(p) &= \frac{n^p \Gamma(p+1/2)\Gamma(n/2)}{\Gamma(1/2)\Gamma(n/2+p)}, \\ H_{\mathrm{ang}}(p) &= \prod_{j=1}^{d} c_{n_j}^{\mathrm{sph}}(p), \\ \varkappa_{\mathrm{ang}} &= \exp\left[\sum_{j=1}^{d}\{\log n_j + \psi(3/2) - \psi(n_j/2+1)\}\right], \end{aligned} \tag{4.20}$$

where $p \geq 1$ is real and $\psi = (\log\Gamma)'$. In particular, $c_n^{\mathrm{sph}}(1) = 1$, and a one-dimensional mode contributes one to $\varkappa_{\mathrm{ang}}$.

Theorem 4.9 (Expected approximation of the full sampled range)**.** *Let A be fixed, let the columns of Ω be independent real Gaussian product probes or independent product probes whose mode-j factor is uniform on the sphere of radius $\sqrt{n_j}$, and put $P = P_{\mathrm{range}(A\Omega)}$. Fix $1 \leq r < \operatorname{rank} A$ and $0 < \varepsilon \leq 1$, and set*

$$T = \|A\|_{\mathrm{F}}^2, \qquad \tau = \|A - A_r\|_{\mathrm{F}}^2, \qquad u_0 = T/\tau - 1.$$

If $u_0 \leq \varepsilon$, the desired bound holds already for the empty range. Otherwise, the explicit condition

$$m \geq \left\lceil \varkappa_{\mathrm{ang}} r \left[\frac{1}{\varepsilon} - \frac{1}{u_0} + \log\frac{u_0}{\varepsilon}\right]\right\rceil \tag{4.21}$$

implies

$$\mathbb{E}\|(I-P)A\|_{\mathrm{F}}^2 \leq (1+\varepsilon)\tau. \tag{4.22}$$

In particular, a simpler sufficient condition is

$$m \geq \left\lceil \varkappa_{\mathrm{ang}} r \left[\varepsilon^{-1} + \log(\varepsilon^{-1}) + \log(T/\tau)\right]\right\rceil. \tag{4.23}$$

For PSD K, the same budgets, with $T = \operatorname{tr} K$ and $\tau = \sum_{i>r}\lambda_i(K)$, give $\mathbb{E}\operatorname{tr}(K - \mathcal{N}_K(\Omega)) \leq (1+\varepsilon)\tau$ for the full Nyström approximation.

The proof is in Appendix B.5. The conclusion concerns the entire sampled range and expectation. Its constant $\varkappa_{\mathrm{ang}}$ can be smaller than the fourth-moment constant, but its budget contains the trace-to-tail ratio. The theorem does not assert the same estimate for a rank-r truncation, for regression, or for the Gram distortion of an unnormalized sketch.

The angular constant obeys

$$\varkappa_{\mathrm{ang}} \leq (e^{2-\gamma_{\mathrm{E}}}/2)^d = (2.0743278106\dots)^d.$$

It improves the spherical fourth-moment product when some mode has dimension greater than one. Appendix B.5 also gives the corresponding Frobenius estimate for the full PSD residual.

4.3 A separation on the same input

For suitable fixed inputs, relative-error approximation and an upper spectral obstruction occur together for the same draw of the sketch. Intersecting the two probability bounds gives a high-probability separation for Gaussian and Rademacher factors, without requiring independence of the two events.

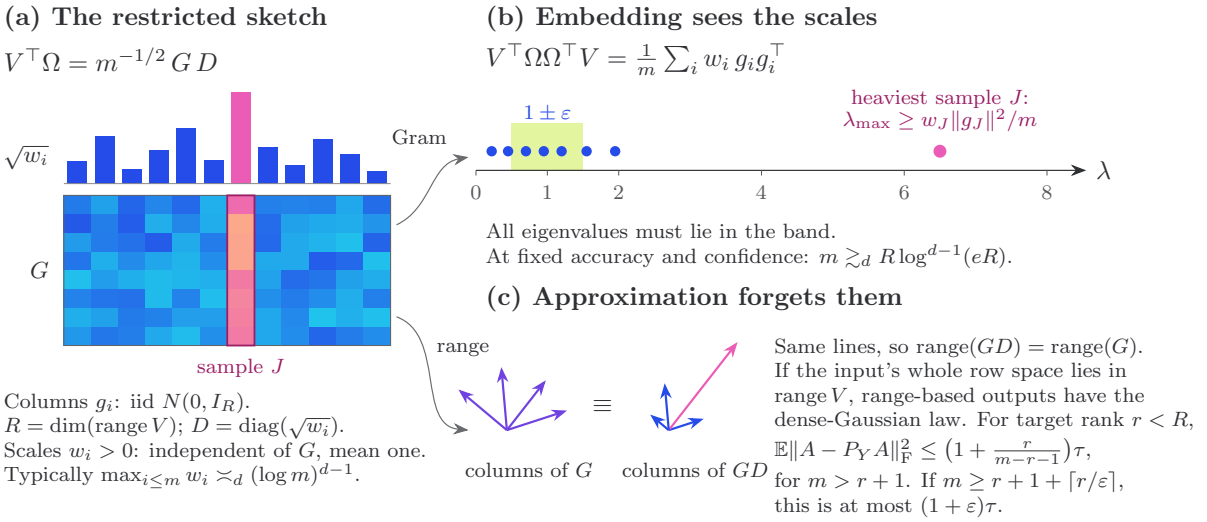


Figure 1: The Gaussian common-factor mechanism. The Gram retains the heavy column scales, while the range discards them (Section 2.2.1; Proposition 4.11). Here V has R orthonormal columns, r is the approximation rank, $Y = A\Omega$, and $\tau = \|A - A_r\|_{\mathrm{F}}^2$. The exact output law requires the input's entire relevant support in range V, not merely its leading subspace. The weight scale is asymptotic at fixed d; all depicted entries, weights, and eigenvalues are schematic.

Corollary 4.10 (Relative approximation with an upper spectral obstruction)**.** *Fix $d \ge 2$ and $0 < \varepsilon, \delta < 1$. For either independent Gaussian or independent-coordinate Rademacher factors there are fixed inputs $A^{(r)}$ and sketch sizes $m_r = O_{d,\varepsilon,\delta}(r)$ such that, writing $M_r = U_r^\top \Omega\Omega^\top U_r$,*

$$\mathbb{P}\big\{\|A^{(r)} - \widehat{A}_r^{(r)}\|_{\mathrm{F}}^2 \le 1 + \varepsilon,\\ \lambda_{\min}(M_r) \ge \tfrac{1}{2}, \quad \lambda_{\max}(M_r) \ge 2\big\} \ge 1 - \delta - o(1).$$

Here U_r spans the leading right singular space, the optimal squared rank-r tail is one, and the input rank exceeds m_r. The same error bound holds for the untruncated rangefinder on this event. In contrast, the zero output has squared error $r+1$.

Proof. Both factor laws have directional fourth-moment constant at most 3^d. Choose m_r by (4.12) and U_r from the corresponding lower-bound construction. Enlarge unused mode coordinates so that $N_r > \max\{m_r, r+1\}$, without changing the projected probe law. With $P_r = U_r U_r^\top$, put

$$A^{(r)} = P_r + (N_r - r)^{-1/2}(I_{N_r} - P_r).$$

Its first r singular values are one, its remaining singular values are $(N_r - r)^{-1/2} < 1$, and its squared tail is one. In the proof of Theorem 4.5, the lower-injection event and the cross moment event jointly have probability at least $1 - \delta$; both error bounds and $M_r \succeq I_r/2$ hold on this same event. Thus adding the lower spectral edge does not spend further confidence. Since $m_r/[r(\log r)^{d-1}] \to 0$ and $m_r \ge r$, the Gaussian and Rademacher lower-bound proofs each give $\lambda_{\max}(M_r) \ge 2$ with probability $1 - o(1)$. Intersect the events by a union bound. □

For regression, the same separation follows by taking design U_r and response $\sqrt{r}\,U_r e_1 + v_r$, where v_r is a fixed unit vector orthogonal to range U_r: the optimal squared residual is one and the

zero-coefficient residual is $r+1$. For PSD trace approximation, take $K_r = P_r + (N_r - r)^{-1}(I - P_r)$ and apply the range result to $K_r^{1/2}$. Each is a separate fixed-input probability statement.

A complementary condition-number estimate at linear sketch size is recorded in Appendix A.6.

4.4 Exact laws from random lines

Nonzero column scales leave sampled ranges unchanged. This observation, also used for Gaussian and spherical probes in [10, Section 5.4], gives exact output laws when the input's entire relevant support has the required distribution of random lines. The comparison is with a dense Gaussian sketch applied to the same input, whose geometry still shapes the sampled range. In this section the right sketch continues to use $\Omega = m^{-1/2}[z_1 \ \cdots \ z_m]$.

Proposition 4.11 (Exact Gaussian range laws)**.** *Let $V : \mathbb{R}^R \to S \subseteq \mathbb{R}^N$ be an isometry. Suppose $x = V^\top z \neq 0$ almost surely and its unoriented line is uniform on real projective space, meaning the image of the uniform sphere under $u \sim -u$. Independent probes then admit a coupling*

$$V^\top \Omega = m^{-1/2} G D, \tag{4.24}$$

where G has iid standard Gaussian entries and D is an almost surely invertible diagonal matrix, possibly dependent on G. For every fixed input with $\operatorname{range}(A^\top) \subseteq S$, its sampled range, all fixed powered ranges, and their block Krylov range have the dense-Gaussian joint law. Rangefinding and optimal fixed-rank outputs have their Gaussian laws when selected as fixed measurable functions of the input and the relevant sampled subspace, equivalently its orthogonal projector. The same selection rule is used for both sketch laws, including at ties; it does not inspect raw column scales. Equal ranges always give equal optimal errors. The same convention gives the output laws for Nyström and its fixed-rank truncation when $K \succeq 0$ and $\operatorname{range} K \subseteq S$.

A sufficient condition, for a random environment h, is

$$V^\top z \mid h \sim N(0, w(h) I_R), \qquad w(h) > 0 \quad \text{almost surely.} \tag{4.25}$$

Then $D = \operatorname{diag}(\sqrt{w(h_i)})$ can be chosen independent of G. The Gaussian supports in Definition 3.1 satisfy this condition.

Proof. For independent fair signs ϵ_i and independent Gaussian radii χ_i, each with the norm distribution of a standard Gaussian in $\mathbb{R}^R$, put $g_i = \chi_i \epsilon_i x_i / \|x_i\|$ and $d_i = \epsilon_i \|x_i\| / \chi_i$. Uniformity of the unoriented line makes g_i standard Gaussian and $g_i d_i = x_i$. Independence across augmented probes proves the factorization. Every deterministic map annihilating $S^\perp$ then has the same right-hand invertible factor D, which preserves each range and the concatenated block range. Their projectors therefore coincide under the coupling, so the same measurable function of the input and the projector gives the same output, including at ties. For PSD outputs use $\mathcal{N} = K^{1/2} P_{K^{1/2}\Omega} K^{1/2}$. Under the conditional condition, $g_i = x_i / \sqrt{w(h_i)}$ has a constant Gaussian conditional law and hence is independent of the environment, giving the stronger independence statement. □

Other factor laws. The proposition transfers existing Gaussian range guarantees [18, 27]; it gives neither a new Gaussian bound nor a floating-point conditioning law. The entire relevant input support must satisfy the hypothesis. More generally, on a common-factor support with independent factor laws,

$$V^\top \Omega = m^{-1/2} X \operatorname{diag}(s_i), \qquad X_i = B^\top g_{i1}, \quad s_i = \prod_{j=2}^{d} \langle b_j, g_{ij} \rangle,$$

where B collects the orthonormal first-mode vectors. The range equals that of the first-factor columns with $s_i \neq 0$; conditional on these indices the retained columns are iid with their original

law, and their count is binomial. A coordinate first-mode support with sign factors gives dense sign columns. A general orthonormal projection of sign factors need not have independent coordinates. Taking the flat common-factor length odd in the Rademacher witness makes every s_i nonzero, so no columns are lost, while preserving the logarithmic obstruction for sufficiently large r.

Gaussian relative-error coefficients. Under Proposition 4.11 on the entire relevant support, the Gaussian moment identities [18, Propositions 10.1 and A.5] and the rank-r competitor in the proof of Theorem 4.5 give, for $m > r + 1$,

$$\mathbb{E}\|A - P_{A\Omega}A\|_{\mathrm{F}}^2, \quad \mathbb{E}\|A - \widehat{A}_r\|_{\mathrm{F}}^2 \le \left(1 + \frac{r}{m-r-1}\right)\|A - A_r\|_{\mathrm{F}}^2, \tag{4.26}$$

$$\mathbb{E}\,\mathrm{tr}(K - \mathcal{N}), \quad \mathbb{E}\,\mathrm{tr}(K - \mathcal{N}_r) \le \left(1 + \frac{r}{m-r-1}\right)\tau_r^{\mathrm{tr}}(K). \tag{4.27}$$

Thus $m = r + 1 + \lceil r/\varepsilon \rceil$ gives expected relative error at most $1 + \varepsilon$ for all four objectives; the PSD cases use the $K^{1/2}$ correspondence in Corollary B.7.

The common-factor case gives an exact distributional counterpart to Corollary 4.10: the range-based approximation outputs retain their dense-Gaussian laws while the probability of violating any fixed upper embedding tolerance tends to one. The relative-error bounds in the following proposition are expectation bounds, distinct from the high-probability conclusion of that corollary.

Proposition 4.12 (Exact approximation laws despite leading-subspace embedding failure)**.** *Fix tensor order $d \ge 2$, accuracy $\varepsilon \in (0,1)$, and upper distortion tolerance $\alpha > 0$. For each r, let $m = r + 1 + \lceil r/\varepsilon \rceil$, choose $R \ge m$, and let a fixed input A have full row support equal to an R-dimensional common-factor support in Definition 3.1. Choose its leading r right singular vectors to be the first r vectors of that basis. Gaussian product probes have exactly the dense-Gaussian approximation laws and satisfy (4.26). Nevertheless, for that actual leading isometry V_1, the normalized sketch obeys*

$$\mathbb{P}\{\lambda_{\max}(V_1^\top \Omega\Omega^\top V_1) > 1 + \alpha\} \longrightarrow 1 \qquad (r \longrightarrow \infty).$$

The same construction applies to a PSD input with this positive eigenspace, so (4.27) coexists with the same head obstruction. One may take $R > m$ to keep the approximation genuinely nonexact.

Proof. Exact laws and the approximation coefficients follow from the preceding results. On the leading support the columns have form $m^{-1/2}\sqrt{w_i}g_i$, with iid $g_i \sim N(0, I_r)$ independent of iid $w_i = \prod_{j=2}^d \xi_{ij}^2$, where all ξ_{ij} are standard normal. Put $c_{\mathrm{cap}} = 3 + \varepsilon^{-1}$, so $m/r \le c_{\mathrm{cap}}$. The random weight has unbounded support, hence

$$p_* = \mathbb{P}\{w_i > 2c_{\mathrm{cap}}(1 + \alpha)\} > 0.$$

Also $\mathbb{P}\{\|g_i\|^2 \ge r/2\} \to 1$, by the variance $2r$ of its squared norm. Therefore the probability q_r that a column satisfies both inequalities tends to $p_* > 0$. Such a column alone gives

$$\lambda_{\max}(V_1^\top \Omega\Omega^\top V_1) \ge \frac{w_i\|g_i\|^2}{m} > 1 + \alpha.$$

The column events are independent, and $(1 - q_r)^m \to 0$. This is upper distortion on the leading rank-r space even though $m > r$; no ambient rank-deficiency obstruction is involved. □

For comparison, a dense Gaussian sketch with the same $m/r \to c_\varepsilon = 1 + \varepsilon^{-1}$ has

$$\lambda_{\max}(V_1^\top \Omega\Omega^\top V_1) \longrightarrow (1 + c_\varepsilon^{-1/2})^2$$

almost surely [39]. Thus failure of the upper tolerance $1+\alpha$ is not specific to Khatri–Rao sketches when $\alpha < (1 + c_\varepsilon^{-1/2})^2 - 1$. The proposition distinguishes the product law from the dense law for every fixed tolerance above that edge: the product upper eigenvalue diverges, whereas the dense one stays bounded. At $\varepsilon = 1/2$ the dense edge is $(1 + 1/\sqrt{3})^2 \approx 2.488$.

4.5 Regression under Gaussian scale mixtures

Column scales remain relevant to sketch-and-solve and generalized Nyström, as noted in [10, Section 5.4]. The weighted-least-squares interpretation is classical for sampling sketches [25]; related weighted covariance matrices arise from elliptical data in [16]. Here the sketch supplies the weights. If $A = V\bar{A}$ and $b = V\bar{b}$ both lie in a support satisfying (4.25), then

$$\|\Omega^\top(Ax - b)\|^2 = \frac{1}{m}\sum_i w_i \langle g_i, \bar{A}x - \bar{b}\rangle^2,$$

with weights independent of the Gaussian g_i. Changing the weights preserves sampled ranges but changes this objective. Consequently the range-containment confidence argument does not apply to raw regression.

Proposition 4.13 (Conditional regression law and a confidence certificate)**.** *Draw iid probes with $V^\top z = \sqrt{w}\, g$, where g is standard Gaussian, $w > 0$ independent of g, $\mathbb{E}w = 1$, and $\mu_2 = \mathbb{E}w^2 < \infty$. Fix A, b with $\operatorname{span}(\operatorname{range} A, b) \subseteq S$, $\operatorname{rank} A = r$, and nonzero optimal residual $e = b - Ax_*$. Let $\widetilde{x}$ be the sketched least-squares solution and put*

$$\mathcal{E}_m^{\mathrm{reg}} = \frac{\|A\widetilde{x} - b\|^2}{\|e\|^2} - 1.$$

For $m \geq r$, there are iid $\xi_i \sim N(0, I_r)$ and independent $\eta_i \sim N(0,1)$, independent of the weights, such that

$$\mathcal{E}_m^{\mathrm{reg}} = \|a_m\|^2, \quad a_m = S_m^{-1}\sum_i w_i \xi_i \eta_i, \quad S_m = \sum_i w_i \xi_i \xi_i^\top. \tag{4.28}$$

Conditional on $(w_i, \xi_i)_{i \leq m}$, a_m is centered Gaussian with

$$C_m = S_m^{-1}\Big(\sum_i w_i^2 \xi_i \xi_i^\top\Big) S_m^{-1}, \qquad \mathbb{E}[\mathcal{E}_m^{\mathrm{reg}} \mid (w_i, \xi_i)] = \operatorname{tr} C_m. \tag{4.29}$$

For every $t > 0$, the conditional Gaussian quadratic-form bound gives

$$\mathbb{P}\left\{\mathcal{E}_m^{\mathrm{reg}} > \operatorname{tr} C_m + 2\sqrt{t \operatorname{tr}(C_m^2)} + 2t\|C_m\| \;\middle|\; (w_i, \xi_i)_{i \leq m}\right\} \leq e^{-t}. \tag{4.30}$$

For fixed r and a fixed weight law, as $m \to \infty$,

$$mC_m \longrightarrow \mu_2 I_r \quad \text{almost surely,} \qquad m\mathcal{E}_m^{\mathrm{reg}} \;\Rightarrow\; \mu_2 \chi_r^2.$$

In particular, the scaled conditional mean tends to $r\mu_2$.

For $0 < \varepsilon, \delta < 1$, let $b_m(\delta)$ be deterministic with $\mathbb{P}\{\max_i w_i > b_m(\delta)\} \leq \delta/4$, and put $C_w = 3\mu_2$. Then $\mathbb{P}\{\mathcal{E}_m^{\mathrm{reg}} > \varepsilon\} \leq \delta$ whenever

$$m \geq \max\left\{\mathfrak{m}_{\mathrm{inj}}(C_w, r; 1/2, \delta/2), \frac{2b_m(\delta)}{\varepsilon}\big(\sqrt{r} + \sqrt{2\log(4/\delta)}\big)^2\right\}. \tag{4.31}$$

The normal equations identify C_m; the Gaussian quadratic-form bound retains its full spectrum. Bounding C_m by $2b_m(\delta)I_r/m$ gives the deterministic certificate above. Appendix B.7 proves these claims and the limit.

For the common-factor Gaussian law, the certificate uses

$$\mu_2 = 3^{d-1}, \qquad b_m(\delta) = \big[2\log(8(d-1)m/\delta)\big]^{d-1}.$$

For order-two common-Schmidt weights, with $\rho = \sum_b \lambda_b^2$ and $L = \max_b \lambda_b$, it uses

$$\mu_2 = 1 + 2\rho, \qquad b_m(\delta) = 1 + 2\sqrt{\rho\log(4m/\delta)} + 2L\log(4m/\delta).$$

Both give polylogarithmic confidence cost; for the common-factor law, a sufficient explicit order is

$$m \geq C_d \frac{r + \log(1/\delta)}{\varepsilon}\left[\log\frac{e(r+\log(1/\delta))}{\varepsilon\delta}\right]^{d-1}.$$

The common-Schmidt spectrum therefore has three distinct roles: range-based outputs have the Gaussian law, fixed-rank regression fluctuations carry the factor $1 + 2\rho$, and embedding width depends on L. The regression limit fixes r and does not assert the finite-sample identity $\mathbb{E}\mathcal{E}_m^{\mathrm{reg}} = r\mu_2/m$ or convergence of unconditional means. Its confidence certificate also needs weight-tail control; arbitrary mixing weights need not give polylogarithmic cost. The general fourth-moment obstruction in Proposition B.10 remains applicable outside these Gaussian support classes.

5 Numerical illustration

The experiments compare the exact range-law mechanism with approximation on inputs outside those support classes. All approximation inputs have rank R, with r singular values equal to one and the remaining $R - r$ equal to $(R-r)^{-1/2}$. Thus the optimal squared rank-r tail is one. We plot $\|A - \widehat{A}_r\|_{\mathrm{F}}^2/\tau_r^{\mathrm{F}}(A)$, where $\widehat{A}_r$ is the optimal rank-r output in the sampled range. Lines are medians and bands are empirical 10–90% trial quantiles.

5.1 Exact range-law separation

Figure 2 uses a common-factor support, $m = 3r + 1$, and $R = 2m$. The upper Gram edge increases with tensor order while the coupled Gaussian approximation outputs coincide, as Proposition 4.11 predicts. At $r = 2048$, median upper eigenvalues are 2.48 for dense Gaussian and $6.34, 22.77, 51.09$ for product orders two, three, and four. The approximation ratios remain close to 1.30 over the larger tested ranks. The sign experiment uses the nonzero common-factor construction in the paragraph following Proposition 4.11; its range has the dense-sign law.

5.2 Beyond exact range-law supports

For Figure 3, the leading right singular space is the common-factor witness, and the tail is a uniformly sampled subspace of its orthogonal complement, fixed before sketching. All laws share this input, and each sketch is evaluated separately. The full support therefore need not satisfy the exact Gaussian-law hypothesis.

Panel (a) increases r at $m = 3r + 1$ and $R = 2m$. At $r = 1024$, median relative errors are 1.300, 1.522, 1.805, and 1.948 for dense Gaussian and product orders two, three, and four. Panel (b) instead fixes the input and increases m, always with $m/R \leq 1/4$. On its tested grid, the empirical 90th-percentile error first falls below 1.5 at $m/r = 3, 6, 6, 8$, respectively. The panels use different mode dimensions and input ranks, specified in the caption; their error levels are not directly interchangeable. These are fixed-input observations, not uniform sample-complexity bounds.

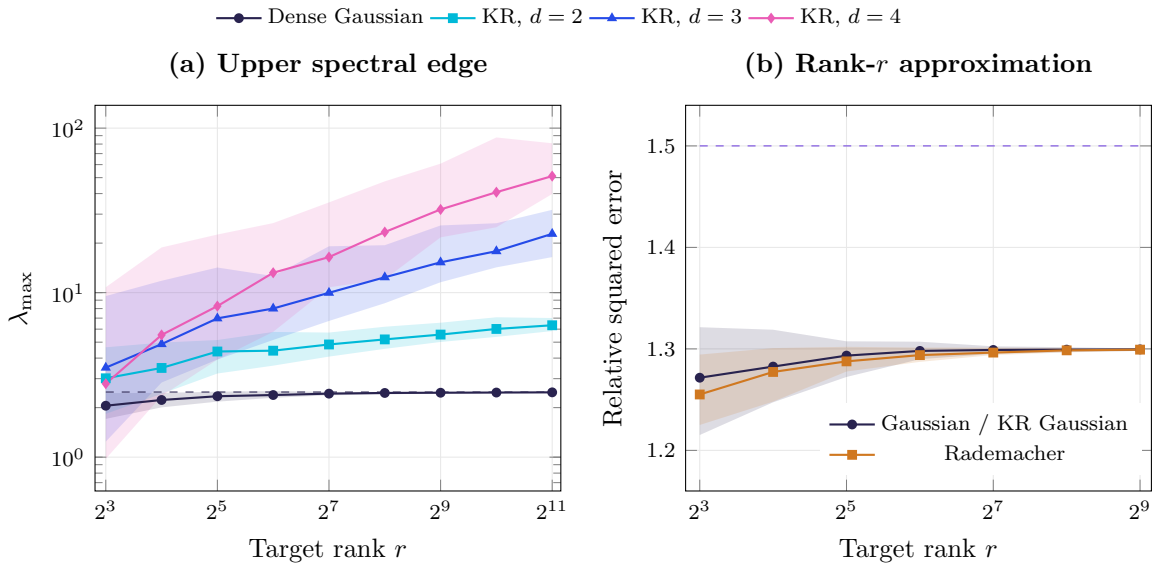


Figure 2: Common-factor supports at $m = 3r + 1$. The Gaussian product laws share their first-mode Gaussian matrix, so their range outputs coincide. The Rademacher curve uses nonzero common signs. Dashed lines mark the asymptotic dense-Gaussian upper edge [39] and the Gaussian expected-error bound 1.5. Panel (a) uses 96 trials through $r = 512$, 64 at $r = 1024$, and 32 at $r = 2048$; panel (b) uses 64 through $r = 256$ and 32 at $r = 512$.

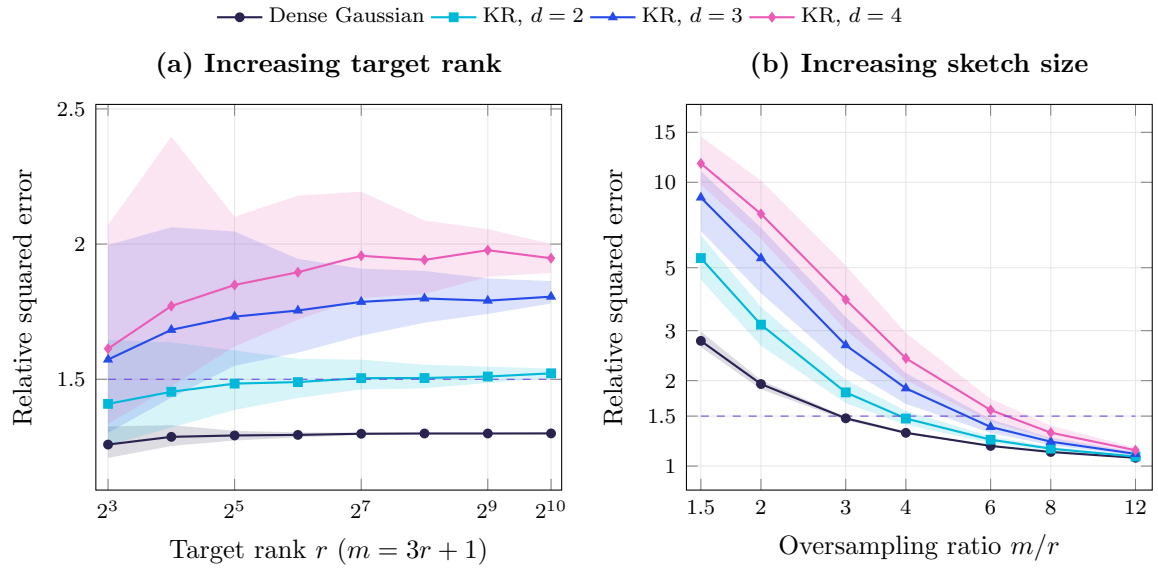


Figure 3: Approximation with a common-factor head and a generic orthogonal tail. Panel (a) uses $N = 16r$, $R = 2m$, and common modes (16), (4, 4), and (2, 2, 4) for orders two, three, and four. There are 128 trials through $r = 64$, 96 at 128, 64 at 256, and 32 at 512 and 1024. Panel (b) fixes $r = 32$, $N = 16384$, and $R = 1536$, with common modes (512), (16, 32), and (8, 8, 8); 96 trials use nested sketch widths. The dashed level 1.5 is an error target; it also bounds Gaussian expected error in panel (a).

Finite certificates. Table 2 compares sufficient embedding widths at $r = n_j = 8$, $\varepsilon = 1/2$, and $\delta = 1/10$. The closed bound uses Proposition 2.6 with ${K_{\mathrm{sg}}}^2 = 8/3$, selecting $1 \le q \le 200$. The finite-profile calculation uses the law-specific moments in Appendix A.5.3 and the centered recurrence (A.26), selecting $1 \le q \le 6$. Rational budgets and upward rounding certify the latter counts, which are rounded up to multiples of ten. These widths guarantee embedding of every fixed subspace; they are not empirical thresholds.

Certificate / factor law	$d = 2$	$d = 3$
Closed, ${K_{\mathrm{sg}}}^2 = 8/3$	14 715	149 890
Finite profile, Gaussian	1 540	5 930
Finite profile, spherical	910	2 570
Finite profile, independent signs	1 360	4 090

Table 2: Sufficient embedding widths at fixed accuracy and confidence. Moment orders are selected deterministically.

Code and data availability. Code, raw trials, and certificate calculations are available at https://github.com/DiarHaidary/khatri-rao-numerics.

6 Discussion and open questions

The sharp embedding cost and the linear-size approximation guarantee reflect different requirements on the same sampled matrix. The remaining questions concern the parameters and subspace geometries for which these guarantees can be sharpened.

Accuracy and confidence. The matching bounds determine rank dependence at fixed parameters; they do not determine the jointly optimal dependence on $r, \varepsilon, \delta, d$, and K_{sg}. Theorem 2.3 removes the variance rank logarithm when $\delta \ge 2/r$. At smaller failure probabilities, the two upper bounds trade a variance logarithm against an additional confidence logarithm in the large-row term. Closing this gap is open. The Gaussian obstruction in Proposition A.12, general OSE lower bounds [28], and sign-product fixed-vector lower bounds [26] constrain different regimes and do not yet give a matched joint theorem. For raw regression, Proposition B.10 gives the obstruction under fourth moments, while Theorem 4.8 gives polylogarithmic confidence for sub-Gaussian product laws on every fixed input. The jointly sharp dependence on confidence, accuracy, rank, and tensor order remains open.

Subspace geometry. Theorem 3.2 identifies the largest Schmidt weight as the parameter governing the order-two common-Schmidt transition. The all-confidence bound retains this geometric parameter, while Corollary 4.6 isolates the mixed moment relevant to approximation. A broader characterization of subspaces admitting $O(r)$ embedding width, with constants independent of large mode dimensions, remains open. Useful geometric criteria should be computable from the input and explain when the extreme-row obstruction disappears.

Weighted regression. The conditional bound (4.30) suggests controlling $\operatorname{tr} C_m$, $\operatorname{tr}(C_m^2)$, and $\|C_m\|$ jointly with the weighted Gram inverse. A large weight affects both factors in C_m; replacing it by its maximum can lose this compensation. The fixed-rank limit identifies $\mathbb{E}w^2$ as the limiting variance factor, but does not establish a uniform finite-sample bound as rank grows. Removing unnecessary logarithms from the confidence certificate requires such an estimate. The general Hilbert comparison in Theorem 4.8 provides another sufficient certificate; its minimum with the weighted certificate is valid without spending additional failure probability.

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A Embedding proofs and quantitative certificates

A.1 Accuracy refinement by sample covariance

Proof of Theorem 2.3. Write $p = d-1$ and $\Phi(s) = rs^p + s^{p+1}$. Constants denoted $C_{d,K_{\mathrm{sg}}}$ depend only on d, K_{sg} and may change between displays. We first assume $m \geq 2r$, as both final budgets ensure after increasing their constants.

Row norms. Let $G \sim N(0, I_r)$ be independent of y. Integrating $|\langle G, y\rangle|^{2q}$ in the two orders and using (2.6)–(2.7) gives $\mathfrak{g}_q \mathbb{E}\|y\|^{2q} \leq (q!)^d {K_{\mathrm{sg}}}^{2dq} D_{r,q}$ for integers $q \geq 2$, where we dropped $\theta^d \leq 1$. Since $\mathfrak{g}_q \geq q!$, $q! \leq q^q$, and $D_{r,q} \leq (r+2q)^q$,

$$(\mathbb{E}\|y\|^{2q})^{1/q} \leq {K_{\mathrm{sg}}}^{2d} q^p (r+2q). \tag{A.1}$$

Markov's inequality with $q = \lceil \log(m/\beta) \rceil$, increased to two if necessary, and a union bound give, with failure at most β,

$$\max_{i \leq m} \|y_i\|^2 \leq C_{d,K_{\mathrm{sg}}} \Phi(\log(2m/\beta)). \tag{A.2}$$

Taking $q = \lceil \log(em) \rceil$ also gives

$$\mathbb{E} \max_{i \leq m} \|y_i\|^2 \leq C_{d,K_{\mathrm{sg}}} \Phi(\log(em)), \tag{A.3}$$

because the L_q norm of the maximum is at most $m^{1/q}$ times the L_q norm of one squared row norm.

Published covariance input. The y_i need not be centered. Introduce independent Rademacher signs η_i and apply [35, Corollary 2] to $\eta_i y_i$ at moment order six. These vectors are centered and isotropic; (2.6) bounds their directional sixth moments by a constant depending only on d, K_{sg}. Their Gram matrices and norms are unchanged. For $r \geq 2$, outside an event $\mathcal{B}$ of probability at most $1/r$,

$$\|Z\| \leq C_{d,K_{\mathrm{sg}}} \left[\sqrt{\frac{r}{m}} + \frac{\max_i \|y_i\|^2}{m} \right]. \tag{A.4}$$

Use (A.2) with $\beta = \delta/2$. If $\delta \geq 2/r$, then, with failure at most δ,

$$\|Z\| \leq C_{d,K_{\mathrm{sg}}} \left[\sqrt{\frac{r}{m}} + \frac{\Phi(\log(4m/\delta))}{m} \right]. \tag{A.5}$$

Independence and (2.13) also imply $\mathbb{E}\|Z\|_{\mathrm{F}}^2 = m^{-1} \operatorname{tr} \mathbb{E}(H - I_r)^2 \leq C_{d,K_{\mathrm{sg}}} r^2/m$. Cauchy–Schwarz bounds the contribution of $\mathcal{B}$ to $\mathbb{E}\|Z\|$ by $C_{d,K_{\mathrm{sg}}} \sqrt{r/m}$. Integrate (A.4) on its complement and use (A.3) to obtain

$$\mathbb{E}\|Z\| \leq C_{d,K_{\mathrm{sg}}} \left[\sqrt{\frac{r}{m}} + \frac{\Phi(\log(em))}{m} \right]. \tag{A.6}$$

For $r = 1$, the scalar variance bound gives this directly.

Arbitrary confidence. Set $q = \lceil \log(8m/\delta) \rceil$ and $R = e{K_{\mathrm{sg}}}^{2d} q^p (r+2q)$. Equation (A.1) gives

$$\mathbb{P}\{\|y\|^2 > R\} \leq e^{-q} \leq \frac{\delta}{8m}, \qquad b := \mathbb{E}[\|y\|^2 \mathbf{1}_{\{\|y\|^2 > R\}}] \leq Re^{-q} \leq \frac{\delta R}{8m}.$$

Define $\widetilde{H} = yy^\top \mathbf{1}_{\{\|y\|^2 \leq R\}}$ and $\widetilde{Z} = m^{-1} \sum_i \widetilde{H}_i - \mathbb{E}\widetilde{H}$. With failure at most $\delta/8$, no sample is removed, and $\|Z\| \leq \|\widetilde{Z}\| + b$. Also,

$$\mathbb{E}\|\widetilde{Z}\| \leq \mathbb{E}\|Z\| + 2b \leq C_{d,K_{\mathrm{sg}}} \left[\sqrt{r/m} + R/m \right].$$

Truncation occurs only in this proof; the sketch remains unchanged. Apply [6, Theorem 2.3] to the centered quadratic forms $a^\top \widetilde{H} a - \mathbb{E}(a^\top \widetilde{H} a)$ and their negatives, on a countable dense subset of the unit sphere. Their absolute values are at most R, and their variances are bounded by $\mathbb{E}|\langle a, y\rangle|^4 \le C_{d,K_{\mathrm{sg}}}$. With $u = \log(8/\delta)$, the inequality and $2\sqrt{xy} \le x + y$ give, with failure at most $\delta/8$,

$$\|\widetilde{Z}\| \le 2\mathbb{E}\|\widetilde{Z}\| + C_{d,K_{\mathrm{sg}}}\sqrt{u/m} + \frac{4Ru}{3m}.$$

Combining the bounds yields, with failure at most $\delta/4$,

$$\|Z\| \le C_{d,K_{\mathrm{sg}}}\left[\sqrt{\frac{r+u}{m}} + \frac{u\Phi(\log(8m/\delta))}{m}\right]. \tag{A.7}$$

Closing the sample-size logarithm. Use $(a, w) = (r, 1)$ for (A.5) and $(a, w) = (r + u, u)$ for (A.7). Put

$$t = \log(4r/\delta), \qquad F = \frac{a}{\varepsilon^2} + \frac{w\Phi(t)}{\varepsilon}, \qquad M = AF, \qquad \ell = \log(1/\varepsilon),$$

where A is a sufficiently large constant depending only on d, K_{sg}. In both cases $a \ge r$, $1 \le w \le 2t$, and $\log(8M/\delta) \le C_d[t + \ell + \log(eA)]$. For each fixed integer $k \ge 1$, splitting into $\ell \le t$ and $\ell > t$ gives

$$w\ell^k \le wt^k + 2\ell^{k+1} \le wt^k + C_k/\varepsilon.$$

It follows that

$$w\Phi(\log(8M/\delta)) \le C_d[1 + \log^d(eA)]\,[w\Phi(t) + r/\varepsilon] \le C_d[1 + \log^d(eA)]\varepsilon F.$$

The square-root term at M is at most $\varepsilon/\sqrt{A}$ and the linear term at most $C_{d,K_{\mathrm{sg}}}[1+\log^d(eA)]\varepsilon/A$. Choose A so their sum is at most ε. Finally, $\log^k(8m/\delta)/m$ decreases once $\log(8m/\delta) \ge k$; increasing the same constant ensures this for $k \le d$ and every $m \ge M$. Both conclusions therefore hold for every integer sample size above the stated budgets. □

A.2 Radialization and the factorial envelope

Lemma A.1 (Direction-preserving radialization)**.** *Let $y \in \mathbb{R}^r$ satisfy $\mathbb{E}|\langle u, y\rangle|^{2p} \le A_p\|u\|^{2p}$ for all $u \in \mathbb{R}^r$, where $p \ge 2$ is an integer. Then*

$$\mathbb{E}(yy^\mathsf{T})^p \preceq \frac{c_p^{\mathrm{rad}} A_p D_{r,p}}{r\mathfrak{g}_p} I_r, \qquad c_p^{\mathrm{rad}} = 3\sqrt{3}\left(\frac{p-1}{p}\right)^{p-1}. \tag{A.8}$$

Proof. Fix a unit vector x and, independently of y, take $z \sim N(0, I_r + \lambda xx^\mathsf{T})$, with $\lambda > 0$. Conditional Gaussian integration gives

$$\mathbb{E}_z\langle z, y\rangle^{2p} = \mathfrak{g}_p(\|y\|^2 + \lambda\langle x, y\rangle^2)^p \ge \mathfrak{g}_p\frac{p^p}{(p-1)^{p-1}}\lambda\|y\|^{2p-2}\langle x, y\rangle^2.$$

The last step is weighted AM–GM: $(a + b)^p \ge p^p a^{p-1} b/(p-1)^{p-1}$ for $a, b \ge 0$. Integrating in the opposite order therefore yields

$$x^\mathsf{T}\mathbb{E}(yy^\mathsf{T})^p x \le \frac{(p-1)^{p-1}A_p}{\mathfrak{g}_p p^p \lambda}\mathbb{E}\|z\|^{2p}. \tag{A.9}$$

For $G \sim N(0, I_r)$, $\|z\|^2$ has the law $\|G\|^2 + \lambda\langle x, G\rangle^2$. Gaussian polar decomposition gives

$$\mathbb{E}\|z\|^{2p} = D_{r,p}\sum_{k=0}^{p}\binom{p}{k}\lambda^k\frac{\mathfrak{g}_k}{D_{r,k}}.$$

Here $\mathbb{E}\|G\|^{2p} = D_{r,p}$ follows by integrating the chi-square density, and the spherical moment $\mathfrak{g}_k/D_{r,k}$ follows by factoring $\mathbb{E}\langle x, G\rangle^{2k} = \mathfrak{g}_k$ into independent radius and direction. Using $D_{r,k} \geq r^k$ and $\binom{p}{k} \leq p^k/k!$ gives

$$\mathbb{E}\|z\|^{2p} \leq D_{r,p} \sum_{k\geq 0} \frac{\mathfrak{g}_k}{k!} \left(\frac{p\lambda}{r}\right)^k = \frac{D_{r,p}}{\sqrt{1-2p\lambda/r}} \quad \left(\frac{p\lambda}{r} < \frac{1}{2}\right).$$

The series identity is the binomial expansion of $(1-2a)^{-1/2}$. Set $\lambda = r/(3p)$ in (A.9) to prove the result. □

Lemma A.2 (Sharp fourth moment from the exponential normalization)**.** *Let $\xi_{\mathrm{sg}} > 0$ be the unique solution of*

$$\frac{e^{\xi_{\mathrm{sg}}} - 1}{\xi_{\mathrm{sg}}} = {K_{\mathrm{sg}}}^2, \qquad c_{\mathrm{sg}} = {K_{\mathrm{sg}}}^2 \xi_{\mathrm{sg}}. \tag{A.10}$$

Every factor in Lemma 2.2 satisfies

$$\mathbb{E}|\langle u, g\rangle|^4 \leq c_{\mathrm{sg}}\|u\|^4.$$

Consequently its tensor product has directional fourth-moment constant c_{sg}^d. The scalar constant is sharp under isotropy and (1.2), and $1 \leq c_{\mathrm{sg}} < 2\theta {K_{\mathrm{sg}}}^4$.

Proof. For unit u, put $T = |\langle u, g\rangle|^2/{K_{\mathrm{sg}}}^2$ and $\mu = \mathbb{E}T = {K_{\mathrm{sg}}}^{-2}$. Define the probability measure $d\mathbb{Q} = T\, d\mathbb{P}/\mu$. The function

$$f(t) = \frac{e^t - 1}{t} = \int_0^1 e^{st}\, ds, \qquad f(0) = 1,$$

is strictly increasing and convex on $[0, \infty)$. Jensen under $\mathbb{Q}$ gives

$$f\left(\frac{\mathbb{E}T^2}{\mu}\right) \leq \mathbb{E}_{\mathbb{Q}} f(T) = \frac{\mathbb{E}e^T - 1}{\mu} \leq {K_{\mathrm{sg}}}^2.$$

Thus $\mathbb{E}T^2 \leq \mu\xi_{\mathrm{sg}}$, proving the fourth-moment bound. Since ${K_{\mathrm{sg}}}^2 \geq 1/\log 2 > 1$, the defining positive root exists uniquely; also $f(\log 2) = 1/\log 2$ gives $\xi_{\mathrm{sg}} \geq \log 2$ and $c_{\mathrm{sg}} \geq 1$.

For sharpness, let a scalar X equal $\sqrt{c_{\mathrm{sg}}}$ and $-\sqrt{c_{\mathrm{sg}}}$ with probability $1/(2c_{\mathrm{sg}})$ each, and zero otherwise. Then $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, $\mathbb{E}e^{X^2/{K_{\mathrm{sg}}}^2} = 2$, and $\mathbb{E}X^4 = c_{\mathrm{sg}}$. The strict inequality $e^b - 1 - b > b^2/2$ at $b = \xi_{\mathrm{sg}} > 0$ gives $\xi_{\mathrm{sg}} < 2({K_{\mathrm{sg}}}^2 - 1)$ and hence $c_{\mathrm{sg}} < 2\theta {K_{\mathrm{sg}}}^4$. Convexity over normalized eigenvalues also gives $\mathbb{E}(g^\top B g)^2 \leq c_{\mathrm{sg}}(\operatorname{tr} B)^2$ for $B \succeq 0$. The factorwise induction in Lemma 2.2 therefore gives the product constant. □

At ${K_{\mathrm{sg}}}^2 = 8/3$, the scalar fourth-moment constant is $c_{\mathrm{sg}} = 4.588619425\ldots$, compared with $80/9$ from the exponential remainder alone. This improves the finite variance budgets and fourth-moment approximation results. The universal factorial normalization below continues to use $[2\theta {K_{\mathrm{sg}}}^4]^d$. This distinction is necessary for the stated all-order envelope. For the three-point extremizer with $\xi_{\mathrm{sg}} = 3$, take $r = 1$ and $d = 2$. Then $\mathbb{E}H^3 = c_{\mathrm{sg}}^4$, whereas replacing the factorial normalization by c_{sg}^2 while retaining radius coefficient $1/2$ would require $c_{\mathrm{sg}}^4 \leq 6c_{\mathrm{sg}}^2 {K_{\mathrm{sg}}}^4$. The ratio of the two sides is $\xi_{\mathrm{sg}}^2/6 = 3/2$.

Apply Lemma A.1 to (2.6) with $a = Uu$. This retains a separate explicit matrix budget at every order:

$$\mathbb{E}H^p \preceq a_p^{\mathrm{dir}} I_r, \qquad a_p^{\mathrm{dir}} = \frac{c_p^{\mathrm{rad}} [\theta \beta_p {K_{\mathrm{sg}}}^{2p}]^d D_{r,p}}{r\mathfrak{g}_p} \quad (p \geq 2). \tag{A.11}$$

At second order the finite tilted Gaussian calculation can be minimized explicitly. Fix a unit test vector x and let $z \sim N(0, I_r + \lambda x x^\top)$. The inequality $(a+b)^2 \geq 4ab$ gives

$$x^\top \mathbb{E} H^2 x \leq \frac{A_2}{12\lambda}\left[r(r+2) + 2\lambda(r+2) + 3\lambda^2\right], \qquad A_2 = [2\theta {K_{\mathrm{sg}}}^4]^d.$$

The choice $\lambda = \sqrt{r(r+2)/3}$ yields

$$\mathbb{E} H^2 \preceq v I_r, \qquad V := \mathbb{E}(H - I_r)^2 \preceq (v-1) I_r.$$

Moreover $s_r \geq 1$ and $2\theta {K_{\mathrm{sg}}}^4 > 1$, so $v > 1$. The latter strict inequality follows from isotropy and $e^x - 1 - x > x^2/2$ for $x > 0$, by taking expectations in any unit factor direction. The covariance comparison is easiest to apply when the higher moments share a factorial envelope. We first record two scalar estimates for the full exponential remainder, then derive the envelope while keeping the same second-moment normalization.

For $p \geq 3$, the supremum defining β_p is attained at the unique positive root x_p of

$$(p-x)e^x = p + (p-1)x, \qquad p - 1 < x_p < p.$$

Indeed, the derivative has the sign of $D_p(x) = (p-x)e^x - p - (p-1)x$, whose second derivative is $(p-2-x)e^x$. Starting from $D_p(0) = D_p'(0) = 0$, this shows that D_p increases before decreasing through a unique positive zero. Moreover, $D_p(p) < 0$ and $D_p(p-1) > 0$, since $e^n > 1 + n + n^2$ for $n \geq 2$. The last inequality follows from

$$\frac{n^2}{2} + \frac{n^3}{6} + \frac{n^4}{24} - n^2 = \frac{n^2(n-2)(n+6)}{24} \geq 0$$

and the remaining positive exponential terms. Consequently,

$$\beta_3 \leq 3, \qquad \beta_p \leq \frac{(p/e)^p}{1 - p e^{1-p}} < \frac{5}{4}(p/e)^p \quad (p \geq 4). \tag{A.12}$$

For the first inequality, Taylor expansion and AM–GM give $e^x - 1 - x \geq x^3[1/6 + 1/(2\sqrt{3})] \geq x^3/3$. For the second, use $x_p > p-1$, maximize $x^p e^{-x}$ at p, and note $pe^{1-p} \leq 4e^{-3} < 1/5$. For reference, $\beta_3 \approx 1.82864$ and $\beta_4 \approx 5.24911$, compared with $3! = 6$ and $4! = 24$.

Proposition A.3 (Factorial matrix envelope)**.** *For every integer $p \geq 2$,*

$$\mathbb{E} H^p \preceq \frac{p!}{2} v \left[\tfrac{1}{2} {K_{\mathrm{sg}}}^{2d} p^{d-2}(r+p)\right]^{p-2} I_r. \tag{A.13}$$

Proof of Proposition A.3. The case $p = 2$ is (2.13). For $p \geq 3$, the full exponential remainder and weighted radialization give

$$\mathbb{E} H^p \preceq c_p^{\mathrm{rad}} [\theta \beta_p {K_{\mathrm{sg}}}^{2p}]^d \frac{D_{r,p}}{r \mathfrak{g}_p} I_r, \qquad c_p^{\mathrm{rad}} = 3\sqrt{3}\left(\frac{p-1}{p}\right)^{p-1}.$$

The elementary inequality $p! \leq 2p^{p-2}$ follows inductively, since the ratio of successive normalized quantities is $(p/(p+1))^{p-2} \leq 1$. Since $\beta_p \leq p!$, the ratio of this raw bound to the asserted factorial envelope at order d is the corresponding ratio at $d = 2$ multiplied by

$$\left(\frac{\beta_p}{2p^{p-2}}\right)^{d-2} \leq 1.$$

It is therefore enough to prove the claim for $d = 2$.

Write

$$F_p(r) = \frac{D_{r,p}}{r^2 (r+p)^{p-2}}.$$

Retain the outer factors $r+2$ and $r+2p-2$ and apply AM–GM to the remaining factors. Their arithmetic mean is $r+p$, so

$$F_p(r) \le \frac{(r+2)(r+2p-2)}{r(r+p)} = 1 + \frac{p}{r+p} + \frac{4(p-1)}{r(r+p)} \le \frac{3(2p-1)}{p+1} < 6.$$

The ratio at $d=2$, with radius coefficient $1/2$, is

$$R_p = \frac{c_p^{\mathrm{rad}}\beta_p^2}{2p!\mathfrak{g}_p} F_p(r)\frac{r}{s_r}\, 2^{p-2}.$$

For $p=3$, one has

$$F_3(r)\frac{r}{s_r} = \frac{6(r+2)(r+4)}{(r+3)(r+2+\sqrt{3r(r+2)})} \le 4.$$

After rearrangement and squaring, this inequality follows from

$$12r(r+3)^2 - (r+6)^2(r+2) = 11r^3 + 58r^2 + 48r - 72 \ge 0 \qquad (r \ge 1).$$

The scalar estimate $\beta_3 \le 3$ and $c_3^{\mathrm{rad}} = 4\sqrt{3}/3$ now give $R_3 \le 8\sqrt{3}/15 < 1$.

For $p \ge 4$, the scalar remainder estimate gives $\beta_p \le (5/4)(p/e)^p$. Also $c_p^{\mathrm{rad}} \le c_3^{\mathrm{rad}}$; indeed, the derivative of $(x-1)\log(1-1/x)$ is $\log(1-1/x) + 1/x < 0$. Using $p!\mathfrak{g}_p = (2p)!/2^p$, Stirling's lower bound, and $r/s_r \le 6/(1+\sqrt{3})$ gives

$$R_p \le \frac{25 c_p^{\mathrm{rad}} F_p(r)}{128\sqrt{4\pi p}}\frac{r}{s_r} \le \frac{75\sqrt{3}}{32\sqrt{\pi}(1+\sqrt{3})} < 1.$$

This proves the envelope for all $p \ge 2$ and $d \ge 2$. □

A.3 The centered step and closed moment comparison

Rank one makes each cyclic word a product of scalar matrix coefficients. Spectral Hölder controls higher insertions; the two-insertion estimate is the classical trace AM–GM inequality [37].

Lemma A.4 (Centered one-step inequality)**.** *Let $H \succeq 0$ have rank at most one, $\mathbb{E}H = I_r$, and finite moments through order $2p$. Put $X = H - I_r$. For every deterministic symmetric A,*

$$\begin{aligned}\mathbb{E}\,\mathrm{tr}(A+X)^{2p} \le\ & \mathrm{tr}\,A^{2p} + \binom{2p}{2}\mathrm{tr}(A^{2p-2}\mathbb{E}X^2) \\ & + \sum_{j=3}^{2p}\binom{2p}{j}\mathrm{tr}\big(|A|^{2p-j}\mathbb{E}(H+I_r)^j\big). \end{aligned} \tag{A.14}$$

Proof of Lemma A.4. Indeed, the one-insertion terms vanish since $\mathbb{E}X = 0$. For two insertions, if $a+b = n_A$ is even, diagonalizing A and applying weighted AM–GM to each matrix entry gives

$$\mathrm{tr}(A^a X A^b X) \le \mathrm{tr}(|A|^{n_A} X^2).$$

For higher insertions, expand each X as $H - I_r$. Write $H = \tau uu^{\mathsf{T}}$ with $\tau \ge 0$ and $\|u\| = 1$. A cyclic trace containing $\ell \ge 1$ copies of H and $n_A > 0$ copies of A is

$$\tau^\ell \prod_{a=1}^{\ell}\langle u, A^{k_a}u\rangle, \qquad \sum_a k_a = n_A.$$

Spectral Hölder bounds its absolute value by $\tau^\ell\langle u, |A|^{n_A}u\rangle = \mathrm{tr}(|A|^{n_A}H^\ell)$. The same bound holds when $n_A = 0$; the term with no H is bounded by $\mathrm{tr}\,|A|^{n_A}$. Summing over choices of the ℓ copies of H supplies $\mathrm{tr}(|A|^{n_A}(H+I_r)^j)$ for a word with j insertions. This proves (A.14). □

Theorem A.5 (Closed finite-moment covariance bound)**.** *Let $H_1, \dots, H_m$ be iid positive semidefinite matrices of rank at most one, with $\mathbb{E}H_i = I_r$. Write $X_i = H_i - I_r$, $Z = m^{-1}\sum_i X_i$, and let $q \geq 1$ be an integer. Suppose*

$$\mathbb{E}X_i^2 \preceq \nu I_r, \qquad \mathbb{E}(H_i + I_r)^j \preceq \frac{j!}{2}\nu\mathfrak{r}^{j-2}I_r \quad (3 \leq j \leq 2q),$$

where $\nu, \mathfrak{r} > 0$. With the Gaussian moments from (2.7), define

$$h_1 = 0, \qquad h_q = \frac{2(q-1)}{\sqrt{2q-3}} \quad (q \geq 2).$$

Then

$$(\mathbb{E}\operatorname{tr} Z^{2q})^{1/(2q)} \leq r^{1/(2q)}\mathfrak{g}_q^{1/(2q)}\left(\sqrt{\frac{\nu}{m}} + \frac{h_q\mathfrak{r}}{m}\right). \tag{A.15}$$

In particular, the $2q$-th power of the right side divided by ε^{2q} bounds $\mathbb{P}\{\|Z\| > \varepsilon\}$.

Proof of Theorem A.5. Apply Lemma A.4. Let $S_n = \sum_{i=1}^n X_i$, and put $M_{n,p} = r^{-1}\mathbb{E}\operatorname{tr} S_n^{2p}$, with $M_{n,0} = 1$. Cauchy–Schwarz in the joint probability and normalized spectral measure yields

$$r^{-1}\mathbb{E}\operatorname{tr}|S_n|^{2k+1} \leq \sqrt{M_{n,k}M_{n,k+1}}.$$

For $u = (u_0, \dots, u_q) \geq 0$ define

$$\mathcal{I}_{2k}(u) = u_k, \qquad \mathcal{I}_{2k+1}(u) = \sqrt{u_k u_{k+1}}, \qquad \Phi_p(u) = \sum_{j=2}^{2p}\binom{2p}{j}b_j\mathcal{I}_{2p-j}(u), \tag{A.16}$$

where $b_2 = \nu$ and $b_j = (j!/2)\nu\mathfrak{r}^{j-2}$ for $j \geq 3$; put $\Phi_0 = 0$. Conditioning (A.14) on S_{n-1} gives

$$M_{n,p} \leq M_{n-1,p} + \Phi_p(M_{n-1}). \tag{A.17}$$

Every Φ_p is componentwise nondecreasing.

For the pairing comparison, set

$$\omega_{2k} = \mathfrak{g}_k, \quad \omega_{2k+1} = \sqrt{\mathfrak{g}_k\mathfrak{g}_{k+1}}, \quad C_{p,j} = \frac{(2p)!\,\omega_{2p-j}}{2p(2p-j)!\mathfrak{g}_p}. \tag{A.18}$$

For $q \geq 2$, these coefficients satisfy

$$C_{p,j} \leq h_q^{j-2} \qquad (1 \leq p \leq q,\ 2 \leq j \leq 2p). \tag{A.19}$$

Here $C_{p,2} = 1$. For $j = 2\ell$,

$$C_{p,2\ell} = \frac{2^{\ell-1}(p-1)!}{(p-\ell)!},$$

so its successive ratios are at most $2(p-1)$. For $j = 2\ell+1$, writing $k = p - \ell \geq 1$ gives

$$C_{p,2\ell+1} = \frac{2^\ell(p-1)!}{(k-1)!\sqrt{2k-1}}.$$

Thus $C_{p,3} = h_p \leq h_q$ for $p \geq 2$, and the successive odd ratios, when $k \geq 2$, are $2(k-1)\sqrt{(2k-1)/(2k-3)} \leq 2k \leq 2(p-1)$. The inequality $h_p \leq h_q$ follows from the increasing sequence $h_p^2 = 2p - 1 + (2p-3)^{-1}$ for $p \geq 2$. Finally, $2(p-1) \leq 2(q-1) \leq h_q^2$. Induction on each parity proves (A.19).

For $q \geq 2$, write $a = h_q\mathfrak{r}$, $s(t) = \sqrt{\nu t} + a$, and

$$B_0(t) = 1, \qquad B_p(t) = \mathfrak{g}_p s(t)^{2p} \quad (1 \leq p \leq q).$$

For $t > 0$, $\mathcal{I}_k(B(t)) = \omega_k s(t)^k$, and the pairing bound gives

$$\frac{\Phi_p(B(t))}{2p\mathfrak{g}_p s(t)^{2p-1}} = \sum_{j=2}^{2p} \frac{C_{p,j} b_j}{j!} s(t)^{1-j} \leq \frac{\nu}{2s(t)} \sum_{j=2}^{2p} \left(\frac{a}{s(t)}\right)^{j-2} \leq \frac{\nu}{2(s(t)-a)} = s'(t).$$

Hence $B_p'(t) \geq \Phi_p(B(t))$. Since B is componentwise nondecreasing, integration on $[n, n+1]$ yields

$$B_p(n+1) \geq B_p(n) + \Phi_p(B(n)) \quad (n \geq 0).$$

At $n = 0$ this integration remains valid: B_p is absolutely continuous on every compact interval and its derivative has only an integrable $t^{-1/2}$ singularity. The initial data satisfy $M_{0,p} \leq B_p(0)$. Induction in n using (A.17) and monotonicity of Φ now proves $M_{m,q} \leq B_q(m)$, which is (A.15). For $q = 1$, independence and centering give directly $\mathbb{E}\operatorname{tr} S_m^2 = m \operatorname{tr} \mathbb{E} X_1^2 \leq mr\nu$, yielding the asserted bound with $h_1 = 0$. Finally, $\|Z\|^{2q} \leq \operatorname{tr} Z^{2q}$ and Markov's inequality give the tail statement. □

A.4 Shifting raw factorial budgets

Lemma A.6 (Shifted factorial budgets)**.** *Suppose $H \succeq 0$ has rank at most one, $\mathbb{E}H = I_r$, and $\mathbb{E}H^j \preceq (j!/2)v\mathcal{R}^{j-2} I_r$ for $2 \leq j \leq 2q$, with $v, \mathcal{R} > 0$. Then the hypotheses of Theorem A.5 hold with*

$$\nu = v, \qquad \mathfrak{r} = \mathcal{R} + 1 + \frac{4}{3v}. \tag{A.20}$$

Proof of Lemma A.6. The variance is $V = \mathbb{E}H^2 - I_r \preceq vI_r$. For $j \geq 3$, binomial expansion gives

$$\mathbb{E}(H + I_r)^j \preceq \left[j + 1 + \frac{j!}{2} v \sum_{h=0}^{j-2} \frac{\mathcal{R}^{j-2-h}}{h!} \right] I_r.$$

Set $n = j - 2$ and $c = 1 + 4/(3v)$. Every coefficient of $(\mathcal{R} + c)^n$ dominates the corresponding $1/h!$, since $c \geq 1$. Its excess constant coefficient is at least $c^n - 1/n! \geq c - 1 = 4/(3v) \geq 2(j+1)/(j!v)$. This also absorbs the first term and proves the asserted budgets. □

Proof of Proposition 2.6. The scale $\frac{1}{2} K_{\mathrm{sg}}{}^{2d} p^{d-2}(r+p)$ increases with p. Apply Proposition A.3 through order $2q$, followed by Lemma A.6. Theorem A.5 then applies with $\nu = v$ and $\mathfrak{r} = \mathfrak{r}_q$. Markov's inequality proves (2.16); solving the resulting quadratic in $m^{-1/2}$ gives (2.17). □

A.5 Sharper certificates

The closed bound is sufficient for the sharp dimension order. For a fixed moment order or a structured covariance, one can retain the individual budgets, zero initial moments, and exact terminal insertions instead. The resulting computation is finite and monotone in every input budget.

The following comparison is independent of the tensor model. Let $H_1, \ldots, H_m$ be independent identically distributed positive semidefinite matrices of rank at most one, with $\mathbb{E}H_i = I_r$ and finite moments through order $2q$. Write $X_i = H_i - I_r$, $S_n = \sum_{i=1}^n X_i$, and $Z = S_m/m$. Assume finite scalar budgets

$$V = \mathbb{E}X_1^2 \preceq \nu I_r, \qquad b_2 = \nu > 0, \qquad \mathbb{E}(H_1 + I_r)^j \preceq b_j I_r \quad (3 \leq j \leq 2q). \tag{A.21}$$

The budgets need not be moments of a scalar probability law. We compare two ways to propagate them: a direct update of even trace moments after each sample, and a single continuous scale

that yields an integral certificate. Both begin at a zero partial sum. The discrete bound is tighter; the integral form avoids updating all moments after each of the m samples.

Use the interpolation maps and insertion operators in (A.16), now with the individual budgets b_j, and the pairing weights in (A.18). Starting at $u^{(0)} = (1, 0, \dots, 0)$, the discrete certificate is

$$u^{(n)} = u^{(n-1)} + \Phi(u^{(n-1)}). \tag{A.22}$$

At order $2q$ these coefficients define the growth rate ϕ_q of the common scale. Integrating its reciprocal gives the time needed to reach a scale, and inversion evaluates the scale after m samples. For $s > 0$, define

$$\phi_q(s) = \sum_{j=2}^{2q} \frac{C_{q,j} b_j}{j!} s^{1-j}, \qquad \Gamma_q(a) = \int_0^a \frac{ds}{\phi_q(s)}, \qquad T_q(m) = \Gamma_q^{-1}(m). \tag{A.23}$$

Theorem A.7 (Finite-profile covariance certificate)**.** *Under (A.21), the inverse in (A.23) is uniquely defined for every $m > 0$, and*

$$\mathbb{E}\,\mathrm{tr}\, Z^{2q} \le \frac{r}{m^{2q}} u_q^{(m)} \le \frac{r\mathfrak{g}_q}{m^{2q}} T_q(m)^{2q}. \tag{A.24}$$

Consequently,

$$\mathbb{P}\{\|Z\| > \varepsilon\} \le \frac{r u_q^{(m)}}{m^{2q}\varepsilon^{2q}} \le \frac{r\mathfrak{g}_q T_q(m)^{2q}}{m^{2q}\varepsilon^{2q}}. \tag{A.25}$$

If upper bounds $\widehat{\tau}_{2p}$ on $\tau_{2p} = r^{-1}\mathbb{E}\,\mathrm{tr}\, X_1^{2p}$ are available, with $\widehat{\tau}_2 \le \nu$ and $\widehat{\tau}_{2p} \le b_{2p}$, the stronger discrete certificate replaces $u^{(n)}$ by

$$\begin{aligned} &\widehat{u}_0^{(n)} = 1, \qquad \widehat{u}_1^{(n)} = n\widehat{\tau}_2, \\ &\widehat{u}_p^{(n)} = \widehat{u}_p^{(n-1)} + \binom{2p}{2} \nu \widehat{u}_{p-1}^{(n-1)} + \widehat{\tau}_{2p} + \sum_{j=3}^{2p-2} \binom{2p}{j} b_j \mathcal{I}_{2p-j}(\widehat{u}^{(n-1)}) \quad (2 \le p \le q), \end{aligned} \tag{A.26}$$

where $\widehat{u}_p^{(0)} = 0$ for $p \ge 1$.

Proof. Apply Lemma A.4 to obtain (A.14). Set $M_{n,p} = r^{-1}\mathbb{E}\,\mathrm{tr}\, S_n^{2p}$ and $M_{n,0} = 1$. Cauchy–Schwarz over the joint sample and normalized spectral measure gives

$$r^{-1}\mathbb{E}\,\mathrm{tr}\,|S_n|^{2k+1} \le \sqrt{M_{n,k} M_{n,k+1}}.$$

Conditioning (A.14) on S_{n-1} now yields $M_{n,p} \le M_{n-1,p} + \Phi_p(M_{n-1})$. Each Φ_p is nonnegative, componentwise nondecreasing, and depends only on coordinates $0, \dots, p-1$. Induction proves $M_{n,p} \le u_p^{(n)}$.

For the refined update, the all-X term is exactly τ_{2p}. When $p \ge 2$, a word with $2p-1$ copies of X has one copy of S_{n-1} and cyclic trace $\mathrm{tr}(S_{n-1}X^{2p-1})$. Its expectation is zero because the two matrices are independent and $\mathbb{E}S_{n-1} = 0$. The second moment is $M_{n,1} = n\tau_2$. These observations prove (A.26) by the same induction, and show $\widehat{u}_p^{(n)} \le u_p^{(n)}$.

To prove the integral comparison, first record

$$C_{p,j} \le C_{p+1,j}, \qquad C_{p,j} \le (2p)^{(j-2)/2}. \tag{A.27}$$

The coefficient formulas in the proof of Theorem A.5 give these statements directly. Even coefficients increase with p. For an odd index $j = 2\ell + 1$, write $k = p - \ell \ge 1$; the ratio on increasing p is $[(k+\ell)/k]\sqrt{(2k-1)/(2k+1)} \ge 1$. The smallest case $\ell = 1$ reduces after squaring to $2k^2 - 1 \ge 0$. The second bound follows from (A.19) and $h_p \le \sqrt{2p}$, with $p = 1$ immediate.

Since $b_2 = \nu > 0$, $\phi_q(s) > 0$ and $1/\phi_q(s)$ is integrable at zero. At infinity $\phi_q(s) \sim \nu/(2s)$. Thus Γ_q increases continuously from zero to infinity and has a unique inverse. With $T_q(0) = 0$, it satisfies $T_q'(t) = \phi_q(T_q(t))$ for $t > 0$.

Construct continuous functions in increasing degree by

$$H_0(t) = 1, \qquad H_p(0) = 0, \qquad H_p'(t) = \Phi_p(H(t)).$$

The triangular structure makes this a recursive integration of known continuous functions. All components are nondecreasing, so $H(n+1) \geq H(n) + \Phi(H(n))$ and $u_p^{(n)} \leq H_p(n)$. Put $B_0 = 1$ and $B_p(t) = \mathfrak{g}_p T_q(t)^{2p}$ for $1 \leq p \leq q$. The common scale gives $\mathcal{I}_k(B) = \omega_k T_q^k$. Hence

$$\frac{\Phi_p(B)}{2p\mathfrak{g}_p T_q^{2p-1}} = \sum_{j=2}^{2p} \frac{C_{p,j} b_j}{j!} T_q^{1-j} \leq \phi_q(T_q) = T_q'.$$

Thus $B_p' \geq \Phi_p(B)$. Induction in p and integration from zero give $H_p \leq B_p$. Integration is justified even if derivatives are unbounded at zero: on $[a,t]$ it is ordinary integration, and $B_p(a) \to 0$ as $a \downarrow 0$; the positive derivative has integral $B_p(t)$. This proves both bounds in (A.24). Finally, $\|Z\|^{2q} \leq \operatorname{tr} Z^{2q}$ and Markov's inequality prove (A.25). □

Lemma A.8 (Centered trace budgets from raw moments)**.** *Suppose $H = yy^{\mathsf{T}}$, $\mathbb{E}H = I_r$, and $\mathbb{E}H^h \preceq a_h I_r$ for $2 \leq h \leq 2q$, with $a_0 = a_1 = 1$. Then valid centered trace budgets for (A.26) are*

$$\widehat{\tau}_{2p} = 1 + \sum_{\substack{2 \leq h \leq 2p \\ h \text{ even}}} \binom{2p}{h} a_h - \sum_{\substack{1 \leq h \leq 2p-1 \\ h \text{ odd}}} \binom{2p}{h} r^{h-1} \qquad (1 \leq p \leq q). \tag{A.28}$$

These satisfy $0 \leq \tau_{2p} \leq \widehat{\tau}_{2p} \leq b_{2p}$ for the positive shifted budgets $b_j = \sum_{h=0}^{j} \binom{j}{h} a_h$ at $j \geq 3$ and $b_2 = a_2 - 1$. In particular, $\widehat{\tau}_2 = a_2 - 1$ and $\widehat{\tau}_4 = a_4 + 6a_2 - 4r^2 - 3$.

Proof. Put $t = \operatorname{tr} H = \|y\|^2$. The rank-one identity and the even power give

$$\tau_{2p} = 1 + \sum_{h=1}^{2p} (-1)^h \binom{2p}{h} \frac{\mathbb{E}t^h}{r}.$$

For even h, the raw budget gives $\mathbb{E}t^h/r \leq a_h$. For odd h, Jensen and $\mathbb{E}t = r$ give $\mathbb{E}t^h/r \geq r^{h-1}$. Applying these bounds with their respective signs proves (A.28). The resulting expression is nonnegative because it bounds the nonnegative actual moment. It is no larger than the positive binomial budget at even orders at least four; at second order it is exactly $a_2 - 1$. □

A.5.1 Finite radial choices

Let A_p bound all directional $2p$ moments of y. Define

$$P_{r,p}(\lambda) = \sum_{k=0}^{p} \binom{p}{k} \lambda^k \frac{\mathfrak{g}_k}{D_{r,k}}, \qquad L_p(\lambda) = \begin{cases} (1+\lambda)^p, & 0 \leq \lambda \leq 1/(p-1), \\ p^p \lambda/(p-1)^{p-1}, & \lambda \geq 1/(p-1). \end{cases}$$

The exact finite Gaussian calculation in Lemma A.1 gives, for every deterministic $\lambda \geq 0$,

$$\mathbb{E}H^p \preceq a_p^{\mathrm{tilt}}(\lambda) I_r, \qquad a_p^{\mathrm{tilt}}(\lambda) = \frac{A_p D_{r,p}}{\mathfrak{g}_p} \frac{P_{r,p}(\lambda)}{L_p(\lambda)}. \tag{A.29}$$

Indeed, $t = \langle x, y\rangle^2/\|y\|^2 \in [0,1]$ and $L_p(\lambda) = \min_{0<t\leq 1}(1+\lambda t)^p/t$. This retains all terms in the finite Gaussian polynomial. Any evaluated tilt is a valid upper bound; finding its global minimum is unnecessary for validity. Two further explicit candidates are

$$a_p^{\mathrm{Höl}} = A_p (D_{r,p}/\mathfrak{g}_p)^{(p-1)/p}, \qquad a_2^{\mathrm{quad}} = A_2 \frac{r + 2 + \sqrt{3r(r+2)}}{6}. \tag{A.30}$$

For the first, Hölder bounds $\mathbb{E}\|y\|^{2p-2}\langle x,y\rangle^2$ using $\mathbb{E}\|y\|^{2p} \le A_p D_{r,p}/\mathfrak{g}_p$ and the directional moment. For the second, minimize the quadratic tilted-Gaussian polynomial after the weighted AM–GM step. At $r=1$ it equals A_2; for every p the scalar bound $a_p = A_p$ is valid at that rank.

A.5.2 Explicit tensor budgets

Let $A_p = [\theta\beta_p {K_{\mathrm{sg}}}^{2p}]^d$ for the sub-Gaussian model. Lemma 4.3 gives

$$v_{\mathrm{orth}} = A_2 + (r-1)\gamma(A_2) \le A_2 r. \tag{A.31}$$

Lemma A.2 supplies the additional candidate

$$a_2^{\mathrm{sg}} = \min\left\{c_{\mathrm{sg}}^d s_r,\, c_{\mathrm{sg}}^d + (r-1)\gamma(c_{\mathrm{sg}}^d)\right\}. \tag{A.32}$$

The two entries follow, respectively, from the quadratic radial bound and the orthogonal fourth-moment bound using the sharper scalar constant. Retain the separate raw and variance budgets by choosing

$$\begin{aligned}
a_0 = a_1 = 1, \qquad & a_2 = \min\{v, v_{\mathrm{orth}}, a_2^{\mathrm{sg}}, a_2^{\mathrm{dir}}, a_2^{\mathrm{quad}}, \inf_{\lambda\ge 0} a_2^{\mathrm{tilt}}(\lambda)\},\\
& a_k = \min\left\{a_k^{\mathrm{dir}}, a_k^{\mathrm{Höl}}, \inf_{\lambda\ge 0} a_k^{\mathrm{tilt}}(\lambda), \frac{k!}{2} v[\tfrac{1}{2}{K_{\mathrm{sg}}}^{2d} k^{d-2}(r+k)]^{k-2}\right\}, \quad 3\le k\le 2q,\\
& \nu = a_2 - 1, \qquad b_2 = \nu, \qquad b_j = \sum_{k=0}^{j}\binom{j}{k} a_k \quad (3\le j\le 2q).
\end{aligned} \tag{A.33}$$

At $r=1$, also take the minimum with A_k. Any finite list of chosen tilts may replace the infima. Each candidate is an upper bound, so minimizing and positive binomial expansion preserve validity. If $\nu = 0$, then $\mathbb{E}\|H - I\|_{\mathrm{F}}^2 = 0$ and $Z = 0$ almost surely; otherwise Theorem A.7 applies directly.

All these raw budgets are no larger than the factorial envelope with radius coefficient $1/2$. Its positive shift has parameters $(v, \mathfrak{r}_q)$. Decreasing any budget decreases ϕ_q, increases Γ_q, and decreases $T_q(m)$. Therefore the finite-profile allowance

$$\eta_q c_q \frac{T_q(m)}{m} \tag{A.34}$$

is no larger than (2.16). Indeed, the dominating factorial profile satisfies $\phi_q^{\mathrm{fact}}(s) \le v/[2(s - h_q\mathfrak{r}_q)]$ above $h_q\mathfrak{r}_q$, so integration gives $T_q(m) \le \sqrt{vm} + h_q\mathfrak{r}_q$. For $q=1$, the respective scales are $\sqrt{\nu m}$ and $\sqrt{vm}$. The discrete certificate and centered terminal budgets in Lemma A.8 can improve this further.

A.5.3 Exact scalar profiles for standard factor laws

The sub-Gaussian parameter is unnecessary when the factor law has an explicit directional moment profile. The Gaussian and spherical formulas are classical; the Rademacher flat extremizer is due to Eaton and Komorowski [43, 44]. We include an elementary proof at even integer moments, the orders needed here. For an integer $p \ge 1$, set

$$\begin{aligned}
c_{n,p}^{\mathrm{G}} &= \mathfrak{g}_p,\\
c_{n,p}^{\mathrm{sph}} &= \mathfrak{g}_p \frac{n^p}{D_{n,p}},\\
c_{n,p}^{\mathrm{Rad}} &= \frac{1}{2^n n^p}\sum_{a=0}^{n}\binom{n}{a}(n-2a)^{2p}.
\end{aligned} \tag{A.35}$$

These are the sharp directional $2p$-moment constants for, respectively, a standard Gaussian factor, a factor uniform on the sphere of radius $\sqrt{n}$, and a factor with independent Rademacher coordinates. In each case $c_{n,1} = 1$. In particular,

$$c_{n,2}^{\mathrm{Rad}} = 3 - \frac{2}{n}, \qquad c_{n,3}^{\mathrm{Rad}} = 15 - \frac{30}{n} + \frac{16}{n^2}, \qquad c_{n,2}^{\mathrm{sph}} = \frac{3n}{n+2}.$$

Thus dimension-one sign and spherical modes contribute no moment loss.

Lemma A.9 (Sharp product scalar budgets)**.** *Choose any of the three laws independently in each mode, let $A_p = \prod_{j=1}^{d} c_{n_j,p}^{(j)}$, and form the product probe z. Then*

$$\mathbb{E}|\langle a, z\rangle|^{2p} \le A_p \|a\|^{2p}.$$

The constant A_p is attained by an appropriate rank-one unit tensor. Consequently the direction-preserving radialization lemma can be applied with this A_p in place of $[\theta\beta_p {K_{\mathrm{sg}}}^{2p}]^d$ at every required integer order.

Proof. The Gaussian identity is classical. The spherical identity follows by factoring a Gaussian into its independent radius and direction: $\mathbb{E}R^{2p} = D_{n,p}$ and $\mathbb{E}G_1^{2p} = \mathfrak{g}_p$.

For the sign law, signs of deterministic coefficients can be absorbed into the independent Rademacher variables. Fix two nonnegative coefficients a, b with $c = a^2 + b^2$, and condition on the other coordinates, whose sum is s. With independent signs ϵ, η,

$$\mathbb{E}(a\epsilon + b\eta)^{2k} = \frac{(c + 2ab)^k + (c - 2ab)^k}{2}.$$

This is a polynomial in a^2b^2 with nonnegative coefficients, and is strictly increasing when $k \ge 2$ and $c > 0$. Expanding $\mathbb{E}(s + a\epsilon + b\eta)^{2p}$ retains only even local moments, with nonnegative coefficients $\binom{2p}{2k} s^{2p-2k}$. For $p \ge 2$, averaging unequal squared coefficients therefore strictly increases the moment while preserving the squared norm. A maximizer exists by compactness; every squared coefficient at a maximizer must equal $1/n$. The binomial law of $\sum_{a=1}^{n} \epsilon_a$ gives the last formula in (A.35). The case $p = 1$ is isotropy.

For tensorization, slice a coefficient tensor in its last factor. The local directional bound leaves a power of the sum of squared partially contracted coefficients. Minkowski in L^p, followed by induction, bounds this by $A_p\|a\|^{2p}$. A rank-one tensor made of local maximizing directions attains the product bound by independence. □

Proposition A.10 (Gaussian and sign factorial radius)**.** *Suppose $d \ge 2$ and each factor is standard Gaussian or has independent Rademacher coordinates, independently across modes. With $v_{\mathrm{G}} = 3^d r$, for every integer $p \ge 2$,*

$$\mathbb{E}H^p \preceq \frac{p!}{2} v_{\mathrm{G}} \left[\frac{11}{5}\, 2^{d-2} p^{d-2} (r + p)\right]^{p-2} I_r. \tag{A.36}$$

For every integer $q \ge 1$, the closed certificates (2.16)–(2.17) therefore apply with $(v, \mathfrak{r}_q)$ replaced by $(v_{\mathrm{G}}, \mathfrak{r}_q^{\mathrm{G}})$, where

$$\mathfrak{r}_q^{\mathrm{G}} = \frac{11}{5}\, 2^{d-2} (2q)^{d-2} (r + 2q) + 1 + \frac{4}{3v_{\mathrm{G}}}. \tag{A.37}$$

The raw right side of (A.36) is also an additional candidate in the finite profile. Spherical factors satisfy the same envelope; their exact radial multipliers below can improve it.

Proof. All the stated laws have directional $2p$ moments bounded by $A_p = \mathfrak{g}_p^d$. For the sign comparison, expand an even moment of a linear form. Only even coordinate powers survive, and

every corresponding Gaussian coordinate moment is at least one. The factorwise induction then gives the product bound. This argument at arbitrary orders requires full coordinate independence in the sign case; four-wise independence alone supplies the fourth-moment bounds. For spheres, $n^p/D_{n,p} \le 1$ gives the same comparison.

The case $p = 2$ follows from the directional fourth moment and $\mathbb{E}H^2 \preceq 3^d r I_r$. For $p \ge 3$, use the Hölder candidate in (A.30) and $D_{r,p} \le r(r+p)^{p-1}$, which is AM–GM on the final $p-1$ factors of $D_{r,p}$. Relative to the right side of (A.36), with 11/5 replaced temporarily by one, the resulting ratio is at most

$$\left[\frac{\mathfrak{g}_p}{3(2p)^{p-2}}\right]^{d-2} \frac{2\mathfrak{g}_p^{1+1/p}}{9p!}\left(\frac{r+p}{r}\right)^{1/p}.$$

The first factor is at most one: retain $1 \cdot 3$ in $\mathfrak{g}_p$ and bound each of its remaining $p-2$ factors by $2p$. Since $r \ge 1$, the second factor is bounded by

$$G_p = \frac{2}{9}\frac{\mathfrak{g}_p}{p!}[\mathfrak{g}_p(p+1)]^{1/p}.$$

At $p = 3, 4$, the needed inequalities are the exact rational comparisons

$$G_3^3 = \frac{2500}{243} \le (11/5)^3, \qquad G_4^4 = 525(35/36)^4 \le (11/5)^8.$$

For $p \ge 4$, we have $[\mathfrak{g}_p(p+1)]^{1/p} \le 6p/5$. Indeed, $\mathfrak{g}_p(p+1)/p^p$ decreases for $p \ge 2$, because its consecutive ratio is

$$\frac{(2p+1)(p+2)p^p}{(p+1)^{p+2}} \le 1.$$

To see this, the first three binomial terms give $(1+1/p)^p \ge 2 + (p-1)/(2p)$, whereas $(2p+1)(p+2)/(p+1)^2 = 2 + p/(p+1)^2$ and $(p-1)/(2p) \ge p/(p+1)^2$ for $p \ge 2$. The bound at $p = 4$ is $525 \le (24/5)^4$; it persists since $(6/5)^p$ increases.

Consequently $G_p \le (4/15)p\mathfrak{g}_p/p!$ for $p \ge 4$. At $p = 5$ this upper bound is $21/2 \le (11/5)^3$, and its consecutive ratio is $2 + 1/p \le 11/5$ for $p \ge 5$. Thus $G_p \le (11/5)^{p-2}$ for all $p \ge 3$. This proves (A.36); Lemma A.6 gives (A.37). □

For a purely spherical product, the exact Gaussian coupling also improves the whole uncentered matrix profile. With

$$\varsigma_p = \prod_{j=1}^{d} \frac{n_j^p}{D_{n_j,p}}, \tag{A.38}$$

and the same fixed isometry U, let $H_{\mathrm{sph}} = y_{\mathrm{sph}}y_{\mathrm{sph}}^\top$ and $H_{\mathrm{G}} = y_{\mathrm{G}}y_{\mathrm{G}}^\top$. Independent Gaussian radii give the exact matrix identity

$$\mathbb{E}H_{\mathrm{sph}}^p = \varsigma_p \mathbb{E}H_{\mathrm{G}}^p. \tag{A.39}$$

Indeed, $y_{\mathrm{G}} = a y_{\mathrm{sph}}$ with an independent scalar $a = \prod_j R_j/\sqrt{n_j}$, and $\mathbb{E}a^{2p} = \varsigma_p^{-1}$. In particular, a Gaussian raw matrix budget a_p^{G} gives the spherical raw budget $\varsigma_p a_p^{\mathrm{G}}$, while at second order

$$\mathbb{E}(H_{\mathrm{sph}} - I_r)^2 = \varsigma_2 \mathbb{E}H_{\mathrm{G}}^2 - I_r.$$

The positive binomial budgets and the centered trace-profile formulas can then be rebuilt using these improved raw inputs. There is no claim that the same scalar factor multiplies every centered moment or the centered sum; those contain several different homogeneous degrees.

For Gaussian/sign factors, Lemma B.3 permits the additional raw second-moment candidate $3^d + (r-1)(3^d-1)/2$, whose centered version is $[(3^d-1)/2](r+1)$. At tensor order two, Lemma B.5 gives the additional raw candidate $3r + 4 + 2\sqrt{r}$. Thus one may use

$$a_2 \le \min\{4r+5,\, 3r+4+2\sqrt{r}\}, \qquad \nu = a_2 - 1 \le \min\{4r+4,\, 3r+3+2\sqrt{r}\}.$$

Here a_2 denotes the minimum of the applicable raw candidates, including any smaller radial or exact geometric profile already available. For a known Gaussian subspace, (B.7) gives the raw matrix exactly. For spherical factors multiply each applicable Gaussian raw candidate by ς_2 before subtracting the identity. The coordinate-class fourth-moment conclusions need only four-wise independence; arbitrary-order sign profiles still require full coordinate independence. All higher radial formulas hold with these law-specific A_p; the universal sub-Gaussian factorial envelope remains an optional additional candidate when a valid K_{sg} is supplied.

A.6 Spectral consequences and accuracy-sensitive lower bounds

Proposition A.11 (Spectrum at linear sketch size)**.** *Assume the factor model of Theorem 2.1, with $d \geq 2$. Let $C_4 = c_{\mathrm{sg}}^d$, with c_{sg} from Lemma A.2, a valid product directional fourth-moment constant. For a fixed isometry $U \in \mathbb{R}^{N\times r}$ put $M = U^\top \Omega\Omega^\top U$, $m = \lceil Cr \rceil$, $C \geq 1$, and $t = \log(8r/\delta)$, where $0 < \delta < 1$. There is a constant $B_{d,K_{\mathrm{sg}}}$ such that, whenever*

$$m \geq \mathfrak{m}_{\mathrm{inj}}(C_4, r; 1/2, \delta/2),$$

with probability at least $1-\delta$,

$$\lambda_{\min}(M) \geq \tfrac{1}{2}, \qquad \lambda_{\max}(M) \leq 1 + B_{d,K_{\mathrm{sg}}} \left[\sqrt{\frac{t}{C}} + \frac{t^{d-1}}{C} + \frac{t^d}{Cr} \right].$$

Writing $\mathrm{cond}(B) = \sigma_{\max}(B)/\sigma_{\min}(B)$ for a full-column-rank matrix, for fixed d, K_{sg} and sufficiently large fixed C, this implies $\mathrm{cond}(\Omega^\top U) = O_{d,K_{\mathrm{sg}},C}((\log r)^{(d-1)/2})$ with probability tending to one as $r \to \infty$. For Gaussian factors (with ${K_{\mathrm{sg}}}^2 \geq 8/3$), the common-factor witness attains this order:

$$\mathrm{cond}(\Omega^\top U_r) = \Theta_{d,K_{\mathrm{sg}},C}((\log r)^{(d-1)/2}) \quad \text{with probability } 1 - o(1).$$

Proof. Apply (2.16) with failure $\delta/2$ and $q = \lceil \log(8r/\delta) \rceil$. The proof of the upper OSE theorem gives the displayed spectral allowance without imposing that it be below one. The lower-injection bound spends at most $\delta/2$, so their intersection has probability at least $1-\delta$. For the asymptotic statement use failure r^{-1} in the upper estimate. For the lower edge, use (4.3) with $\alpha = 1/2$, $\eta = e^{-r}$, and $s = r$. Its sufficient count is at most $120C_4 r$ for $r \geq 1$. Thus a fixed $C \geq 120C_4$ makes $M \succeq I_r/2$ with failure at most e^{-r}. The resulting upper edge is $O((\log r)^{d-1})$ for every $d \geq 2$. On the Gaussian common-factor witness, the existing one-heavy-column argument gives $\lambda_{\max}(M) \geq c_d(\log r)^{d-1}/C$ with probability $1 - o(1)$. Also $r^{-1} \operatorname{tr} M \to 1$ in probability, since its variance is $[3^{d-1}(1 + 2/r) - 1]/m$; hence $\lambda_{\min}(M) \leq 2$ with probability $1 - o(1)$. Take square roots of the ratio of spectral edges. □

Proposition A.12 (Accuracy-sensitive Gaussian upper obstruction)**.** *Fix $d \geq 2$ and put $p = d-1$. On the Gaussian common-factor subspace let $M = m^{-1} \sum_{i=1}^m w_i \xi_i \xi_i^\top$, with independent $\xi_i \sim N(0, I_r)$ and $w_i = \prod_{j=1}^p \eta_{ij}^2$, where all η_{ij} are standard normal. For $r \geq 4$, $m \geq r$ sufficiently large, and $L_m = (\log m/p)^p$, there is a constant $a_d > 0$ such that*

$$\mathbb{P}\left\{ \lambda_{\max}(M) - 1 \geq \frac{rL_m}{8m} \right\} \geq 1 - \exp\left[-a_d \frac{\sqrt{m}}{(\log m)^{p/2}} \right] - e^{-r/16} - \frac{64(3^d - 1)m}{r^2 L_m^2}.$$

Consequently, for $\varepsilon_r \in (0,1)$ with $\varepsilon_r r(\log r)^p \to \infty$, uniformly over integer sizes

$$r \leq m \leq \frac{r(\log r)^p}{16 p^p \varepsilon_r}, \qquad \mathbb{P}\{\|M - I_r\| > \varepsilon_r\} \longrightarrow 1.$$

Sizes $m < r$ fail deterministically. Thus in this accuracy regime any fixed-confidence OSE on this witness needs $m = \Omega_d(r(\log r)^{d-1}/\varepsilon_r)$.

Proof. Choose J maximizing w_i using only the weights and set $x = \xi_J/\|\xi_J\|$. An exact identity is

$$m[x^\top Mx - 1] = w_J(\|\xi_J\|^2 - 1) + Q, \quad Q = \sum_i w_i - m + \sum_{i\neq J} w_i(\langle \xi_i, x\rangle^2 - 1).$$

Conditional on all weights and ξ_J, the second sum defining Q has mean zero and variance $2\sum_{i\neq J} w_i^2$. The first sum has variance $m(3^p-1)$, and the cross covariance vanishes by conditional expectation. Therefore $\mathbb{E}Q = 0$ and $\mathbb{E}Q^2 \le (3^d-1)m$. Chebyshev gives $\mathbb{P}\{Q < -rL_m/8\} \le 64(3^d-1)m/(r^2L_m^2)$. The existing Gaussian extreme-weight argument gives $w_J \ge L_m$ outside the stated first failure term. Selection by the weights leaves ξ_J standard Gaussian, so its norm is at least $\sqrt{r/2}$ outside probability $e^{-r/16}$. Since $r/2 - 1 \ge r/4$, intersecting these three events proves the finite bound. In the specified size range the spectral increase is at least $2\varepsilon_r$, and the last failure term is $O_d([\varepsilon_r r(\log r)^p]^{-1})$. The other two terms tend to zero uniformly: $m \ge r$, and the accuracy condition bounds m above by $o(r^2(\log r)^{2p})$, so $\log m = O_d(\log r)$. □

A.7 Common-Schmidt transition and spherical comparison

The arguments here use classical Gaussian moments, scalar exponential estimates, an Euclidean net, and Gaussian radius–direction independence. Weighted Gaussian covariance estimates have a broader theory in [9]. We give the elementary special case needed for Section 3; the Gaussian/spherical range invariance in [10, Section 5.4] is consistent with the polar comparison below.

Common-Schmidt row law and exact variance. For an order-two support with orthonormal basis

$$v_a = \sum_{b=1}^{s} \sqrt{\lambda_b}\, e_{a,b} \otimes f_b, \quad a = 1, \dots, r,$$

the rs vectors $e_{a,b}$ are jointly orthonormal and the f_b are orthonormal. Zero weights may be omitted. Conditional on the second Gaussian factor, the projections have independent centered Gaussian coordinates with common variance $w = \sum_b \lambda_b h_b^2$. Dividing by $\sqrt{w}$ shows that the resulting g is independent of w, since its conditional law is constant. Here $w > 0$ almost surely and

$$\mathbb{E}w = 1, \qquad \mathbb{E}w^2 = 1 + 2\rho, \qquad \rho = \sum_b \lambda_b^2, \quad L = \max_b \lambda_b.$$

For a standard r-dimensional Gaussian, $\mathbb{E}(gg^\top)^2 = (r+2)I_r$: its diagonal entries are $\mathbb{E}[g_a^2 \sum_b g_b^2] = 3+(r-1)$ and its off-diagonal entries vanish. Expansion of $(wgg^\top - I_r)^2$ proves (3.1). In particular,

$$\mathbb{E}\left\| m^{-1}\sum_i y_i y_i^\top - I_r \right\|_{\mathrm{F}}^2 = \frac{r}{m}\big[r + 1 + 2(r+2)\rho\big], \qquad \mathbb{E}\langle u, y\rangle^4 = 3(1+2\rho)\|u\|^4. \tag{A.40}$$

These are exact formulas, including $r = 1$ and all Schmidt spectra.

A conditional weighted-Gaussian certificate.

Lemma A.13. *Let $a_1, \dots, a_m \ge 0$ be deterministic, and let g_i be independent standard r-dimensional Gaussians. Define $A_2 = \sum_i a_i^2$, $B = \max_i a_i$, $\bar{a} = m^{-1}\sum_i a_i$, and $t = r\log 9 + \log(2/\alpha)$ for $0 < \alpha < 1$. Then, with probability at least $1-\alpha$,*

$$\left\| m^{-1}\sum_i a_i g_i g_i^\top - \bar{a} I_r \right\| \le \frac{4\sqrt{A_2 t} + 4Bt}{m}. \tag{A.41}$$

Proof. For any fixed unit u, $\langle u, g_i\rangle$ are independent standard real Gaussians. The Gaussian density gives $\mathbb{E}e^{zG^2} = (1-2z)^{-1/2}$ for $z < 1/2$. Expansion of the logarithm therefore gives, for $|z| < (2B)^{-1}$,

$$\log \mathbb{E}\exp\left[z\sum_i a_i(\langle u, g_i\rangle^2 - 1)\right] \le \frac{A_2 z^2}{1-2B|z|}.$$

If $B = 0$ the conclusion is deterministic. Otherwise the exponential Markov inequality, using $z = \sqrt{t}/(\sqrt{A_2} + 2B\sqrt{t})$ and its negative, gives

$$\mathbb{P}\left\{\left|\sum_i a_i(\langle u, g_i\rangle^2 - 1)\right| > 2\sqrt{A_2 t} + 2Bt\right\} \le 2e^{-t}.$$

A 1/4-net of the unit sphere has at most 9^r points, by disjoint radius-1/8 balls and volume comparison. For any symmetric matrix D, approximating a maximizing unit vector by a net point gives $\|D\| \le 2\max_{u \text{ in net}} |u^\top D u|$. A union bound proves (A.41). This proof does not require a matrix exponential bound. □

Proof of Theorem 3.2. Upper bound. Put

$$u = \log(8/\delta), \quad v = \log(4m/\delta), \quad t = r\log 9 + u, \quad b = 1 + 2\sqrt{\rho v} + 2Lv.$$

The Gaussian exponential moment, applied to the weights λ_b, gives

$$\mathbb{P}\{\max_i w_i > b\} \le \delta/4,$$
$$\mathbb{P}\{|m^{-1}\textstyle\sum_i w_i - 1| > 2\sqrt{\rho u/m} + 2Lu/m\} \le \delta/4.$$

Let $\mu_2 = 1 + 2\rho$. Convexity gives $\mu_4 := \mathbb{E}w^4 \le \sum_b \lambda_b \mathbb{E}h_b^8 = 105$. The independent variables $w_i^2 \mathbf{1}_{\{w_i \le b\}}$ are bounded by b^2, have means at most μ_2, and variances at most μ_4. Scalar Bernstein therefore gives, except with probability $e^{-u} = \delta/8$, and on the maximum-weight event,

$$\sum_i w_i^2 \le A_m := m\mu_2 + \sqrt{2m\mu_4 u} + \frac{2}{3}b^2 u. \tag{A.42}$$

Condition on the weights and apply Lemma A.13 with $\alpha = \delta/4$. A union bound gives, with probability at least $1-\delta$,

$$\left\| m^{-1}\sum_i y_i y_i^\top - I_r \right\| \le 2\sqrt{\rho u/m} + \frac{2Lu}{m} + \frac{4\sqrt{tA_m} + 4bt}{m}. \tag{A.43}$$

This is a numerical certificate using the actual second and fourth moments of w. With $s_k = \sum_b \lambda_b^k$, the latter is

$$\mu_4 = 1 + 12\rho + 12\rho^2 + 32s_3 + 48s_4.$$

To obtain (3.2), assume $m \ge u$ and use $\mu_2 \le 3$, $\mu_4 \le 105$, $u \le t$, and $b \le 2 + 3L\log(4m/\delta)$. The right side of (A.43) is at most

$$C_0\left[\sqrt{\frac{\tau_\delta}{m}} + \frac{\tau_\delta[1 + L\log(4m/\delta)]}{m}\right], \qquad \tau_\delta = r + \log(8/\delta),$$

for a universal C_0. Let $a = \log(e\tau_\delta/(\varepsilon\delta))$ and $M = C\tau_\delta(\varepsilon^{-2} + L\varepsilon^{-1}a)$. Since $a \ge 1$ and $L\varepsilon \le 1$, $\log(4M/\delta) \le C_1 a + \log C$ for a universal C_1. At $m = M$ the last display is bounded by $\varepsilon[C_2/\sqrt{C} + C_3(1 + \log C)/C]$. Choose C sufficiently large. Both terms decrease for $m \ge M$, which proves the claimed all-confidence bound. Its fixed-parameter specialization gives the upper transition uniformly over the weights.

Lower bound. Take $c = 1/16$. If $m < r$, rank deficiency gives error at least one. Otherwise choose a largest Schmidt coefficient and denote its Gaussian coordinate in sample i by $h_{i,*}$. Then $w_i \geq L h_{i,*}^2$. For $m \geq r$ sufficiently large, integration of the Gaussian density on $[\sqrt{\log m}, \sqrt{\log m} + 1/\sqrt{\log m}]$ yields

$$\mathbb{P}\{h_{i,*}^2 \geq \log m\} \geq c_0 m^{-1/2} (\log m)^{-1/2}.$$

Consequently, except with probability $\exp[-c_0\sqrt{m}/\sqrt{\log m}]$, at least one such coordinate occurs. Choose an index J attaining $\max_i h_{i,*}^2$, using only the weights' underlying h variables. The Gaussian vector g_J is still independent and standard; exponential Markov gives $\mathbb{P}\{\|g_J\|^2 < r/2\} \leq e^{-r/16}$. Positivity then implies

$$\lambda_{\max}\left(m^{-1}\sum_i y_i y_i^\top\right) \geq \frac{rL\log m}{2m}.$$

In the nonempty range $r \leq m \leq cr[1 + L\log(er)]$, necessarily $L\log(er) \geq 1/c - 1 = 15$. For large r, $\log r \geq \frac{1}{2}\log(er)$, and hence

$$\frac{rL\log m}{2m} \geq \frac{L\log(er)}{4c[1 + L\log(er)]} \geq \frac{15}{4c \cdot 16} > 2.$$

The error exceeds one. Both exceptional probabilities tend to zero uniformly in this range, since $m \geq r$ and $m \leq cr[1 + \log(er)]$. This proves the lower part, uniformly even when the weights and their number vary with r. For flat weights, $L = \rho = 1/s$, giving the claimed transition. □

Spherical moment transfer. Let $x_i^{(j)} = \sqrt{n_j}\,u_i^{(j)}$, with all $u_i^{(j)}$ independent and uniform on their real unit spheres. Draw independent $R_i^{(j)} \sim \chi_{n_j}$, independent of all directions, and put $g_i^{(j)} = R_i^{(j)} u_i^{(j)}$. Gaussian polar decomposition makes these independent standard Gaussian factors. For a fixed isometry $U \in \mathbb{R}^{(\prod_j n_j)\times r}$, define

$$y_i^{\mathrm{sph}} = U^\top \bigotimes_j x_i^{(j)}, \qquad y_i^{\mathrm{G}} = U^\top \bigotimes_j g_i^{(j)}, \qquad a_i = \prod_j \frac{R_i^{(j)}}{\sqrt{n_j}}.$$

Then $y_i^{\mathrm{G}} = a_i y_i^{\mathrm{sph}}$ and $\mathbb{E}[a_i^2 \mid (u)] = 1$. For the centered sample covariance errors $Z_{\mathrm{sph}}, Z_{\mathrm{G}}$,

$$\mathbb{E}[Z_{\mathrm{G}} \mid (u)] = Z_{\mathrm{sph}}, \qquad \mathbb{E}\,\mathrm{tr}\, Z_{\mathrm{sph}}^{2p} \leq \mathbb{E}\,\mathrm{tr}\, Z_{\mathrm{G}}^{2p} \quad (p \in \mathbb{N}). \tag{A.44}$$

The second assertion is conditional Jensen for the convex function $D \mapsto \mathrm{tr}\,|D|^{2p} = \|D\|_{S_{2p}}^{2p}$ on symmetric matrices, followed by integration. All moments exist. Thus the Gaussian centered trace-moment proof and Markov step underlying Theorem 2.1 apply without changing constants. Independence of coordinates within a spherical factor is neither true nor needed.

For a single factor $x = \sqrt{n}u$, polar independence and $\mathbb{E}R^{2p} = \prod_{j=0}^{p-1}(n + 2j)$ give

$$\mathbb{E}\langle a, x\rangle^{2p} = \left(\prod_{j=0}^{p-1} \frac{n}{n+2j}\right) \mathbb{E}\langle a, g\rangle^{2p}. \tag{A.45}$$

In particular $\mathbb{E}xx^\top = I_n$ and $\mathbb{E}\langle a, x\rangle^4 = \frac{3n}{n+2}\|a\|^4$. For an individual tensor row, the same homogeneous calculation gives an exact multiplicative correction $\prod_j \prod_{k=0}^{p-1} n_j/(n_j + 2k)$ for its pth uncentered covariance moment. For the centered sum the comparison is Jensen's inequality (A.44); there is no assertion of that same exact correction for the entire centered sum.

Spherical factors retain the worst-case logarithm. Fix $d \geq 2$ and, for large r, put $h = \lfloor \log r/(d-1) \rfloor$. Choose mode dimensions $n_1 = r$, $n_2 = \cdots = n_d = h$ and the common-factor support $Ue_a = e_a \otimes e_1 \otimes \cdots \otimes e_1$. The projected row is

$$y_i = \left(\prod_{j=2}^{d} \sqrt{h}\, u_{i,1}^{(j)} \right) \xi_i, \qquad \xi_i = \sqrt{r}\, u_i^{(1)}, \qquad \|\xi_i\|^2 = r.$$

We give an elementary lower bound for a large spherical coordinate. Write $u = G/\|G\|$ in dimension h. The independent events $G_1^2 \geq h$ and $\sum_{a=2}^{h} G_a^2 \leq 2h$ imply $u_1^2 \geq 1/3$. The second has probability at least $1/2$, by Markov's inequality. Integrating the first on $[\sqrt{h}, \sqrt{h} + 1/\sqrt{h}]$ gives

$$\mathbb{P}\{u_1^2 \geq 1/3\} \geq c_1 h^{-1/2} e^{-h/2}.$$

With independence across common modes, a row has all $d-1$ such coordinates with probability at least $c_d h^{-(d-1)/2} e^{-(d-1)h/2} \geq c_d h^{-(d-1)/2} r^{-1/2}$. Across $m \geq r$ independent rows, the probability none occurs is at most

$$\exp[-c_d \sqrt{r}/h^{(d-1)/2}],$$

which tends to zero. For an occurring row, $\|y_i\|^2 \geq r(h/3)^{d-1}$. As $h \geq \log r/[2(d-1)]$ for large r, every

$$r \leq m \leq \frac{r(\log r)^{d-1}}{2[6(d-1)]^{d-1}}$$

then has sample-covariance top eigenvalue at least two. For $m < r$ the rank argument applies. The failure probability for distortion $1/2$ tends to one uniformly throughout this entire range. Together with (A.44), this proves the spherical worst-case order $\Theta_{d,\varepsilon,\delta}(r \log^{d-1}(er))$ as $r \to \infty$, with tensor order, distortion, and confidence fixed.

B Approximation details

B.1 Lower injection under fourth moments

Lemma B.1 (Gaussian-smoothed lower covariance)**.** *Let $v_1, \ldots, v_m \in \mathbb{R}^k$ be iid, with $V = \mathbb{E} vv^\top$, $0 \preceq V \preceq I_k$, and*

$$\mathbb{E}\langle \vartheta, v \rangle^4 \leq C(\vartheta^\top V \vartheta)^2 \quad \text{for every } \vartheta \in \mathbb{R}^k, \qquad C \geq 1.$$

Put $s = \operatorname{tr} V$. For $0 < \alpha, \eta < 1$, if $m \geq \mathfrak{m}_{\mathrm{inj}}(C, s; \alpha, \eta)$, then

$$\mathbb{P}\left\{ \frac{1}{m} \sum_i v_i v_i^\top + I_k - V \succeq \alpha I_k \right\} \geq 1 - \eta.$$

Proof. We use Gaussian smoothing for lower covariance tails [30], with Catoni's logarithmic influence function [42]. If $s = 0$, then $v = 0$ almost surely and the assertion is deterministic. Otherwise fix $\beta, \lambda > 0$ before sampling and set

$$\varphi(t) = \log(1 + t + t^2/2), \qquad t \geq 0.$$

Its derivative is $(1+t)/(1+t+t^2/2)$, while $\varphi''(t) = -(t + t^2/2)/(1+t+t^2/2)^2$. Thus φ is concave, $0 \leq \varphi' \leq 1$, and

$$e^{-\varphi(t)} \leq 1 - t + t^2/2, \qquad e^{\varphi(t)} = 1 + t + t^2/2.$$

The first inequality follows because the two quadratic polynomials have product $1 + t^4/4$. Write $\mu_\vartheta = \vartheta^\top V \vartheta$. Independence, the fourth-moment hypothesis, and $1 + x \leq e^x$ imply

$$\mathbb{E} \exp Z_\vartheta \leq 1, \qquad Z_\vartheta = m\lambda\mu_\vartheta - \frac{m\lambda^2 C}{2} \mu_\vartheta^2 - \sum_i \varphi(\lambda \langle \vartheta, v_i \rangle^2).$$

Use the Gaussian prior $\pi = N(0, \beta^{-1} I_k)$. Fubini and Markov give $\int e^{Z_\vartheta}\, d\pi(\vartheta) \le 2/\eta$ with probability at least $1 - \eta/2$. On this event, the change-of-measure inequality

$$\int Z_\vartheta \, d\nu(\vartheta) \le \mathrm{KL}(\nu \| \pi) + \log \int e^{Z_\vartheta} \, d\pi(\vartheta)$$

holds simultaneously for all probability measures $\nu \ll \pi$ for which the displayed integrals are finite. It follows from Jensen's inequality after inserting $d\pi/d\nu$. Take $\pi_u = N(u, \beta^{-1} I_k)$ for any unit u; its relative entropy is $\beta/2$. Gaussian moment identities give

$$\begin{aligned}
\mathbb{E}_{\pi_u} \mu_\vartheta &= u^\top V u + s/\beta, \\
\mathbb{E}_{\pi_u} \mu_\vartheta^2 &= (u^\top V u)^2 + \frac{2s\, u^\top V u + 4 u^\top V^2 u}{\beta} + \frac{s^2 + 2 \operatorname{tr} V^2}{\beta^2} \\
&\le 1 + \frac{2(s+2)}{\beta} + \frac{s(s+2)}{\beta^2}.
\end{aligned}$$

The last step uses $V \preceq I_k$, $u^\top V u \le 1$, $u^\top V^2 u \le 1$, and $\operatorname{tr} V^2 \le s$. Consequently, simultaneously for all unit u,

$$\frac{1}{m\lambda} \sum_i \mathbb{E}_{\pi_u} \varphi(\lambda \langle \vartheta, v_i \rangle^2) \ge u^\top V u + \frac{s}{\beta} - \frac{\lambda C}{2} \left(1 + \frac{2(s+2)}{\beta} + \frac{s(s+2)}{\beta^2} \right) - \frac{\beta/2 + \log(2/\eta)}{m\lambda}.$$

Concavity and $\varphi' \le 1$ remove the Gaussian smoothing:

$$\mathbb{E}_{\pi_u} \varphi(\lambda \langle \vartheta, v_i \rangle^2) \le \varphi\big(\lambda \langle u, v_i \rangle^2 + \lambda \|v_i\|^2/\beta\big) \le \lambda \langle u, v_i \rangle^2 + \varphi(\lambda \|v_i\|^2/\beta).$$

The common radial term admits an upper exponential bound under only fourth moments. For an independent $g \sim N(0, I_k)$,

$$3\mathbb{E}\|v\|^4 = \mathbb{E}_g \mathbb{E} \langle g, v \rangle^4 \le C \mathbb{E}_g (g^\top V g)^2 = C(s^2 + 2 \operatorname{tr} V^2) \le Cs(s+2).$$

For $R_i = \varphi(\lambda \|v_i\|^2/\beta)/\lambda$, the identity for e^φ, independence, and Markov therefore give, with probability at least $1 - \eta/2$,

$$\frac{1}{m} \sum_i R_i \le \frac{s}{\beta} + \frac{\lambda C s(s+2)}{6\beta^2} + \frac{\log(2/\eta)}{m\lambda}.$$

Intersect the two events and subtract the common radial term:

$$\frac{1}{m} \sum_i \langle u, v_i \rangle^2 \ge u^\top V u - \frac{\lambda C}{2} A_s^{\mathrm{inj}}(\beta) - \frac{B_\eta^{\mathrm{inj}}(\beta)}{m\lambda} \quad \text{for every unit } u.$$

Here the β^{-2} coefficient is $1 + 1/3 = 4/3$, incorporating the radial upper bound. Choosing $\lambda = \sqrt{2B_\eta^{\mathrm{inj}}(\beta)/(mCA_s^{\mathrm{inj}}(\beta))}$ gives the uniform deficit $\sqrt{2CA_s^{\mathrm{inj}}(\beta) B_\eta^{\mathrm{inj}}(\beta)/m}$. The minimizing β is deterministic. The stated budget makes this deficit at most $1 - \alpha$, so adding $I_k - V$ proves the conclusion.

All expectations are justified under the stated fourth moments. The radial exponential is a quadratic polynomial in $\|v\|^2$, and Fubini applies to the nonnegative prior integrand. Gaussian posterior integrals are finite by their polynomial moment bounds. Thus no exponential moment of the raw covariance sample is assumed. □

Proof of Lemma 4.2. Apply Lemma B.1 with $v_i = Q^\top z_i$ and $V = I_r$.

For completeness, the minimizing β is the unique positive root of

$$\beta^3 - (b + 2aT)\beta - 4bT = 0, \qquad a = 2(s+2), \quad b = \frac{4}{3} s(s+2), \quad T = 2\log(2/\eta). \tag{B.1}$$

Indeed, $A_s^{\mathrm{inj}} B_\eta^{\mathrm{inj}} = \beta/2 + T + a/2 + (aT + b/2)/\beta + bT/\beta^2$ is strictly convex on $(0, \infty)$, and its derivative equation is the displayed cubic. At fixed η as $s \to \infty$, its minimizer satisfies $\beta/s \to 2/\sqrt{3}$, and the objective divided by s tends to $1 + 2/\sqrt{3}$. This proves the stated leading coefficient. For each fixed β, $A_s^{\mathrm{inj}}(\beta)$ is nondecreasing in s. Minimizing over β preserves this order, so an upper bound on the effective dimension is sufficient. □

B.2 Bilinear moments and order-two Wick geometry

Lemma B.2 (Centered bilinear moment)**.** *For an isotropic random vector z satisfying the directional fourth moment bound with constant $C \geq 1$, put*

$$\gamma(C) = \min\{C/2, C - 1\}, \qquad \gamma_{\|}(C) = C - 1 - \gamma(C) \geq 0.$$

Then for all deterministic vectors u, v,

$$\mathbb{E}\big(\langle u, z\rangle\langle v, z\rangle - \langle u, v\rangle\big)^2 \leq \gamma(C)\|u\|^2\|v\|^2 + \gamma_{\|}(C)\langle u, v\rangle^2. \tag{B.2}$$

Consequently, for matrices U, W with the same number of rows and a normalized iid sketch with m columns,

$$\mathbb{E}\|U^\top(\Omega\Omega^\top - I)W\|_F^2 \leq \frac{\gamma(C)\|U\|_F^2\|W\|_F^2 + \gamma_{\|}(C)\|U^\top W\|_F^2}{m}.$$

Proof. By homogeneity it is enough to take $\|u\| = \|v\| = 1$. Put $A = \langle u, z\rangle$, $B = \langle v, z\rangle$, $\chi = \langle u, v\rangle$, and $h = \mathbb{E}A^2B^2$. The bounds in directions $u + v$ and $u - v$ give

$$\mathbb{E}A^4 + \mathbb{E}B^4 + 6h \leq 4C(1 + \chi^2).$$

Since $\mathbb{E}A^4 + \mathbb{E}B^4 \geq 2h$,

$$\mathrm{Var}(AB) = h - \chi^2 \leq \frac{C}{2} + \left(\frac{C}{2} - 1\right)\chi^2.$$

There is also the bound $\mathrm{Var}(AB) \leq C - 1$. To see this when $|\chi| < 1$, let

$$X = \frac{A + B}{\sqrt{2 + 2\chi}}, \quad Y = \frac{A - B}{\sqrt{2 - 2\chi}}, \quad a = \frac{1 + \chi}{2}, \quad b = \frac{1 - \chi}{2}.$$

Both X and Y have unit second moment and fourth moment at most C, and $AB - \chi = a(X^2 - 1) - b(Y^2 - 1)$. The triangle inequality in L^2 gives $\|AB - \chi\|_2 \leq (a + b)\sqrt{C - 1} = \sqrt{C - 1}$. The endpoint $|\chi| = 1$ is the directional fourth-moment bound itself. For $C \geq 2$ use the preceding affine bound in χ^2; for $C \leq 2$ use $C - 1$. These are exactly the displayed coefficients $\gamma, \gamma_{\|}$. Finally, expand the Frobenius norm into entries, remove distinct-sample terms using centering and independence, and sum this scalar inequality. □

Lemma B.3 (Gaussian and sign product bilinear moments)**.** *Let $z = g^{(1)} \otimes \cdots \otimes g^{(d)}$, where the factors are independent standard Gaussian vectors, and let $\gamma_{\mathrm{G}}(d) = (3^d - 1)/2$. Then for all deterministic u, v,*

$$\mathbb{E}\big(\langle u, z\rangle\langle v, z\rangle - \langle u, v\rangle\big)^2 \leq \gamma_{\mathrm{G}}(d)\big(\|u\|^2\|v\|^2 + \langle u, v\rangle^2\big).$$

The same upper bound holds when any or all Gaussian factors are replaced by isotropic four-wise-independent sign vectors. In particular, for orthogonal unit vectors u, v,

$$\mathbb{E}\langle u, z\rangle^2\langle v, z\rangle^2 \leq \gamma_{\mathrm{G}}(d).$$

This orthogonal constant is optimal for Gaussian factors when every mode dimension is at least two. Consequently, for every fixed isometry $U \in \mathbb{R}^{N \times r}$ and $y = U^\top z$,

$$\mathbb{E}(yy^\top - I_r)^2 \preceq \gamma_{\mathrm{G}}(d)(r + 1)I_r. \tag{B.3}$$

Proof. The needed two-form Gaussian square-product inequality is a special case of chaotic square-product inequalities [41]; the present decoupled case has an elementary induction. If P, Q are scalar forms, linear in each of the d independent Gaussian factors, then

$$\mathbb{E}P^2Q^2 \geq \mathbb{E}P^2\,\mathbb{E}Q^2.$$

The statement is immediate at $d = 0$. For the induction step, condition on all but the first factor and write $P = \sum_i g_i^{(1)} a_i$, $Q = \sum_i g_i^{(1)} b_i$, where a_i, b_i are multilinear forms in the remaining factors. The Gaussian fourth-moment formula gives

$$\mathbb{E}_1[P^2Q^2] = \left(\sum_i a_i^2\right)\left(\sum_j b_j^2\right) + 2\left(\sum_i a_i b_i\right)^2.$$

Drop the last nonnegative term and apply the induction hypothesis to each $\mathbb{E}a_i^2 b_j^2$. The resulting sum is $\mathbb{E}P^2\,\mathbb{E}Q^2$.

Normalize u, v to be unit vectors and put $A = \langle u, z\rangle$, $B = \langle v, z\rangle$, $\chi = \langle u, v\rangle$, and $h = \mathbb{E}A^2B^2$. Apply the square-product inequality to $A + B, A - B$:

$$\mathbb{E}A^4 + \mathbb{E}B^4 - 2h = \mathbb{E}(A + B)^2(A - B)^2 \geq 4(1 - \chi^2).$$

The product directional fourth-moment bound gives

$$\mathbb{E}A^4 + \mathbb{E}B^4 + 6h \leq 4 \cdot 3^d(1 + \chi^2).$$

Subtracting yields $h \leq (3^d - 1)/2 + (3^d + 1)\chi^2/2$; subtracting χ^2 proves the centered inequality.

For a four-wise-independent sign vector ϵ, direct expansion gives

$$\mathbb{E}\langle a, \epsilon\rangle^2\langle b, \epsilon\rangle^2 = \|a\|^2\|b\|^2 + 2\langle a, b\rangle^2 - 2\sum_i a_i^2 b_i^2 \leq \|a\|^2\|b\|^2 + 2\langle a, b\rangle^2.$$

Conditioning on all other factors therefore allows one sign factor to be replaced by a Gaussian factor without decreasing the raw mixed fourth moment. Repeat over all sign factors; the second moment and χ remain unchanged.

For the matrix conclusion, expand $\mathbb{E}\|y\|^2\langle v, y\rangle^2$ in an orthonormal basis beginning with any fixed unit vector v. Its own coordinate contributes at most 3^d, and each other coordinate contributes at most $\gamma_{\mathrm{G}}(d)$. Subtract I_r and use $3^d - 1 = 2\gamma_{\mathrm{G}}(d)$.

For optimality, let $P = \prod_{j=1}^d g_1^{(j)}$ and $Q = \prod_{j=1}^d g_2^{(j)}$, which are independent, have unit second moment, and have fourth moment 3^d. The forms $A = (P + Q)/\sqrt{2}$, $B = (P - Q)/\sqrt{2}$ have orthogonal unit coefficient tensors and satisfy

$$\mathbb{E}A^2B^2 = \frac{1}{4}\mathbb{E}(P^2 - Q^2)^2 = \frac{3^d - 1}{2}.$$

□

Lemma B.4 (Product-coordinate bilinear moments)**.** *Suppose the independent factors have centered, unit-variance coordinates, four-wise independent within each factor, with coordinate fourth moments at most $\mu_{4,j}$. Put*

$$c_j = \max\{3, \mu_{4,j}\}, \qquad C_{\mathrm{coord}} = \prod_{j=1}^d c_j, \qquad \Gamma_{\mathrm{coord}} = (C_{\mathrm{coord}} - 1)/2.$$

Then for all deterministic vectors u, v,

$$\mathbb{E}\big(\langle u, z\rangle\langle v, z\rangle - \langle u, v\rangle\big)^2 \leq \Gamma_{\mathrm{coord}}\big(\|u\|^2\|v\|^2 + \langle u, v\rangle^2\big). \tag{B.4}$$

In particular, for orthogonal unit u, v,

$$\mathbb{E}\langle u, z\rangle^2\langle v, z\rangle^2 \le \Gamma_{\mathrm{coord}}. \tag{B.5}$$

This extends Lemma B.3 beyond Gaussian and sign coordinates. Its Gaussian orthogonal constant is sharp as shown there.

Proof. For a centered, unit-variance coordinate vector ξ, four-wise independence gives

$$\mathbb{E}(a^\top\xi)^2(b^\top\xi)^2 = \|a\|^2\|b\|^2 + 2\langle a, b\rangle^2 + \sum_i(\mathbb{E}\xi_i^4 - 3)a_i^2b_i^2.$$

The expression is nondecreasing in every coordinate fourth moment. For factor j, replace its coordinates by independent copies of σG, where G is standard Gaussian, σ is independent of G, and $\sigma^2 = c_j/3$ with probability $3/c_j$ and zero otherwise. This comparison coordinate has variance one and fourth moment c_j. Condition on the other factors and replace factors one at a time. The raw mixed square cannot decrease, while the second moments and $\langle u, v\rangle$ remain unchanged.

For the comparison product law, all excess coordinate fourth moments are nonnegative. Consequently the induction used in Lemma B.3 remains valid: for any two multilinear forms F, G, $\mathbb{E}F^2G^2 \ge \mathbb{E}F^2\,\mathbb{E}G^2$. Indeed, after conditioning on one factor, discard the nonnegative inner-product square and the nonnegative excess-fourth-moment terms; apply the induction hypothesis to every coefficient pair in the remaining sum. The comparison directional fourth moment is at most C_{coord} by Lemma 4.1. For unit u, v, put $A = \langle u, z\rangle$, $B = \langle v, z\rangle$, $\chi = \langle u, v\rangle$, and $h = \mathbb{E}A^2B^2$, now under that comparison law. The two bounds give

$$\mathbb{E}A^4 + \mathbb{E}B^4 - 2h \ge 4(1-\chi^2), \qquad \mathbb{E}A^4 + \mathbb{E}B^4 + 6h \le 4C_{\mathrm{coord}}(1+\chi^2).$$

Subtract and then subtract χ^2 to obtain $h - \chi^2 \le (C_{\mathrm{coord}} - 1)(1+\chi^2)/2$. Comparison and homogeneity prove the statement for the original law. □

Lemma B.5 (Order-two Wick geometry and variance)**.** *Let $z = g \otimes h$ for independent standard real Gaussian vectors. For an orthogonal rank-r projector P, write $P_A = \mathrm{Tr}_2 P$, $P_B = \mathrm{Tr}_1 P$, and let P^Γ denote partial transpose in the second factor. Then*

$$M_P := \mathbb{E}[(z^\top Pz)zz^\top] = rI + 2(P_A \otimes I + I \otimes P_B) + 2P + 2P^\Gamma. \tag{B.6}$$

In particular, if $P = UU^\top$, $U^\top U = I_r$, and $H = (U^\top z)(U^\top z)^\top$, then

$$\mathbb{E}H^2 = (r+2)I_r + 2U^\top(P_A \otimes I + I \otimes P_B + P^\Gamma)U. \tag{B.7}$$

Consequently,

$$\mathbb{E}H^2 \preceq (3r + 4 + 2\sqrt{r})I_r, \qquad \mathbb{E}(H - I_r)^2 \preceq (3r + 3 + 2\sqrt{r})I_r. \tag{B.8}$$

The two inequalities, but not necessarily the Wick identities, remain valid for independent factors with centered, unit-variance coordinates, four-wise independent within each factor, whose fourth moments are at most three. For two normalized spherical factors, multiply the Gaussian raw fourth-moment bounds by $\varsigma_2 = \prod_{j=1}^2 n_j/(n_j+2)$ before subtracting I_r.

Proof. The nine real Gaussian Wick contractions prove (B.6). The pairing internal to each factor contributes rI; the four pairings internal in one factor contribute twice each partial trace; the remaining four contribute $2P + 2P^\Gamma$. Compression by U proves (B.7).

The double-reduction construction of universal state inversion [40, Eq. (2.11)] has the positivity property

$$rI - P_A \otimes I - I \otimes P_B + P \succeq 0. \tag{B.9}$$

Here is a direct proof of the matrix inequality. Let $J_{ab} = e_a e_b^\top - e_b e_a^\top$. Entrywise expansion gives

$$\mathscr{D}_n(X) := (\operatorname{tr} X)I - X = \sum_{a<b} J_{ab} X^\top J_{ab}^\top.$$

Thus $(\mathscr{D}_{n_1} \otimes \mathscr{D}_{n_2})(P)$ equals

$$\sum_{a<b} \sum_{c<e} (J_{ab} \otimes J_{ce}) P^\top (J_{ab} \otimes J_{ce})^\top \succeq 0.$$

It is a sum of positive congruences of $P^\top = P$, and its expansion is the left side of (B.9). The formula also covers one-dimensional factors, when the appropriate sum is empty.

Compression of (B.9) bounds the sum of the two partial-trace terms by $(r+1)I_r$. Partial transpose preserves the Frobenius norm, so $\|P^\Gamma\| \le \|P^\Gamma\|_{\mathrm{F}} = \sqrt{r}$. Substitute in (B.7), then use $\mathbb{E}(H - I_r)^2 = \mathbb{E}H^2 - I_r$.

For the coordinate class, test $\mathbb{E}H^2$ in any direction and expand the remaining squared norm in an orthonormal basis. Each summand is a mixed square. The conditional fourth-moment formula in Lemma B.4 shows that replacement by Gaussian factors cannot decrease it. This proves Loewner domination. For spheres, Gaussian polar decomposition gives the exact homogeneous relation $\mathbb{E}H_{\mathrm{sph}}^2 = \varsigma_2 \mathbb{E}H_{\mathrm{G}}^2$. □

Corollary B.6 (Aggregate cross moment for order-two probes)**.** *Under the order-two coordinate assumptions of Lemma B.5, for every isometry $Q \in \mathbb{R}^{N\times r}$ and every unit vector $v \perp \operatorname{range} Q$,*

$$\mathbb{E}\|Q^\top z\|^2 \langle v, z\rangle^2 \le \Xi_r, \qquad \Xi_r = \min\{4r,\, 3r + 2\sqrt{r}\}. \tag{B.10}$$

The leading coefficient three is sharp for Gaussian factors. For two normalized spherical factors the bound is $\varsigma_2 \Xi_r$, with ς_2 as above.

Proof. Gaussian comparison in Lemma B.5 reduces the coordinate class to Gaussian factors. Set $P = QQ^\top$. Since $Pv = 0$, double reduction gives $v^\top (P_A \otimes I + I \otimes P_B) v \le r$. Testing (B.6) therefore gives $v^\top M_P v \le 3r + 2\sqrt{r}$. The separate orthogonal mixed-square bound $\Gamma_{\mathrm{coord}} = 4$ gives $v^\top M_P v \le 4r$. The spherical conclusion follows from the same homogeneous relation.

For sharpness of the leading coefficient, take the head space spanned by $e_1 \otimes e_1, \dots, e_r \otimes e_1$ and the orthogonal unit tail direction $v = e_{r+1} \otimes e_1$. Independence gives

$$\mathbb{E}\|Q^\top z\|^2 \langle v, z\rangle^2 = \mathbb{E}\left(\sum_{a=1}^{r} g_a^2\right) g_{r+1}^2\, \mathbb{E}h_1^4 = 3r.$$

Thus no uniform leading coefficient smaller than three is possible. □

B.3 PSD trace and confidence amplification

Corollary B.7 (PSD Nyström trace-relative error)**.** *Let $K \in \mathbb{R}^{N\times N}$ be positive semidefinite, let $\tau_r^{\mathrm{tr}}(K) = \sum_{j>r} \lambda_j(K)$, and define*

$$\mathcal{N} = \mathcal{N}_K(\Omega), \qquad \mathcal{N}_r = (\mathcal{N})_r.$$

Under (4.12), with probability at least $1 - \delta$,

$$\operatorname{tr}(K - \mathcal{N}) \le (1+\varepsilon)\tau_r^{\mathrm{tr}}(K), \qquad \operatorname{tr}(K - \mathcal{N}_r) \le (1+\varepsilon)\tau_r^{\mathrm{tr}}(K). \tag{B.11}$$

Proof. For $A = K^{1/2}$ and $P = P_{A\Omega}$, the standard Nyström representation $\mathcal{N} = APA$ [38, Proposition A.2] identifies the two trace errors with the projection and best rank-r squared Frobenius errors in range P. Apply Theorem 4.5 to A, whose optimal squared tail is $\tau_r^{\mathrm{tr}}(K)$. □

Proof of Corollary 4.7. Partition the columns into b independent groups of size m_0. Theorem 4.5 with confidence parameter $1/4$ makes each group successful with probability at least $3/4$. Overall normalization changes no range. The concatenated range contains every group range, so both the projection error and the optimal rank-r error in it are no larger than the corresponding group errors. The probability that all groups fail is at most $4^{-b} \le \delta$. Apply the same argument to $K^{1/2}$ for the Nyström statements. □

Independent regression solves can instead be amplified by selecting the candidate with smallest original residual, when evaluating that residual is permitted.

B.4 High-confidence Hilbert comparisons

Proof of Theorem 4.8. Let Q be an isometry with r columns and let $Q^\top W = 0$. If $F = \|W\|_{\mathrm{F}} > 0$, regard $Y = (Q^\top z)(W^\top z)^\top / F$ as a vector in the Frobenius Hilbert space. Isotropy gives $\mathbb{E}Y = 0$, and Lemma 4.4 gives $\mathbb{E}\|Y\|_{\mathrm{F}}^2 \le B_r$. Minkowski's inequality applied to a sum of squares shows that

$$\mathbb{E}\|Tz\|^{2j} \le A_j \|T\|_{\mathrm{F}}^{2j}.$$

Cauchy–Schwarz therefore gives $\mathbb{E}\|Y\|_{\mathrm{F}}^j \le r^{j/2} A_j = b_j^H$.

For a centered Hilbert vector Y and a fixed vector x, expand $\|x + sY\|^{2p} = (\|x\|^2 + 2s\langle x, Y\rangle + s^2\|Y\|^2)^p$. The linear coefficient has expectation zero. The absolute value of every other coefficient is at most the corresponding coefficient of $(\|x\| + s\|Y\|)^{2p}$. Thus the even moments of $\sum_{i\le n} Y_i$ satisfy the scalar recurrence (A.17), with $b_2 = \nu_H$ and the budgets b_j^H. The pairing comparison in Appendix A.3 applies without a trace or a rank factor and yields

$$\left(\mathbb{E} \left\| m^{-1} \sum_{i=1}^{m} Y_i \right\|_{\mathrm{F}}^{2q} \right)^{1/(2q)} \le c_q \left(\sqrt{\nu_H / m} + h_q \mathfrak{r}_q^H / m \right).$$

Markov's inequality and (4.18) bound failure of the interaction event by $\delta/2$. Lower injection fails with probability at most $\delta/2$. The same deterministic competitors and normal equations as in Theorem 4.5 now apply. If $W = 0$, the interaction vanishes.

For the displayed rate, use the coarser valid budget $\nu_H = Cr$, where $C = [2\theta {K_{\mathrm{sg}}}^4]^d$, and the original factorial scalar bound $A_j \le \theta^d (j!)^d {K_{\mathrm{sg}}}^{2dj}$. The elementary inequality $j! \le 2j^{j-2}$ gives

$$b_j^H \le \frac{j!}{2} Cr \left[{K_{\mathrm{sg}}}^{2d} \sqrt{r}\, (2q)^{d-1} \right]^{j-2}.$$

Choose $q = \lceil \log(2/\delta) \rceil$. The resulting interaction allowance is at most $C_{d,K_{\mathrm{sg}}} [\sqrt{rq/m} + \sqrt{r}\, q^d / m]$. Making its two terms at most $\sqrt{\varepsilon}/4$ proves (4.19) and also covers the injection budget after increasing $C_{d,K_{\mathrm{sg}}}$. □

Proposition B.8 (Keeping the weak Hilbert variance)**.** *Assume $B_r > 0$ and retain the orthogonal mixed-square constant $\Gamma_\perp > 0$ above. Put $r_H = B_r / \Gamma_\perp > 0$ and, for every real $s > 0$, define*

$$D_{s,p} = \prod_{j=0}^{p-1} (s + 2j), \qquad D_{s,0} = 1.$$

The interaction allowance in (4.18) can be decreased by taking its minimum with

$$(2/\delta)^{1/(2q)} D_{r_H,q}^{1/(2q)} \left(\sqrt{\Gamma_\perp / m} + a_{r_H,q}^H / m \right). \tag{B.12}$$

For $s > 0$, set

$$\omega_{2k}^s = D_{s,k}, \qquad \omega_{2k+1}^s = \sqrt{D_{s,k} D_{s,k+1}}, \qquad C_{p,j}^s = \frac{(2p)!\, \omega_{2p-j}^s}{2p(2p-j)! D_{s,p}},$$

and take

$$a^H_{s,q} = \max_{\substack{2\le p\le q\\ 3\le j\le 2p}} \left(\frac{2C^s_{p,j} b^H_j}{j!\Gamma_\perp}\right)^{1/(j-2)}, \qquad a^H_{s,1} = 0.$$

The higher budgets remain $b^H_j = r^{j/2}A_j$, with the actual head rank r. The real parameter r_H enters only the comparison pairing moments; $\Gamma_\perp$ remains a separate bound for the weak Hilbert covariance. In particular, the aggregate bound Ξ_r does not justify replacing $\Gamma_\perp$ by Ξ_r/r. Both allowances bound the same even moment, so their deterministic minimum needs only one application of Markov's inequality.

Proof. For a test matrix M with $\|M\|_{\mathrm{F}} = 1$, take a singular value decomposition $M = \sum_a \sigma_a u_a v_a^\top$. The orthogonal mixed-square bound and Cauchy–Schwarz give

$$\big(\mathbb{E}\langle M, Y\rangle^2_{\mathrm{F}}\big)^{1/2} \le \frac{\sqrt{\Gamma_\perp}}{F}\sum_a \sigma_a \|W v_a\| \le \sqrt{\Gamma_\perp}.$$

The covariance of Y as a Hilbert vector therefore has operator norm at most $\Gamma_\perp$, while its trace is $\mathbb{E}\|Y\|^2_{\mathrm{F}} \le B_r = \Gamma_\perp r_H$. The quadratic coefficient in the Hilbert expansion is at most

$$p\|x\|^{2p-2}\mathbb{E}\|Y\|^2 + 2p(p-1)\|x\|^{2p-4}\mathbb{E}\langle x, Y\rangle^2 \le p\Gamma_\perp(r_H + 2p - 2)\|x\|^{2p-2}.$$

At $x = 0$, use the polynomial expansion itself, so no negative power is required. All higher coefficients use b^H_j.

Put $s(t) = \sqrt{\Gamma_\perp t} + a^H_{r_H,q}$ and $\mathcal{B}_p(t) = D_{r_H,p}s(t)^{2p}$, with $\mathcal{B}_0 = 1$. The identity $D_{r_H,p}/D_{r_H,p-1} = r_H + 2p - 2$ makes the normalized quadratic term exactly $\Gamma_\perp/(2s)$. By the definition of $a^H_{r_H,q}$, every higher term, divided by $2pD_{r_H,p}s^{2p-1}$, is at most $\Gamma_\perp(a^H_{r_H,q}/s)^{j-2}/(2s)$. The resulting geometric sum is bounded by $\Gamma_\perp/[2(s - a^H_{r_H,q})] = s'(t)$. The same monotone induction and integrable initial singularity as in Appendix A.3 prove the comparison with $\mathcal{B}_q(m)$. Only the displayed product identity for $D_{r_H,p}$ is used, so r_H need not be an integer. For $q = 1$, this is the direct second-moment bound. □

The leading variance coefficient in this comparison is $\sqrt{\Gamma_\perp}D^{1/(2q)}_{r_H,q}$, at most $\sqrt{B_r + \Gamma_\perp(q-1)}$ by AM–GM. The scalar comparison uses $\sqrt{B_r}\,\mathfrak{g}_q^{1/(2q)}$. Take their minimum because the higher-insertion radii need not be ordered. This refinement concerns the residual interaction; it does not remove the variance logarithm from the two-sided embedding theorem.

B.5 Angular moments and full-range error

Proof of Theorem 4.9. Let $z = \bigotimes_j x_j$ have the specified spherical factor law. For one factor, the beta distribution of its first squared coordinate gives

$$\mathbb{E}|\langle u, x_j\rangle|^{2p} = c^{\mathrm{sph}}_{n_j}(p)\|u\|^{2p}.$$

This formula also holds at $n_j = 1$. To tensorize, slice a coefficient tensor as $a = \sum_b a_b \otimes e_b$ and condition on all earlier factors. The last-factor moment formula and Minkowski's inequality in L^p yield

$$\left\|\sum_b |\langle a_b, z_{<d}\rangle|^2\right\|_{L^p} \le \sum_b \big(\mathbb{E}|\langle a_b, z_{<d}\rangle|^{2p}\big)^{1/p}.$$

Induction therefore proves

$$\mathbb{E}|\langle a, z\rangle|^{2p} \le H_{\mathrm{ang}}(p)\|a\|^{2p}, \qquad \mathbb{E}(z^\top B z)^p \le H_{\mathrm{ang}}(p)(\operatorname{tr} B)^p \quad (B \succeq 0).$$

The second inequality follows by diagonalizing B and applying Minkowski once more. Differentiating (4.20) at $p = 1$ gives

$$\lim_{\beta \downarrow 0} H_{\mathrm{ang}}(1+\beta)^{1/\beta} = \varkappa_{\mathrm{ang}}. \tag{B.13}$$

For a nonzero PSD residual R, set $a = z^\top R^2 z$ and $b = z^\top R z$. The numerator vanishes whenever the denominator does. Since $\mathbb{E}a = \operatorname{tr}(R^2)$, the probability measure $dQ = a\, dP/\operatorname{tr}(R^2)$ is well defined and has $b > 0$ almost surely. For $\beta > 0$ and $p = 1 + \beta$, convexity and Hölder give

$$\mathbb{E}\frac{a}{b} = \operatorname{tr}(R^2)\mathbb{E}_Q b^{-1} \geq \operatorname{tr}(R^2)(\mathbb{E}_Q b^\beta)^{-1/\beta}, \qquad \mathbb{E}_Q b^\beta \leq H_{\mathrm{ang}}(p)(\operatorname{tr} R)^\beta.$$

For the last estimate, diagonalize $R^2 = \sum_i \lambda_i v_i v_i^\top$ with unit v_i and apply

$$\mathbb{E}[|\langle v_i, z\rangle|^2 b^\beta] \leq (\mathbb{E}|\langle v_i, z\rangle|^{2p})^{1/p}(\mathbb{E}b^p)^{\beta/p} \leq H_{\mathrm{ang}}(p)(\operatorname{tr} R)^\beta$$

before summing. Letting $\beta \downarrow 0$ proves

$$\mathbb{E}\frac{z^\top R^2 z}{z^\top R z} \geq \frac{\operatorname{tr}(R^2)}{\varkappa_{\mathrm{ang}} \operatorname{tr} R}. \tag{B.14}$$

Zero quotients are interpreted as zero.

Write $K = A^\top A$ and let Z_t contain the first t probe columns. The full Nyström residual is

$$R_t = K - K Z_t (Z_t^\top K Z_t)^\dagger Z_t^\top K.$$

It satisfies $0 \preceq R_{t+1} \preceq R_t \preceq K$, and adding a column gives

$$R_{t+1} = R_t - \frac{R_t z_{t+1} z_{t+1}^\top R_t}{z_{t+1}^\top R_t z_{t+1}}.$$

These identities follow by writing the Nyström approximation as $K^{1/2} P_{\mathrm{range}(K^{1/2} Z_t)} K^{1/2}$ and decomposing the enlarged range into its old part and the orthogonalized new column. For the rectangular problem, cyclicity of trace also gives exactly

$$\operatorname{tr} R_t = \operatorname{tr}\big[A^\top (I - P_{\mathrm{range}(AZ_t)})A\big] = \|(I - P_{\mathrm{range}(AZ_t)})A\|_{\mathrm{F}}^2.$$

Let $f_t = \mathbb{E}\operatorname{tr} R_t$. The rank-$r$ tail of R_t is no larger than τ, because $R_t \preceq K$. Thus

$$\operatorname{tr}(R_t^2) \geq \frac{(\operatorname{tr} R_t - \tau)_+^2}{r}.$$

The function $g(x) = (x-\tau)_+^2/x$, with $g(0) = 0$, is convex: above τ it equals $x - 2\tau + \tau^2/x$, and its derivative agrees with zero at τ. Conditioning (B.14) on R_t and applying Jensen therefore yields

$$f_{t+1} \leq f_t - \frac{(f_t - \tau)_+^2}{\varkappa_{\mathrm{ang}} r f_t}.$$

Before $u_t = f_t/\tau - 1$ reaches ε, this implies

$$u_t - u_{t+1} \geq \frac{u_t^2}{\varkappa_{\mathrm{ang}} r (1 + u_t)}.$$

The decreasing convex function $F(u) = u^{-1} - \log u$ satisfies, as long as $u_{t+1} > \varepsilon$,

$$F(u_{t+1}) - F(u_t) \geq F'(u_t)(u_{t+1} - u_t) \geq \frac{1}{\varkappa_{\mathrm{ang}} r}.$$

If the integer m in (4.21) were reached with $u_m > \varepsilon$, summing this inequality would give $F(u_m) \geq F(\varepsilon)$, a contradiction. Once the target is reached, monotonicity preserves it. This proves (4.22); using $-u_0^{-1} + \log u_0 \leq \log(T/\tau)$ gives (4.23).

Finally, each Gaussian factor is a positive independent radius times its spherical direction. Under this coupling a Gaussian product column is a nonzero scalar multiple of the corresponding spherical product column. All sampled ranges and all full Nyström outputs are therefore unchanged. This establishes the Gaussian assertion as well. □

Corollary B.9 (Expected Frobenius error of the full PSD residual)**.** *Let $K \succeq 0$ be fixed, $T = \operatorname{tr} K$, and $R_m = K - \mathcal{N}_K(\Omega_m)$ for the first m product probes of either law in Theorem 4.9. Then, for every integer $m \geq 0$,*

$$\mathbb{E}\|R_m\|_{\mathrm{F}}^2 \leq \frac{\varkappa_{\mathrm{ang}} T^2}{m+1}. \tag{B.15}$$

This bound concerns the full PSD Nyström residual, with no rank-r truncation.

Proof. The case $T = 0$ is immediate. For $T > 0$, couple the probes to an infinite independent sequence and put $f_t = \mathbb{E} \operatorname{tr} R_t$. Since $\operatorname{tr} R_t \leq T$, the conditional drift (B.14) gives

$$\mathbb{E} \operatorname{tr}(R_t^2) \leq \varkappa_{\mathrm{ang}} T (f_t - f_{t+1}).$$

Moreover $R_t - R_{t+1} \succeq 0$ and $R_t + R_{t+1} \succeq 0$, so

$$\operatorname{tr}(R_t^2) - \operatorname{tr}(R_{t+1}^2) = \operatorname{tr}[(R_t - R_{t+1})(R_t + R_{t+1})] \geq 0.$$

Sum the drift through $t = m$ and use this monotonicity:

$$(m+1)\mathbb{E} \operatorname{tr}(R_m^2) \leq \sum_{t=0}^{m} \mathbb{E} \operatorname{tr}(R_t^2) \leq \varkappa_{\mathrm{ang}} T (f_0 - f_{m+1}) \leq \varkappa_{\mathrm{ang}} T^2.$$

The independent continuation step is used only in the proof; the output R_m uses exactly the first m probes. □

For comparison, $\varkappa_{\mathrm{ang}} \leq \prod_j c_{n_j}^{\mathrm{sph}}(2)$ by log-convexity and $H_{\mathrm{ang}}(1) = 1$. The inequality is strict whenever some $n_j > 1$. At $n_j = 2$, each factor contributes $e/2$ to $\varkappa_{\mathrm{ang}}$ and $3/2$ to the spherical fourth-moment bound. In fact, uniformly over the mode dimensions,

$$\varkappa_{\mathrm{ang}} \leq \left(\frac{e^{2-\gamma_{\mathrm{E}}}}{2}\right)^d, \qquad \frac{e^{2-\gamma_{\mathrm{E}}}}{2} = 2.0743278106\ldots.$$

To see this, put $R^2 \sim \chi_n^2$ and $\mathfrak{g}(p) = 2^p \Gamma(p + 1/2)/\Gamma(1/2)$. Gaussian polar decomposition gives $c_n^{\mathrm{sph}}(p) = \mathfrak{g}(p)/\mathbb{E}(R^2/n)^p \leq \mathfrak{g}(p)$ for $p > 1$. Both sides equal one at $p = 1$, so their right logarithmic derivatives satisfy $(\log c_n^{\mathrm{sph}})'(1) \leq \log 2 + \psi(3/2) = 2 - \gamma_{\mathrm{E}} - \log 2$. Each per-mode factor tends to this upper limit as $n \to \infty$, whereas the Gaussian fourth-moment contribution is 3. These are sufficient constants for the specified full-range error criterion.

B.6 A confidence obstruction for raw regression

Proposition B.10 (Raw regression needs $1/(\varepsilon\delta)$ under fourth moments)**.** *Let $N \geq 2$, $0 < \varepsilon \leq 1/64$, $0 < \delta < 1/40$, and $\Lambda \geq 400$. For every integer m with*

$$1 \leq m < \frac{15\Lambda}{2048\,\varepsilon\delta},$$

there are an isotropic probe law satisfying (1.3) with $C_4 \leq \Lambda + 4$, a unit design $a \in \mathbb{R}^N$, and a unit response $b \perp a$ such that the sketch-and-solve solution of Theorem 4.5 from m iid probes satisfies

$$\mathbb{P}\{\|a\widetilde{x} - b\|^2 > 1 + \varepsilon\} > \delta.$$

The optimal squared residual is one. Hence any sketch size that is sufficient uniformly over isotropic laws with $C_4 \le \Lambda + 4$ must be at least $15\Lambda/(2048\varepsilon\delta)$ in this regime: for raw regression, the term $8\gamma(C_4)r/(\varepsilon\delta)$ in (4.12) cannot be replaced by one logarithmic in $1/\delta$.

Proof. First suppose $m \ge \Lambda/(4\varepsilon)$. Let $a = e_1$, $b = e_2$, and $w = (e_1 + e_2)/\sqrt{2}$, and put

$$B^2 = 8\sqrt{\varepsilon}\, m, \qquad \pi = \Lambda/B^4, \qquad \Sigma = (1-\pi)^{-1}(I_N - \pi B^2 w w^\top).$$

Since $m \ge \Lambda/(4\varepsilon)$, we have $\pi \le \varepsilon/(4\Lambda)$, $\pi B^2 \le 1/2$, and $m\pi \le 1/16$. Each probe equals $\sigma B w$ with probability π, where σ is an independent fair sign, and otherwise equals $\Sigma^{1/2} g$ with $g \sim N(0, I_N)$. Then $\mathbb{E} z z^\top = \pi B^2 w w^\top + (1-\pi)\Sigma = I_N$, and for unit u,

$$\mathbb{E}\langle u, z\rangle^4 \le \pi B^4 + 3(1-\pi)\|\Sigma\|^2 \le \Lambda + \frac{3}{1-\pi} \le \Lambda + 4.$$

Write $a_i = \langle e_1, z_i\rangle$ and $b_i = \langle e_2, z_i\rangle$. The sketched problem gives $\widetilde{x} = \sum_i a_i b_i / \sum_i a_i^2$ whenever the denominator is positive, and the original squared residual is $1 + \widetilde{x}^2$. It therefore suffices to show $\widetilde{x} > \sqrt{\varepsilon}$ with probability greater than δ. Let E be the event that exactly one probe, with index J, is of the first kind. Then $a_J b_J = a_J^2 = B^2/2 = 4\sqrt{\varepsilon}\, m$ and

$$\mathbb{P}(E) = m\pi(1-\pi)^{m-1} \ge m\pi(1 - m\pi) \ge \tfrac{15}{16} m\pi.$$

Conditional on E and J, the other $m-1$ pairs (a_i, b_i) are iid centered Gaussians with $\mathbb{E} a_i^2 = s(1-\alpha)$ and $\mathbb{E} a_i b_i = -s\alpha$, where $s = (1-\pi)^{-1} \le 1 + 2\pi$ and $\alpha = \pi B^2/2$. The sums $S_{ab} = \sum_{i \ne J} a_i b_i$ and $S_{aa} = \sum_{i \ne J} a_i^2$ have variances at most ms^2 and $2ms^2$. With $t = \sqrt{\varepsilon}\, m/2$, Chebyshev's inequality shows that both lie within t of their means, except with conditional probability at most $12 s^2/(\varepsilon m) \le 1/2$. On this event $ms\alpha = s\Lambda/(16\sqrt{\varepsilon}) \le t$, and since $\sqrt{\varepsilon} \le 1/8$,

$$\widetilde{x} = \frac{B^2/2 + S_{ab}}{B^2/2 + S_{aa}} \ge \frac{4\sqrt{\varepsilon}\, m - 2t}{4\sqrt{\varepsilon}\, m + sm + t} \ge \frac{3\sqrt{\varepsilon}}{9/16 + s} > \sqrt{\varepsilon}.$$

Hence the failure probability is at least $\frac{1}{2}\mathbb{P}(E) \ge 15\Lambda/(2048\varepsilon m)$, which exceeds δ in the stated range.

It remains to treat $m < \Lambda/(4\varepsilon)$. Put $p_0 = \max\{3/\Lambda, 1/m\}$, and let a probe equal $p_0^{-1/2} g$, with $g \sim N(0, I_N)$, with probability p_0, and zero otherwise. It is isotropic and its directional fourth-moment constant is $3/p_0 \le \Lambda$. Keep $a = e_1$ and $b = e_2$. The number of active probes is $K \sim \mathrm{Binomial}(m, p_0)$, with $1 \le \mu := \mathbb{E} K < 3/(4\varepsilon)$. Conditional on $K = k \ge 1$, the minimizer is unique almost surely and has the distribution $G/\sqrt{X_k}$, where $G \sim N(0,1)$ and $X_k \sim \chi^2_k$ are independent. Denote the standard Gaussian distribution function by Φ_{N}, and set $f(t) = 2\Phi_{\mathrm{N}}(-\sqrt{\varepsilon t})$ for $t \ge 0$. This function is convex, since, for $t > 0$,

$$f''(t) = \frac{\sqrt{\varepsilon}\, \phi_{\mathrm{N}}(\sqrt{\varepsilon t})}{2}(t^{-3/2} + \varepsilon t^{-1/2}) > 0,$$

where ϕ_{N} is the Gaussian density. Applying Jensen's inequality first to X_k and then to K, and discarding $K = 0$, gives

$$\mathbb{P}\{|\widetilde{x}| > \sqrt{\varepsilon}\} \ge \mathbb{E} f(K) - \mathbb{P}(K = 0) \ge f(\mu) - e^{-\mu}.$$

For $1 \le \mu \le 4$, this is at least $1 - 1/(2\sqrt{2\pi}) - e^{-1} > 1/4$; for $\mu > 4$, it exceeds $1 - \sqrt{3/(2\pi)} - e^{-4} > 1/4$. Here we used $\phi_{\mathrm{N}}(t) \le 1/\sqrt{2\pi}$, $\varepsilon \le 1/64$, and $\varepsilon\mu < 3/4$. Thus failure exceeds δ throughout the remaining range. The event $K = 0$ was not used, so the conclusion holds for every choice of a minimizer on singular samples. □

B.7 Regression under Gaussian scale mixtures

Proof of Proposition 4.13. Use an orthonormal basis for range A and the unit residual $e/\|e\|$. Orthogonality and Gaussian rotational invariance give independent design coordinates ξ_i and residual coordinates η_i. The normal equations and Pythagoras give (4.28); S_m is invertible almost surely for $m \geq r$. Conditioning leaves only the independent Gaussian η_i, proving (4.29). Conditional on (w_i, ξ_i), $\mathcal{E}_m^{\mathrm{reg}}$ has the law $\sum_j \lambda_j(C_m) G_j^2$. Gaussian integration gives, for $0 < s < (2\|C_m\|)^{-1}$,

$$\log \mathbb{E}[e^{s(\mathcal{E}_m^{\mathrm{reg}} - \operatorname{tr} C_m)} \mid (w_i, \xi_i)] \leq \frac{s^2 \operatorname{tr}(C_m^2)}{1 - 2s\|C_m\|}.$$

The exponential Markov inequality proves (4.30). The strong law gives $S_m/m \to I_r$ and $m^{-1}\sum_i w_i^2 \xi_i \xi_i^\top \to \mu_2 I_r$. The asserted covariance limit follows, and the conditional Gaussian characteristic functions give the distributional limit.

The projected scale-mixture row has directional fourth-moment constant $C_w = 3\mu_2$. The injection proof needs only isotropy and directional fourth moments of the projected rows. Apply Lemma 4.2 in $\mathbb{R}^r$, with isometry I_r and rows $\sqrt{w_i}\xi_i$, to obtain $S_m \succeq (m/2) I_r$ except with probability $\delta/2$. On this event and $\max_i w_i \leq b_m(\delta)$,

$$C_m \preceq b_m(\delta) S_m^{-1} \preceq \frac{2b_m(\delta)}{m} I_r.$$

For standard $h \in \mathbb{R}^r$, the Gaussian norm bound $\mathbb{P}\{\|h\| > \sqrt{r} + \sqrt{2u}\} \leq e^{-u}$, with $u = \log(4/\delta)$, gives $\mathcal{E}_m^{\mathrm{reg}} \leq 2b_m(\delta)(\sqrt{r} + \sqrt{2u})^2/m$ on the conditional event. A union bound proves the claim. When $e = 0$, the original residual is zero almost surely for $m \geq r$, so no ratio is needed. □

For the common-factor Gaussian law, $\mu_2 = 3^{d-1}$ and one may take

$$b_m(\delta) = \big[2\log(8(d-1)m/\delta)\big]^{d-1}.$$

Indeed, a union bound over the $(d-1)m$ scalar Gaussian factors using $\mathbb{P}\{G^2 > 2v\} \leq 2e^{-v}$ controls every factor. For the order-two common-Schmidt law, write $\rho = \sum_b \lambda_b^2$ and $L = \max_b \lambda_b$. Then

$$\mu_2 = 1 + 2\rho, \qquad b_m(\delta) = 1 + 2\sqrt{\rho \log(4m/\delta)} + 2L\log(4m/\delta),$$

by the scalar Gaussian tail estimate in Appendix A.7. Thus the certificate has polylogarithmic confidence cost in these two support classes. For example, in the common-factor case the explicit sufficient order is

$$m \geq C_d \frac{r + \log(1/\delta)}{\varepsilon} \left[\log \frac{e(r + \log(1/\delta))}{\varepsilon\delta}\right]^{d-1}.$$

To close the logarithm, insert this expression as M: its logarithm is at most a d-dependent multiple of the bracketed logarithm, plus $\log C_d$. Increasing C_d makes every term in (4.31) at most M; the function $\log^{d-1}(cm/\delta)/m$ is decreasing thereafter. The same argument with exponent one applies to the common-Schmidt law, uniformly in its weights. These are sufficient bounds, with no claim of sharpness in rank, accuracy, or confidence.